

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
JNANA SANGAMA, BELAGAVI - 590018



A Project Report on

“Design And Analysis of Various Swirler Configuration for Non-Premixed Combustion”

Submitted in partial fulfilment of the requirements for the award of the degree of

Bachelor of Engineering

In

Department of Aeronautical Engineering

for the Academic Year: 2025-26

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This is to certify that the project work entitled “*Design And Analysis Of Various Swirler Configuration For Non-Premixed Combustion*” has been carried out by **Vaishnavi Khuba (1NT22AE107)**, **Hari Haran V (1NT23AE404)**, and **Suchinth K G (1NT23AE413)** bonafide students of **Nitte Meenakshi Institute of Technology**, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering** in the **Department of Aeronautical Engineering** under **Visvesvaraya Technological University**, Belagavi, during the academic year **2025–2026**.

The project report has been examined and approved as it meets the academic requirements specified under the autonomous scheme of **Nitte Meenakshi Institute of Technology** for the said degree.

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I hereby declare that the project entitled “**Design and Analysis of Various Swirler Configuration for Non-Premixed Combustion**” was carried out under the guidance of **Dr. Srikanth H V**, Professor & HoD, Department of Aeronautical Engineering, Bengaluru. I also like to thank **Dr. Srikanth H V**, Professor and Head of Aeronautical Engineering Department, NMIT, for providing us with the useful resources. This work is not previously the basis for award of any degree nor has been submitted elsewhere for the award of any degree.

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Abstract

Swirlers are widely used in modern combustion systems to enhance fuel–air mixing, improve flame stabilization, and reduce pollutant emissions. In non-premixed combustion systems, fuel and oxidizer enter the combustion chamber separately, making efficient mixing between them a critical factor for achieving stable and efficient combustion. The introduction of swirl into the airflow produces strong turbulence and forms a central recirculation zone inside the combustor. This recirculation zone plays an important role in stabilizing the flame and improving combustion efficiency.

The present study focuses on the design and computational analysis of axial swirler configurations with different blade counts. Three swirler geometries consisting of 8 blades, 12 blades, and 16 blades were modelled and analysed to evaluate their influence on the internal flow characteristics of the combustion chamber. The investigation was carried out using Computational Fluid Dynamics (CFD) simulations performed in ANSYS Fluent.

The numerical simulations were used to examine important flow parameters such as velocity distribution, pressure variation, temperature distribution, turbulence kinetic energy (TKE), and nitrogen oxide (NO_x) emission formation. The simulation results were compared for each swirler configuration to understand how blade count affects swirl strength, turbulence intensity, and combustion behavior.

The results indicate that increasing the number of blades increases swirl intensity and turbulence within the combustion chamber, which enhances fuel–air mixing. However, higher blade counts also lead to increased pressure losses in the system. Among the configurations studied, the 12-blade swirler demonstrated the most balanced performance, providing improved mixing characteristics while maintaining moderate pressure drop and acceptable emission levels.

Overall, this study provides valuable insights into the influence of swirler geometry on combustion performance and can contribute to the design of more efficient and low-emission combustion systems used in gas turbine engines and industrial burners.

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Chapter 1 Introduction

Fuel mixing and combustion, one of the objectives of a combustion system and oxidizer while stabilizing the flame and minimizing the emissions of pollutants. In gas used in turbine engines and industrial burners, swirlers are commonly employed to enhance the mixing influence and stabilize the flame in the combustion chamber. A swirler adds rotational maybe it is this motion to the incoming airflow that creates a very strong turbulence and makes, in consequence, a very centralized recirculation zone that assists maintaining combustion in the combustor. Swirling flows are critical for non-reactors with premixed combustion where both, fuel and oxidizer are separately introduced. In these systems, the efficiency of combustion is highly dependent on the mixing process.

The swirl improves the mixing between fuel and air, improving both flame stability, this leads to the higher heat release rate and better overall combustion efficiency. But the geometric features of a swirler tremendously impact its performance, input variables like number of blades, angle of the blade, hub diameter. These features influence swirl strength, turbulence intensity, pressure loss and emission generation in the combustor. As such, designing the swirler (and its integration into the flamelet) is critical to realizing efficient and stable combustion. In recent years, Computational Fluid Dynamics (CFD) has emerged as an essential tool to understand complex combustion phenomena. CFD helps researchers visualize patterns of flow, distribution of temperature, intensity of turbulence, and pollution production within combustion chambers. Different swirler designs can be assessed using numerical simulations designs without extensive experimental testing.

The scope of this study lies in the design and evaluation of axial swirlers with different blade configurations using CFD simulations. It will have three configurations: 8 blades, 12 blades and it is analyzed and modelled for 16 blades. & blade count to see how it affects characteristics, velocity distribution, pressure drop, temperature distribution, turbulence kinetic energy and NOx emissions. The analysis is performed with ANSYS Fluent, which allows for a comprehensive analysis of fluid dynamics and chemical reactions processes. The study seeks to identify the best swirling design for maximum mixing effectiveness with the minimum pressure loss. This study provides additional insights into swirler design thereby contributing to its effect on combustion performance. The discovery may help with improving the effectiveness and burner technology, hydrodynamics and combustion of pulverized coal in gas turbine combustors and other industrial systems

1.1 Motivation

Reducing the emissions of pollutants in modern propulsion and power production systems. Swirlers are commonly employed in combustors to promote mixing and stabilize flames, but the underlying performance is significantly influenced by geometric design parameters.

A swirler's number of blades contributes to one of its key design parameters, which then directly affects the swirl intensity and turbulence produced in-side the combustion chamber. Increasing the number of blades tends to create better mixing and turbulence, but it could also give rise to elevated pressure losses and higher aerodynamic drag.

Hence, it is necessary to study the effect of different blade configurations on the flow dynamics and combustion characteristics inside the combustor. Understanding these effects assist engineers in designing swirlers that guarantee efficient combustion to work at efficiency but with still acceptable pressure-counterbalance and exhaust levels.

This study attempts to assess the performance of various swirler simulations to detect the most advantageous configuration

1.2 Role of Swirl in Non-Premixed Flames

A major challenge in diffusion flames is that the flame front is difficult to stabilize because the restricted reactant mixing and potential for blow-off at high velocities. Swirlers impart a high tangential velocity component to the incoming air stream, producing a core cross-sectional recirculation zone (CRZ) below the swirler. This CRZ acts as a constant ignition source by entraining hot combustion products back into the incoming fuel-air region, allowing for continued combustion even under lean conditions.

In non-premixed combustion, this swirl-induced recirculation facilitates not only more enhanced enhancing both flame stability and residence time of reactants for an efficient combustion for increasing combustion efficiency and decreasing unburned hydrocarbons (UHC). However, the extent of swirl must be tuned carefully—too much swirl can cause flame lift-off or too high an inadequate swirl can lead to poor mixing and sloppy combustion, while excess swirl creates pressure losses.

1.3 Importance of Swirler Configuration in Diffusion Combustion

One well-known characteristic of the non-premixed combustor performance is its strong dependence on the geometry and configuration of the swirler. The swirl intensity is influenced by the design of the swirler, flow splitting and dimensions of recirculation regions.

Several configurations are commonly employed:

- **Axial Swirlers:**

Vaned axial swirlers are built from vanes at an angle to the flow that are mounted in the path of the flow, axial direction. By changing vane angles (usually 30° to 60°) and the number of the vanes, from which designers can effectively manipulate the intensity of swirl and flow distribution. In the case of non-premixed combustion, axial swirlers provide good fuel-air interaction and are responsible for connecting diffusion flames in the vicinity of injector exit.

- **Radial Swirlers:**

Radial swirlers direct air radially in (or out) before it leaves axially into the combustion chamber that generates a strong vortex and vigorous mixing gasoline C-H-C region due to its suitability for heavy fuels or systems that need improved atomization. These are often employed in burners for which the air must envelop the fuel jet to get diffusion-controlled combustion.

- **Tangential Swirlers:**

Tangential swirlers inject air through tangential slots or holes on the burner periphery. The resulting swirl motion, gives a high swirl number and a strong central vortex. These swirlers work well with large-scale diffusion flames, such as those in industrial furnaces and boiler burners.

1.4 Flame Stabilization Mechanisms

Swirlers aid flame stabilization in non-premixed systems through three primary mechanisms:

1. **Recirculation of Hot Products:** The central recirculation zone transports high-temperature gases upstream, promoting continuous ignition of incoming fuel-air mixtures.
2. **Shear Layer Formation:** Strong velocity gradients between the recirculating core and surrounding flow generate turbulence, improving mixing between fuel and air.
3. **Pressure Gradient Effects:** The adverse pressure gradient generated by swirl causes flow reversal near the center line, forming a low-pressure core that anchors the flame.

The effectiveness of these mechanisms depends on the swirl number (S), a non-dimensional parameter defined as:

$$S = \frac{G_o}{RG_x}$$

Where, G_θ is the axial flux of angular momentum, G_x is the axial flux of linear momentum, and R is the characteristic radius. In non-premixed combustion, moderate swirl numbers (typically between 0.4 and 1.0) are found to provide the best balance between flame stability and pressure loss.

1.5 Influence on Combustion Efficiency and Emissions

The swirler arrangement influences not only flame shape and stability but also combustion efficiency and pollutant generation:

NOx Formation: The high temperatures in the recirculation zone can lead to thermal NOx formation, but better mixing will reduce peak temperatures and overall NOx emissions.

CO and UHC emissions Low levels of CO and unburned hydrocarbons in a combustor imply proper mixing resulting in uniform combustion at the bus-cycle temperature.

Soot formation: In non-premixed flames, swirl enhances the fast oxidation of soot precursors as more amount of oxygen is available around flame zone.

Chapter 2

Literature Survey

i. **Design And Study on Performance of Axial Swirler for Annular Combustor by Changing Different Design Parameters.**

Authors: Bhupendra Khandelwal, Dong Lili, Vishal Sethi.

Objective: This work is aimed to develop a systematic design approach for axial swirlers by varying the number, and angle of vanes along with mass-flow conditions of a neutral gas in order to determine the effect of these geometric parameters on swirl strength, flow structure and pressure drop as well in an annular gas-turbine combustor.

Methodology: Multiple configurations of axial swirlers are developed using pressure-loss and drag-coefficient methods. Conducted CFD simulations for analysis of flow fields, pressure drop, turbulence and recirculation. Different turbulence model comparison in terms of swirl flows.

Results: Swirler performance varied drastically with respect to vane angle, vane count, and mass flow. Four unique axial and radial velocity distributions were found. The double-peaked structure, which was less pronounced downstream, persisted for varying turbulence distributions. This study established an encompassing theoretical + empirical procedure to design practical axial swirlers.

ii. **CFD Simulation Analysis of Non-Premixed Combustion Using a Novel Axial-Radial Combined Swirler for Emission Reduction Enhancement**

Authors: Nor Afzanizam Samiran, Khalil Firdaus Abu Mansor, Razlin Abd Rashid, Muhammad Suhail Sahul Hamid.

Objective: This work aims to experimentally evaluate a novel axial-radial combined swirler and test whether the memetic arrangement of the swirl types compares favorable for fuel–air mixing intensity and consequently, emission characteristics in non-premixed combustion.

Methodology: Built axial and radial swirlers (8 vanes each) and a mix configuration in SolidWorks. Using standard $k-\epsilon$ turbulence model performed CFD simulations in ANSYS Fluent. Fuels: LPG (30% propane, 70% butane) Flow fields, temperatures and emission outputs (CO , CO_2).

Results: The utilization of combined swirler held the advantage of producing more extensive high-temperature zones and higher axial velocity. Because of the more complete conversion to CO, there was a significant reduction in CO. Peak temperatures were marginally higher with the axial-only swirler. The authors recommend increasing the inclination of axial vanes in combined design, if required, to increase flame temperature.

iii. **Optimized Design of a Swirler for A Combustion Chamber of Non-Premixed Flame Using Genetic Algorithms**

Authors: Daniel Alejandro Zavaleta-Luna, Marco Osvaldo Vigueras-Zúñiga, Agustín L. Herrera-May, Sergio Aurelio Zamora-Castro, María Elena Tejeda-del-Cueto.

Objective: Focusing on genetic algorithms, this study aims to maximize the swirl number at inlet of an axial swirler while achieving a high degree of recirculation to improve combustion efficiency and minimize emissions (CO + NO_x).

Methodology: Use genetic algorithms, to address the optimization for vane angle, number of vanes, area of swirler and mass-flow related parameters. CFD simulations, using an RNG k- ϵ model and a mixture-fraction formulation of non-premixed combustion. Baseline vs. optimized design comparison: flow, temperature fields and recirculation.

Results: Optimized swirler produced a longer and stronger recirculation zone. Airflow simulation results closely matched experimental measurements of temperature distribution, confirming the accuracy of the model. Introduced turbulence and mixing decrease CO and NO_x prediction formation considerably. GA-based optimization techniques are effective for practical combustor design.

iv. **Numerical And Experimental Investigation of a Non-Premixed Double Swirl Combustor**

Authors: Jiming Lin, Ming Bao, Feng Zhang, Yong Zhang, Jianhong Yang

Objective: Explore the effects of outer-swirler vane angle and equivalence ratio on flow behavior, combustor temperature, and NO emission in a double-swirl non-premixed burner.

Methodology: Extreme tests were performed on a full-size combustor with the aid of thermocouples and flue-gas diagnostics. Conducted RANS CFD with realizable k- ϵ turbulence model and methane-air two-step chemistry. Various angles of the

outer-swirler vanes were tested and different tertiary-air equivalence ratios investigated.

Results: Loss of central recirculation zone due to draft-fan-induced in-flow distribution. When the angle of outer-swirler blade increased, NO emissions decreased continuously. Elevating tertiary-air equivalence ratio raised average furnace temperature 100 K and reduced NO emissions 25%. The experimental measurements agreed well with the numerical results.

v. Numerical Analysis of Flow Characteristics of a Swirl-Stabilized Premixed Burner with Different Swirl Vane Configurations

Authors: Hae-ji Ju, Ju Hyeong Cho, Jeong Jae Hwang

Objective: Comparison of flat-vane and curved-vane axial swirlers in a premix burner through study of pressure loss, mixing uniformity, NO_x emissions and jet penetration characteristics.

Methodology: Simulated premixed burner, upstream of flat-vane and curved-vane axial swirlers. CFD simulations for mixing quality, pressure drop, flame structure & NO_x formation. Enumeration of swirling crossflow conditions on jet trajectory behavior.

Results: Curved vane enhanced pressure loss decreased by 35% through suppressed flow separation. NO_x was reduced by 69%, thanks to more consistent mixing and fewer local hotspots. Flat vanes resulted in more uniform outlet velocity profile while produced higher NO_x. Existing crossflow formulas do not work, demanding new models for prediction of jet penetration in swirling flow.

vi. Experimental Analysis of Flow Through Rotating Combustion Swirler with Zero Degree Inlet and Outlet Angle of Guide Vane

Author: Mansha Kumari

Objective: By applying five-hole probe to an experimental approach for the study of flow characteristics and velocity/pressure distributions in a rotating ones every eight blades with no guide vane angles.

Methodology: Designed a rotating 8-blade swirler (45° vane angle) with a guide vane stator at 0°/0°. Carried out two experiments using a five-hole probe at various stations below the diffuser. Analyzed the recirculation area based on measurements of swirl intensity and flow structure.

Results: Flow demonstrated significant swirl with visible toroidal central recirculation. Guide vanes at 0° showed less deformation, which guarantees a stable flow rate entering the incident region. They could then use the data to show how swirl number influenced geometry of recirculation, and flow reversal. Characterization of recirculation and turbulence on this configuration swirler was provided.

2.1 Research Gap

Most studies have considered single swirler configurations, thus few compare multiple swirler types for similar operating conditions. In addition, the relationship between geometric parameters (such as vane angle, blade number and hub ratio), and performance indicators (glow number, pressure drop and recirculation length) is not well-studied. This gap is addressed in this study, which systematically evaluates the performance of different swirler designs through both analytical and numerical analysis.

2.2 Objectives

The main objective of the project is to study the effect of swirler blade configurations on a non-premixed combustion system flow dynamics and time-averaged combustor flame performance. In combustion chambers, swirlers help stabilize flames and promote fuel-air mixing. Hence, the effect of blade count on swirl intensity and flow behavior needs to be investigated so as to optimize combustion efficiency and reduce emissions.

The objective of this work is to find the best combination of the above-mentioned parameters in terms of mixing, stability and pressure losses consequently.

- CAD modelling tools to create axial swirlers with 8 blades, 12 blades, and 16 blades.
- Create a combustion chamber model containing the swirler geometry for numerical analysis.
- To simulate the flow characteristics within a combustor using Computational Fluid Dynamics (CFD) Solver ANSYS Fluent
- To investigate significant flow parameters such as velocity distribution, pressure profile, temperature contours, turbulence kinetic energy (TKE), and contaminant formation.
- Evaluation of swirl intensity, mixing efficiency, and pressure drop for the performance comparison of combinations.
- To determine and characterize the swirler configuration which offers better usage of turbulence generation versus aerodynamic drag properties.

Chapter 3 Design Approach and Methodology

3.1 Methodology

In this study, a computational methodology on the design and analysis of various axial swirling configurations used in non-premixed combustors are discussed. The intention of this methodology is to assess the effect of swirler blade number on the flow patterns and combustion performance within the combustor.

The process in general consists of geometry modelling, mesh generation, CFD simulation and result post-processing. Different swirler configurations 8,12 and 16 blades were designed and analyzed. Using ANSYS Fluent, these configurations were incorporated into a combustion chamber model for simulating the flow behaviors and combustion characteristics.

A systematic methodology was adopted to deliver a proper model and accurate simulation results. The steps in the process involved geometry creation, mesh generation, boundary condition setup at various simulation settings, numerical simulation and finally result analysis.

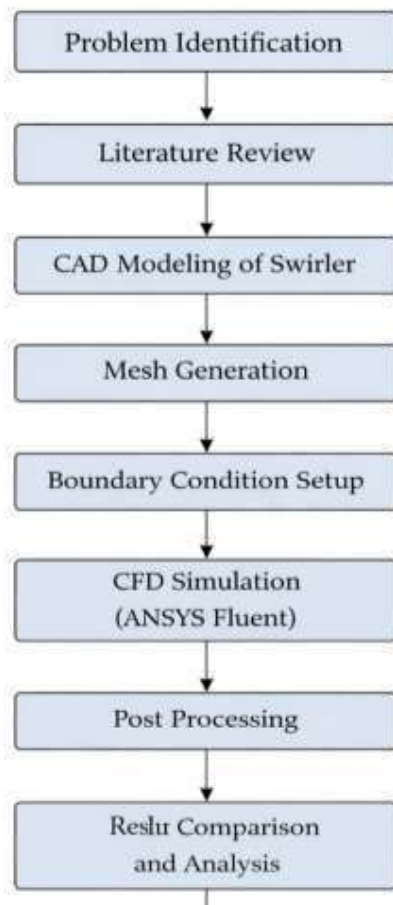


Fig.3.1 Methodology flow chart

3.2 Geometry Modelling

All the designs (including its geometry, integration of swirler with Micro gas turbine combustion chamber model) of Mechanical Parts were designed in SolidWorks software. An axial swirler was designed which consisted in three different configurations; one with 8 blades, another with 12 blades and the last with 16 blades.

Some of the design parameters taken into account during the geometry modelling stage:

- Outer diameter of the swirler
- Hub diameter
- Blade thickness
- Blade spacing
- Number of blades

The CAD models were made with swag so that the geometry closely resembled swirler setups utilized in combustion systems.

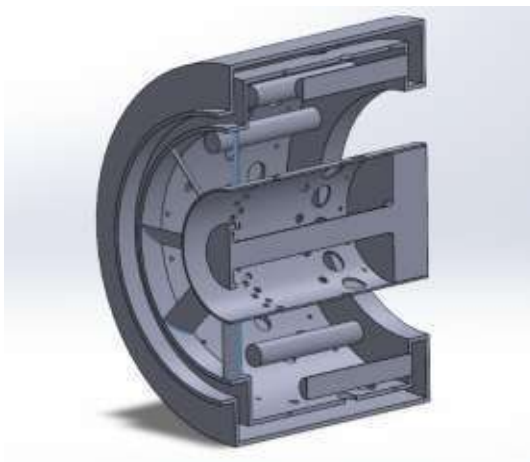


Fig.3.2(a) 8 blade swirler with combustor Cross-sectional view

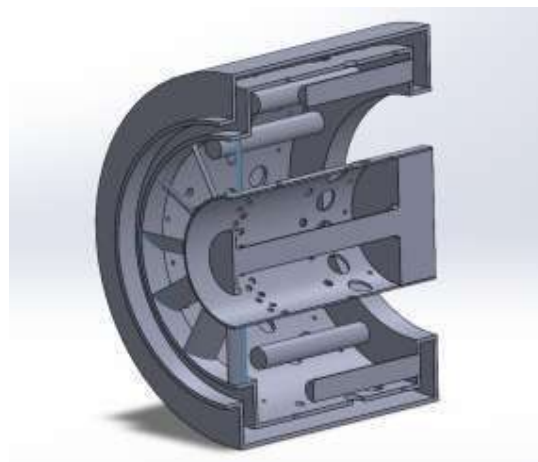


Fig.3.2 (b) 12 blade swirler with combustor Cross-sectional view

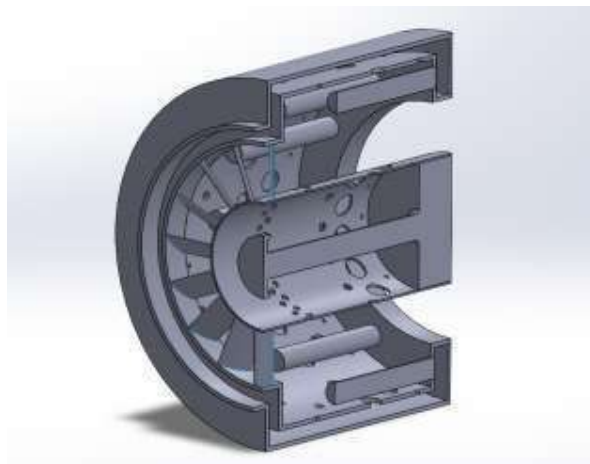


Fig.3.2 (c) 16 blade swirler with combustor Cross-sectional view

3.3 Mesh Generation

Following the design of geometry, ANSYS Meshing was used to discretize the computational domain. First, mesh generation is essential in the CFD simulation because it strongly affects the accuracy of the numerical solution.

A fine mesh was implemented at critical regions like:

- Swirler blades
- Combustion chamber inlet
- Recirculation zones

The mesh was refined to resolve relevant flow features such as swirl motion, turbulence generation, and mixing behavior.

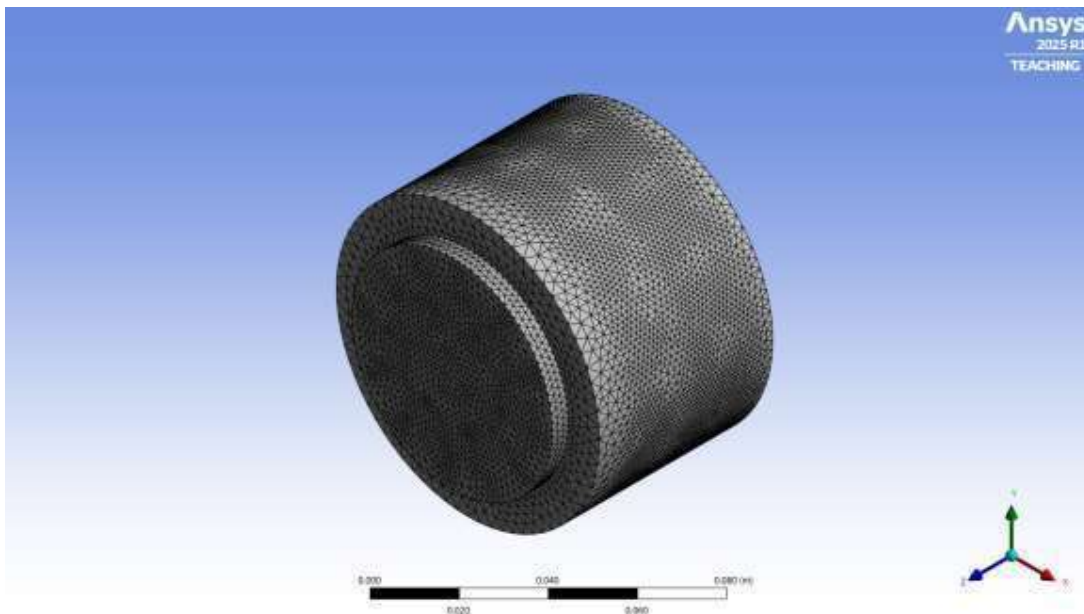


Fig.3.3 Meshing.

Geometry	Type	Cells	Faces	Nodes	Mini Orthogonal quality	Max Aspect Ratio
8 Blade	Tet Cell	777284	1604176	153460	0.00023717044	105.38065
12 Blade	Tet Cell	805308	1662384	159149	9.0590933e-05	196.94703
16 Blade	Tet Cell	834866	1723302	164913	0.0001897783	166.75082

Table 3.3 Mesh size and quality

3.4 Boundary Conditions and Solver Setup

The CFD simulations were performed using ANSYS Fluent and Appropriate boundary conditions were applied to simulate the airflow and combustion environment inside the combustor.

The main simulation parameters used in the analysis are summarized in **Table 3.4**

Table.3.4 Simulation Parameters

Parameter	Value
Solver Type	Pressure-Based
Flow Model	Turbulent Flow
Turbulence Model	k- ω SST Model
Energy Equation	Enabled
Combustion Model	Non-premixed
Simulation Tool	ANSYS Fluent

The inlet boundary condition was defined as a velocity inlet, while the outlet was defined as a pressure outlet.

3.4.1 BOUNDARY CONDITION:

i. Inlet:

- Inlet Type – Mass Flow Inlet - 0.49 kg/S
- Temperature -750 K
- Species- Mean Mixture Fraction - 1
- Turbulence Intensity - 5%

ii. Fuel Inlet:

- Fuel Flow Inlet Type - Mass Flow Inlet - 0.02 kg/S ,
- Turbulence Intensity - 10%
- Total Temperature - 310 K

iii. Outlet:

- Outlet-Gauge Pressure - 0 Pa
- Turbulence Intensity - 5%
- Back Flow Temperature - 300 K

3.4.2 METHODS

- First order – 100 iterations
- Second order – 100 iterations both without pollutants
- Second order – 800 iterations with pollutants Nox and Soot

3.4.3 POLLUTANTS

- a. Soot – Mass Brookes
- b. Nox – Thermal Nox
- c. Radiation -P1
- d. INITIALIZATION – Hybrid
- e. NO. OF ITERATIONS – 1000

3.5 CFD Simulation

CFD simulation was performed to study flow characteristics in combustion chamber. The numerical simulation solved the governing equations of fluid flow, wherein:

- Continuity equation
- Momentum equation
- Energy equation
- Turbulence equations

The numerical results contain extensive information on the flow field and combustion properties in the chamber.

3.6 Post Processing and Result Analysis

Once the simulations were running, the post-processing tools in ANSYS Fluent were used to analyze the results.

Each swirler configuration was tested for the following parameters:

- Velocity distribution
- Pressure variation
- Temperature contours
- Turbulence kinetic energy (TKE)
- NOx emission levels
- Soot accumulation levels

Finally, the performance calculated from CFD simulations for all three configurations was compared to evaluate the effect of number of blades on hardcore swirl and resultant combustion performance.

Chapter 4 Result and Analysis.

4.1 Post Processing CFD Results

4.1.1 Pressure Contour Comparison

4.1.1.1 Pressure Contour 8 Blade

a. Inlet Pressure Contour:

The image shows Pressure contour distribution of the flow using ANSYS .The coloured scale corresponds to pressure values in Pascals, i.e., high-pressure coloured red and orange compared to lower pressure colours (green) and water regions (blue). From the contour, it can be stated that maximum pressure is observed in a central hub region which gets created due to the high compression and rotational value of fluid at centre itself. The air passing through the channels shaped by each of the vanes, as it advances out, there is some loss of pressure, represented by the change in colours from yellow and green to blue shades. It indicates that the pressure across all the vane passages remains uniform, and hence a stable and balanced flow pattern is formed. The change in pressure from the centre to outer end, gradually increasing towards the outer region and centre of fluid toward it is a common characteristic of rotating aerodynamic components where the flow stream expands and accelerates as progress along flow path.

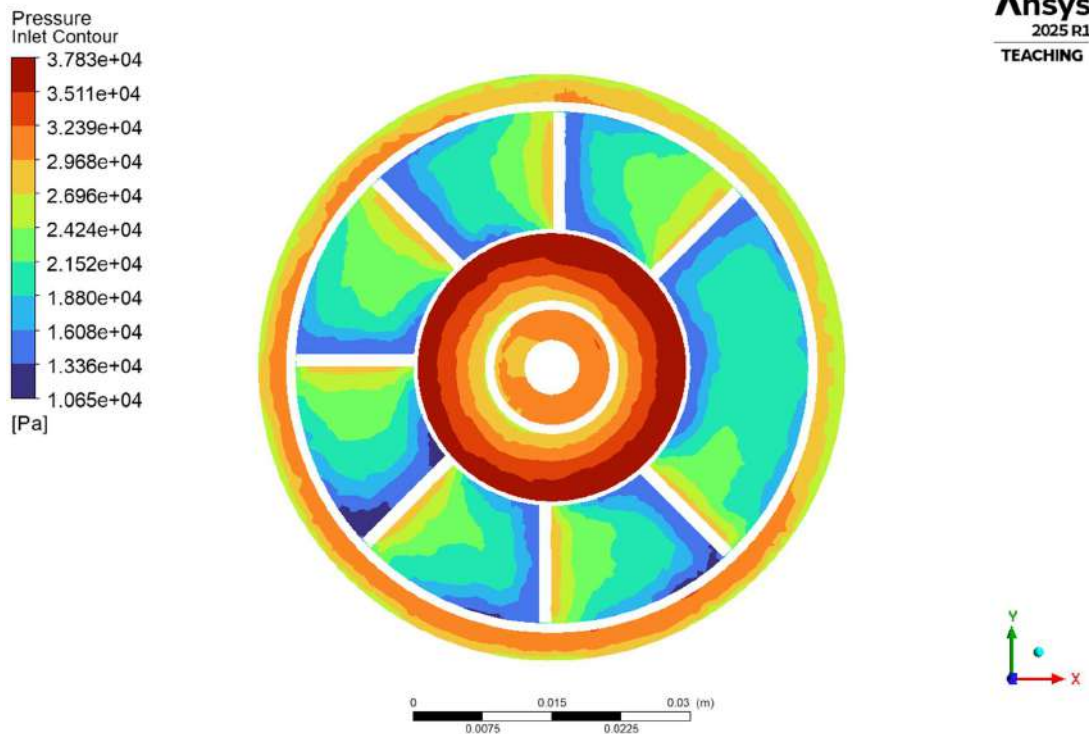


Fig.4.1.1.1(a) Inlet Pressure Contour.

b. Zone 1 Pressure Contour:

We get to know from the colouring scale, red regions have more with pressure than blue ones in Pascals. The maximum pressure data is then represented on the circular contour by displaying a red area in the middle, showing that most of the pressure occupies the hub region. The pressure will drop outward (with blue colours conveying the value in a region surrounding it) as one moves further away from its central core.

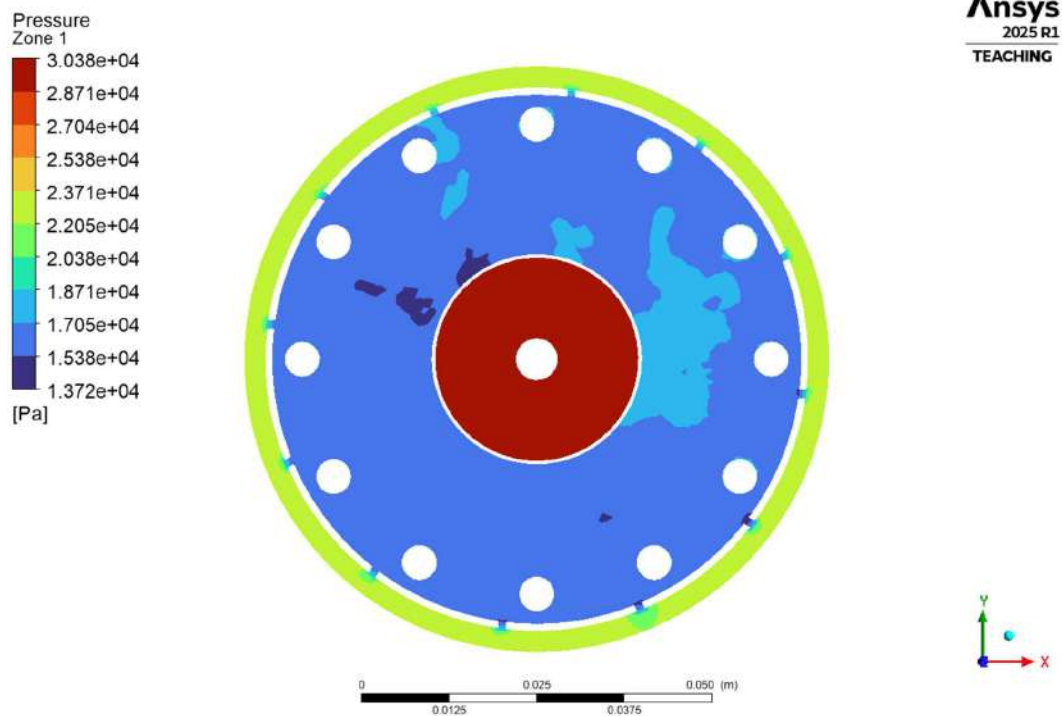


Fig.4.1.1.1(b) Zone 1 Pressure Contour.

These little round holes surrounding the outer rim are openings or passages that control the flow distribution. The blue region has slight fluctuations where pressure is affected by the fluid shape and its movement in such a shape. In general, the contour shows that pressure is maximum in centre and decreases as we move outwards suggesting stable pressure distribution within the system.

c. Zone 2 Pressure Contour:

Pressure contour distribution of the system using ANSYS is depicted in the figure. The colour scale shows pressure in Pascals, with red showing the highest pressures and blue indicating lowest pressures. In this contour, the high-pressure zone formed in the hub region is due to flow being compressed and concentrated around the centre (shown by red colour). A circle of blue and cyan colours appears, surrounding the hub, signalling a drop in pressure due to the fluid accelerating and expanding outward. To the sides of the centre, pressure slowly levels off and is shown as green and yellow colours all over the outer area. This leads to different places for it to flow into, forming cracks and veins in the material. Overall, this contour shows that the pressure drops in the central part of the system more uniform in the outer part.

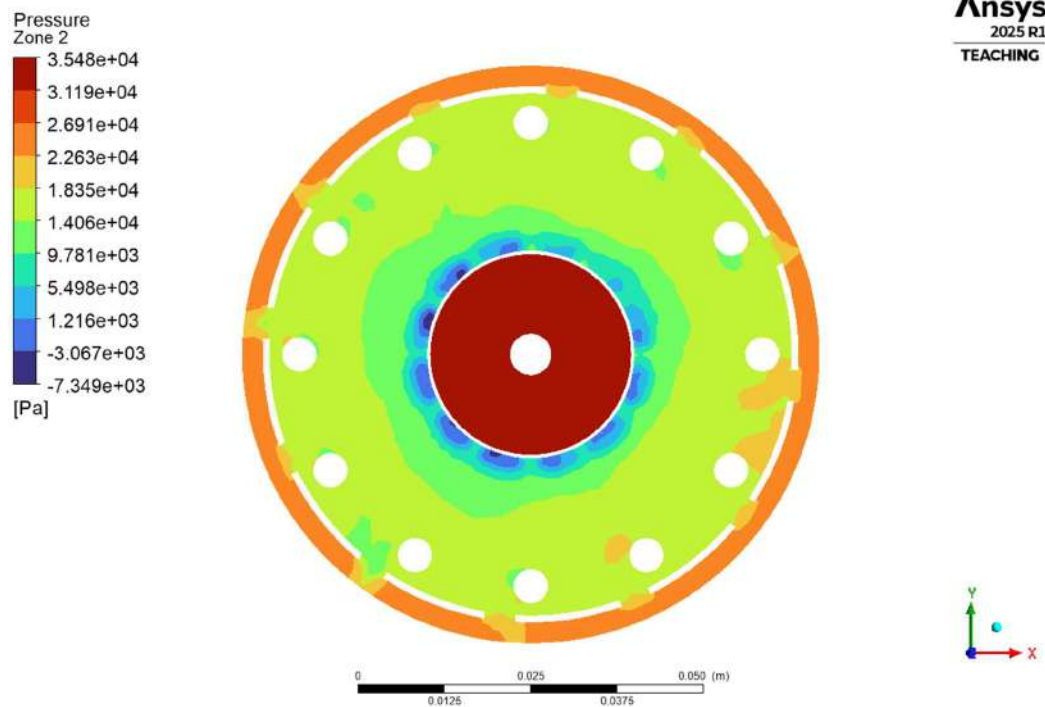


Fig.4.1.1.1(c) Zone 2 Pressure Contour.

d. Outlet Pressure Contour:

The reference from scale of colour, in Pascals showing pressure values, where red and orange colours represent regions with high pressure values while green to blue are low. As can be seen in the contour, high pressure is generated close to the inner circular boundary (orange and red) because of flow concentration and contact with the surface of the inner wall. As the flow moves away from the inner boundary to far outer region pressure decrease gradually and represented with green and blue shape. These colours are not evenly distributed about the ring, offering hints of minor fluctuations (mostly) in pressure due to flow turbulence and circulation interior to the annular pathway. In general, the contour shows that the pressure is greater near the inner surface and decrease from it to the outer edge of the domain.

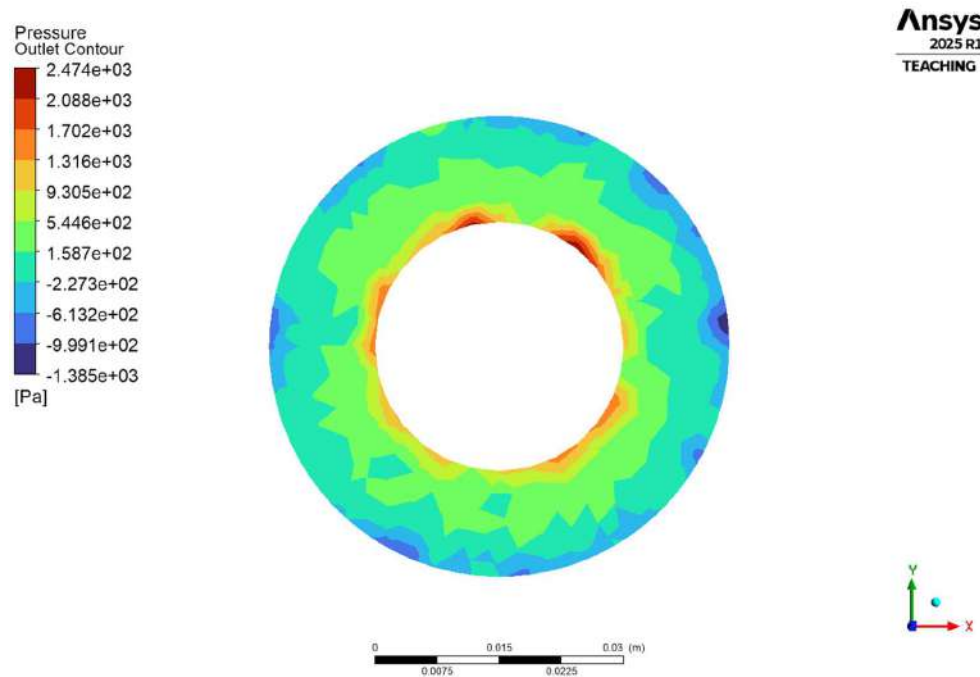


Fig.4.1.1.1(d) Outlet Pressure Contour.

e. Cross-Section Pressure Distribution:

In the image, the colour scale signifies variation in pressure (in Pascals), with red and orange representing high-pressure areas and green, cyan, and blue indicating lower pressure regions. Paying more attention to the contour, we see a strong high-pressure area is being formed at the inlet side, which is indicated by red colour towards the left of the contour because when incoming fluid comes inside and fills the chamber, there will definitely be some pressure difference in that closed volume. The pressure drops and stabilizes into the greenish and cyan regions as the flow moves through internal passages and cavities. Moving towards the outlet region, at right side once again the pressure coming down and shown as blue colour; this represents lowest pressure region while fluid exiting the system. The contour also illustrates how the internal geometry directs flow through different chambers, resulting in subtle pressure variations due to fluid expansion and directional changes. Overall, the pressure profile shows a very clear decrease from inlet to outlet, which is indicative of the normal pressure drop through any flow path in a system.

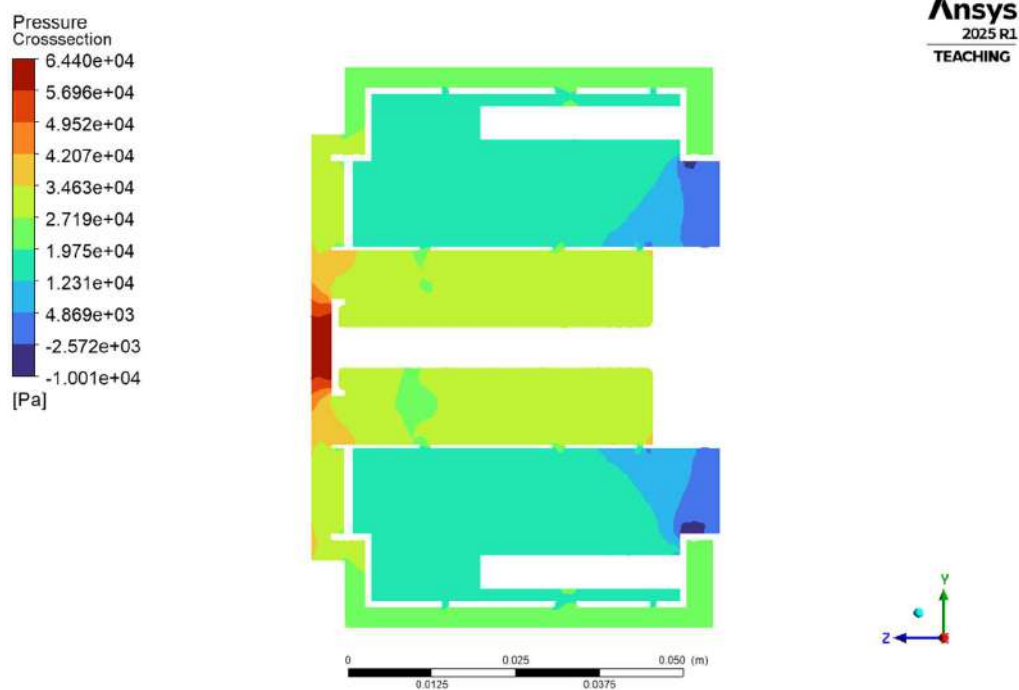


Fig.4.1.1.1(e) Cross-Section Pressure Contour.

4.1.1.2 Pressure contour 12 blade

a. Inlet Pressure Contour:

The colour scale shows pressure (Pa), with red and orange denoting regions of high pressure and green and blue representing areas of lower pressure. It has the maximum pressure indicates in red and orange colours due to convergence of flow, decay demand and swirl mechanism at the core. The pressure gradually decreases and the transition from yellow (high pressure) and green (intermediate high pressure) to blue shades is evident on the blades channels as we move outwards with that liquid, PI2 liquid. Many vane passages have identical patterns showing by symmetric and uniform pressure distribution in surrounding, all these properties indicate a stable flow field. The contour shows that the pressure at centre is highest and decreases as fluid flows from blade passages to outer side.

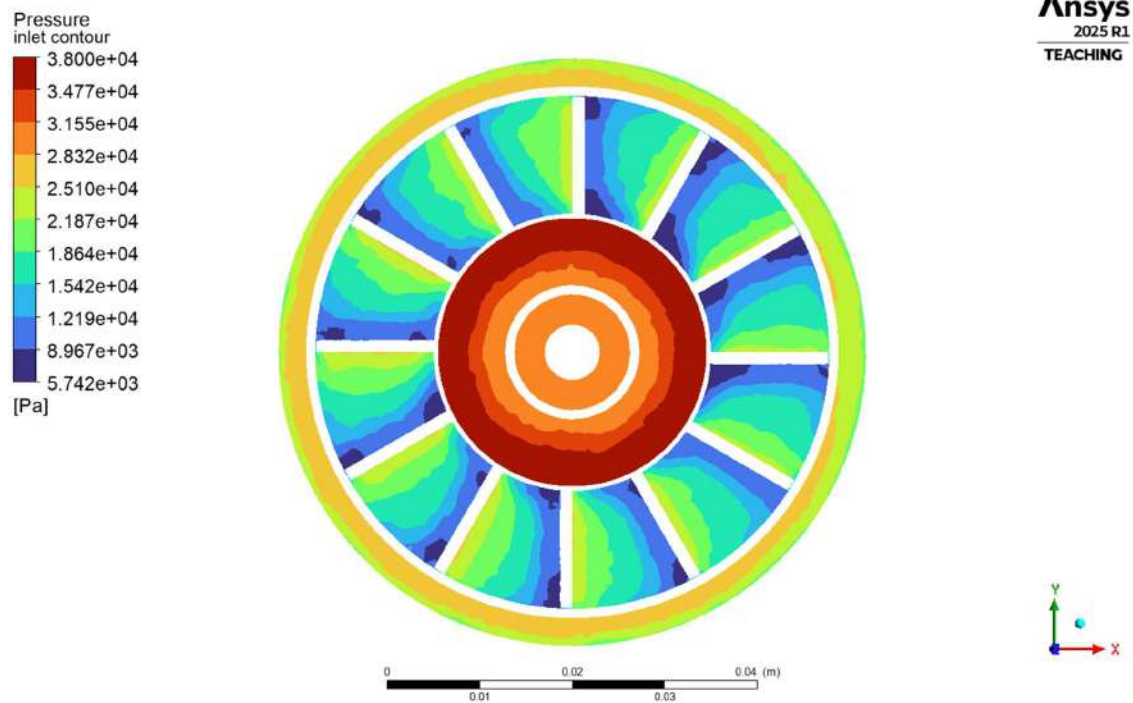


Fig.4.1.1.2(a) Inlet Pressure Contour

b. Zone 1 Pressure Distribution:

The pressure contour distribution was simulated in ANSYS circular flow domain based on which the parameters reported. The colour scale indicates pressure variation in Pascals from approximately 9.34×10^3 Pa (dark blue – low pressure) to 3.19×10^4 Pa (red – high pressure). As can be seen from the contour, this central region turned out to have high pressure zone which appear in red colour likely caused by flow stagnation or compression at the core. Moving outwards from the centre we can see that the pressure reduces and takes up most of the domain where P is represented using blue coloured region. This means that the fluid stretches and flows much faster in a radial direction, resulting in a lower pressure. In addition, you can see the three small white holes around the circular domain; they are likely air or fuel injection ports. These gaps affect the flow and generate localized pressure changes, apparent as darker blue patches in some areas. The green ring at the outer boundary represents somewhat higher pressure than in the inner blue area, indicating an extremely weak recovery of pressure close to the wall as flow is affected by interacting with the boundary. In general, this contour shows a radial pressure gradient with higher pressure at the centre and gradually decreasing as we move toward the outside region of the circular system that we assume here, indicating how pressure behaves and flows in these areas.

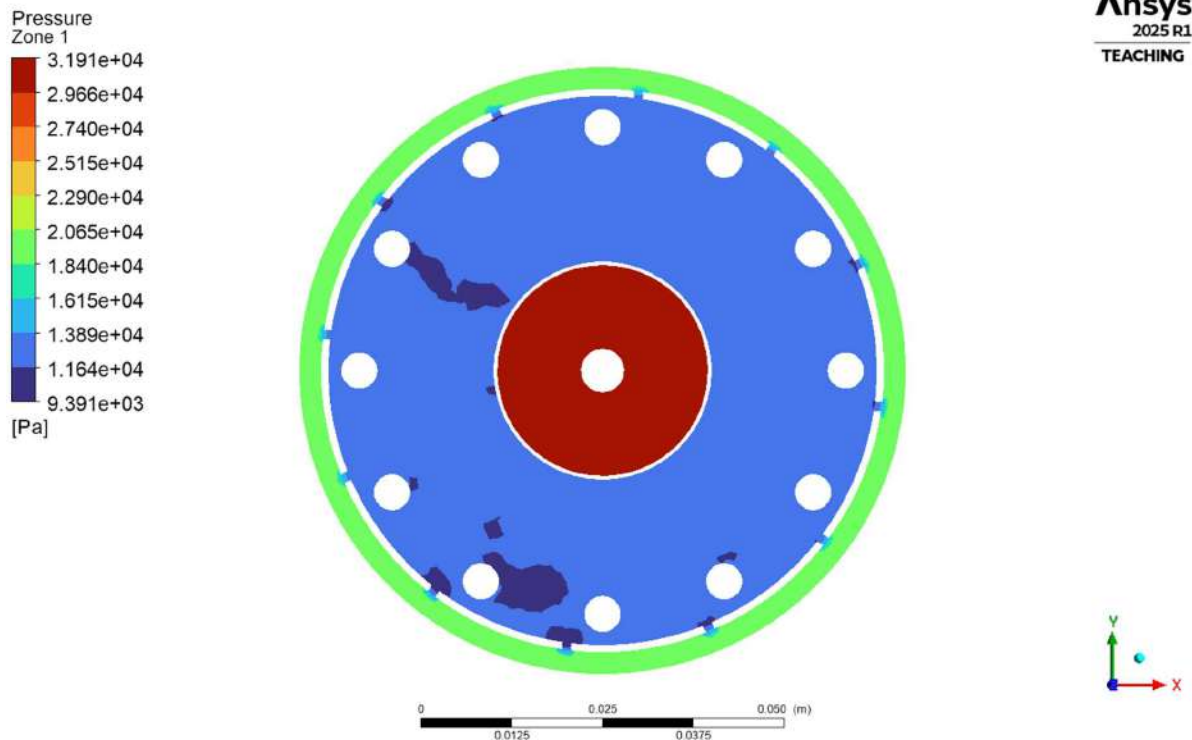


Fig.4.1.1.2(b) Zone 1 Pressure Contour

c. Zone 2 Pressure Distribution:

The pressure contour distribution after ANSYS simulation, as it represents how pressure is distributed in a circular domain. Colour scale represents pressure values in Pascals: low pressure -2.35×10^3 Pa (dark blue) and high pressure, 3.68×10^4 Pa (red). The maximal pressure region, depicted in red as the central area of the system, likely represents a consequence of flow stagnation or robust compression at its centre. The thin blue loop around this area indicating a drop in pressure suggests the presence of high velocity flow or mixing effects at a much higher rate around the core area. Further out, the pressure builds up to greater values again (green and yellow) where we see moderate pressures in the core flow region. Track up towards the outer boundary orange contour, indicative of the compressed regions at higher pressure due to flow/wall interaction or domain confinement. The small white circular holes that pepper the outer circumference probably represent air or fuel injection ports, slightly perturbing pressure around them and appearing as regions of uneven colour. The overall pressure contour shows a steep gradient from centre to edge that indicates the positive contribution of flow expansion, mixing, and boundary effects to pressure within the circular system.

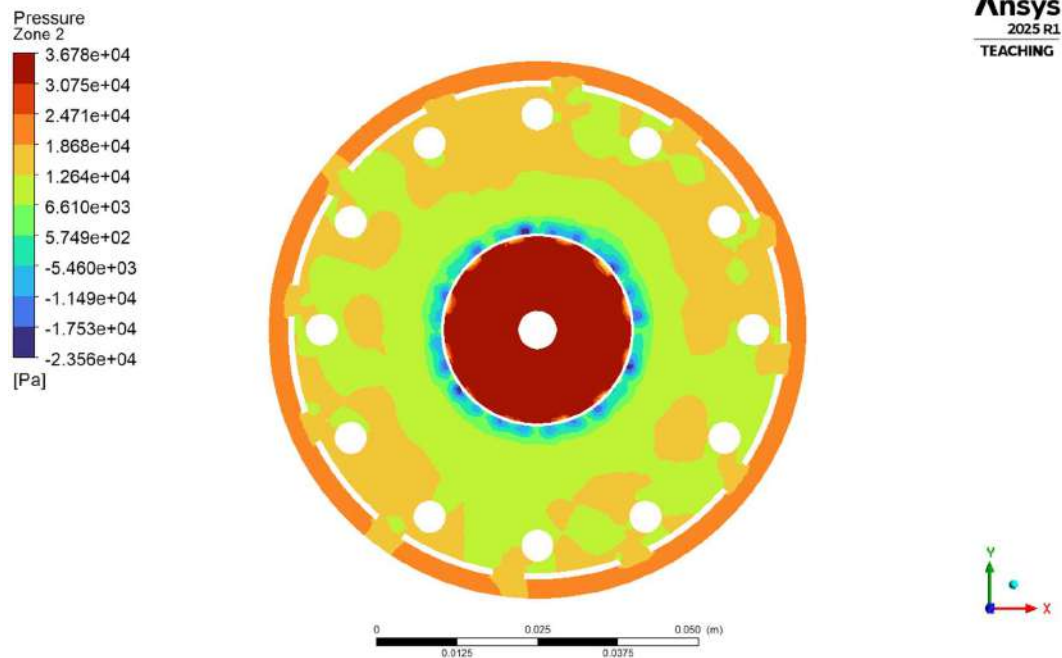


Fig.4.1.1.2(c) Zone 2 Pressure Contour

d. Outlet Pressure Contour:

The colour scale shows pressure variation in Pascals, from -1.25×10^3 Pa (dark blue) to 1.93×10^3 Pa (red), which corresponds to low and high pressure respectively. The maximal pressure was found near to the inner circular limit (shown by red and orange in contour). This means that the flow has higher pressure in close proximity to the inner wall (potential because of flow impingement, recirculation or stagnation effects within that region). The pressure drops gradually from the inner wall toward the outer boundary, indicated by the shift from yellow and green colour to light blue and dark blue. This implies that as the latter flows outwards through the annular passage, its volume also increases which causes acceleration and decrease in pressure. Contour patterns are somewhat uneven around the circumference, as a result of non-uniform pressure distribution resulting from vortices or turbulence as well as uneven flow mixing within the domain. In summary, the plot reveals a radial pressure gradient; that is, high pressure near the inner boundary and low pressure approaching the outer boundary are indicative of expanding flow across the annular region.

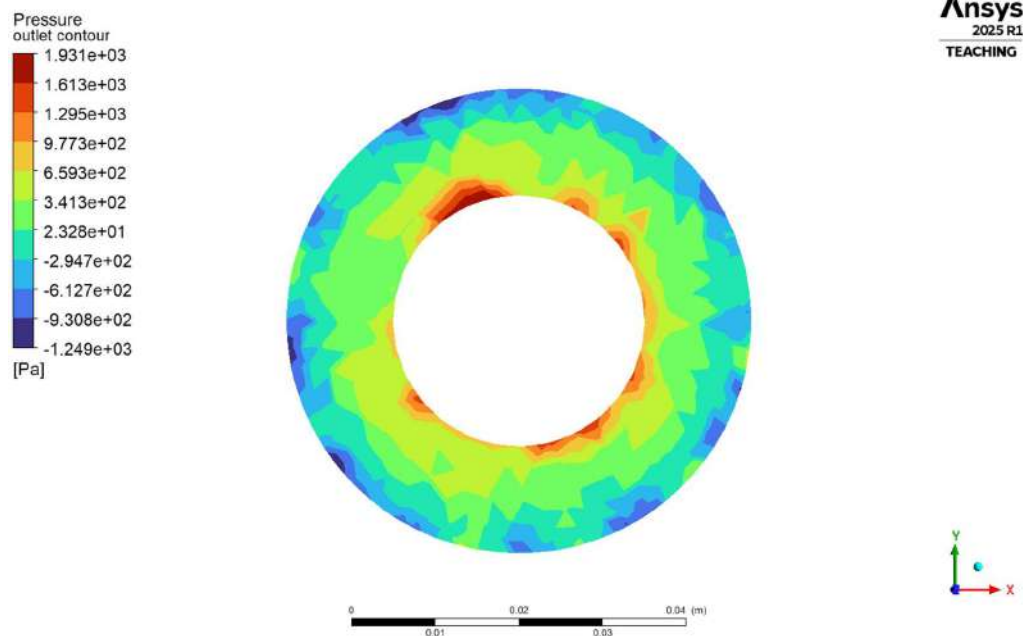


Fig.4.1.1.2(d) Outlet Pressure Contour.

e. Cross-Section Pressure Distribution:

This contour of a cross-section flow domain (combustor / internal flow chamber) The colour bar represents the variation in pressure (in Pascals) varying from -6.09×10^3 Pa (dark blue – very low pressure) to around 6.38×10^4 Pa (red – very high pressure). From the contour, it is evident that high pressures are observed near to left side inlet represented as red and orange colours which are in indication of first pressure given to fluid entering into chamber. The contour starts to change at the right-side outlet areas, where it turns dark blue and demonstrates the lowest pressure zones when fluid accelerates and leaves the area of chamber. Key features of the design include a symmetric pattern between the top and bottom sections which indicate even flow distribution between both pathways. In general, the contour provides a good indication of the pressure difference from inlet to outlet, and how much fluent pressure is lost along the flow path due to expansion.

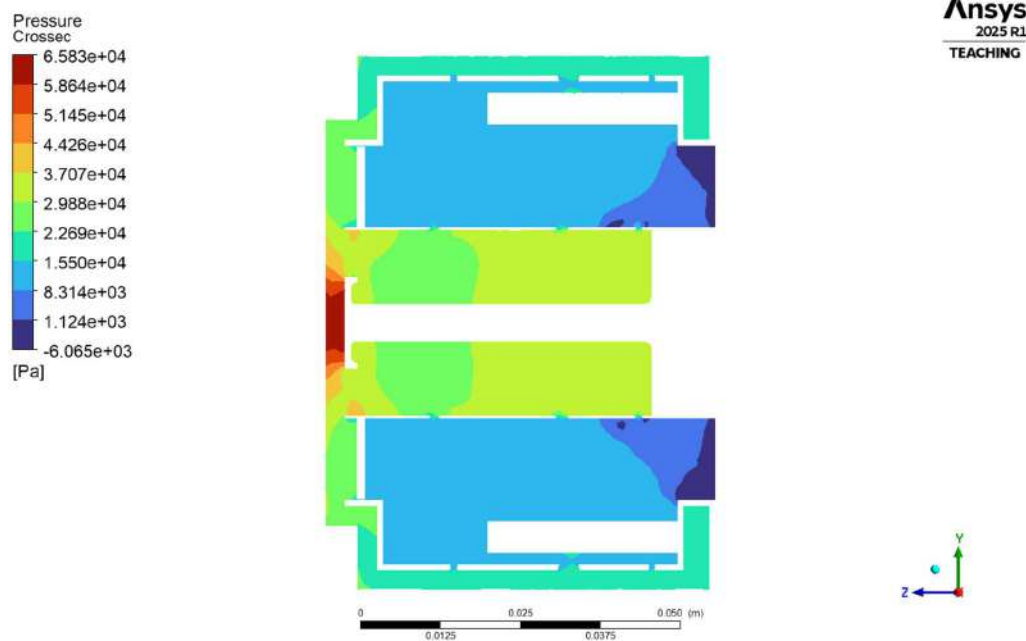


Fig.4.1.1.2(e) Cross-Section Pressure Contour.

4.1.1.3 Pressure Contour 16 Blade

a. Inlet Pressure Contour:

The colour scale denoting pressure changes (Pa) covering approx. -1.84×10^4 Pa (dark blue – lowest pressure) to ca 4.99×10^4 Pa (red – highest pressure). The central circular area is largely orange, suggesting a relatively high and stable pressure region within close proximity to the system body. The radially arranged channels or blades are the areas where the pressure starts changing; The flow goes from centre to amend. Within these passages the colours shift from yellow and green to light blue, indicating a gradual decrease in pressure as fluid flows outward. In some segments, there are slightly different colour patterns, suggesting a non-uniform pressure distribution due to turbulence, mixing or rotation flow effects. At the outer wall boundary, new orange and red areas emerge as we move lower, indicating localized flow interaction with the outer wall or recirculation regions causing pressure buildup near this part of the circular boundary. The contour as a whole gives an idea of how the pressure varies from the near centre area, through the radial paths to the outer casing and this is indicative of pressurization at points within the overall circular structure.

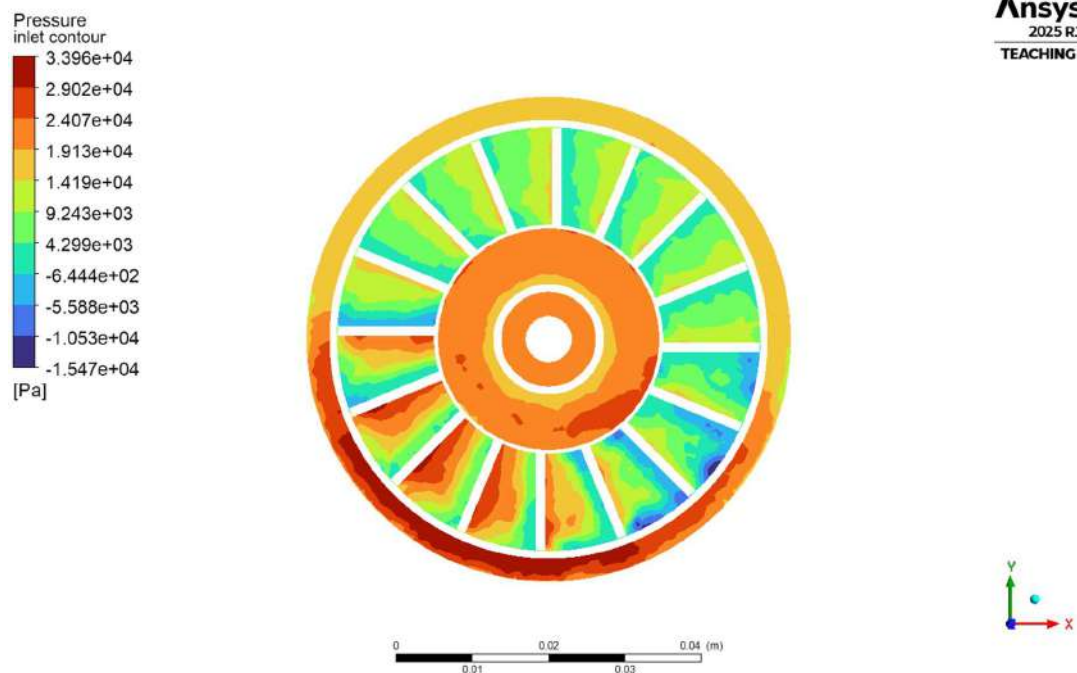


Fig.4.1.1.3(a) Inlet Pressure Contour

b. Zone 1 Pressure Distribution

The colour scale denoting pressure changes (Pa) covering approx. -1.84×10^4 Pa (dark blue – lowest pressure) to ca 4.99×10^4 Pa (red – highest pressure). The central circular area is largely orange, suggesting a relatively high and stable pressure region within close proximity to the system body. The radially arranged channels or blades are the areas where the pressure starts changing; The flow goes from centre to amend. Within these passages the colours shift from yellow and green to light blue, indicating a gradual decrease in pressure as fluid flows outward. In some segments, there are slightly different colour patterns, suggesting a non-uniform pressure distribution due to turbulence, mixing or rotation flow effects. At the outer wall boundary, new orange and red areas emerge as we move lower, indicating localized flow interaction with the outer wall or recirculation regions causing pressure buildup near this part of the circular boundary. The contour as a whole gives an idea of how the pressure varies from the near centre area, through the radial paths to the outer casing and this is indicative of pressurization at points within the overall circular structure.

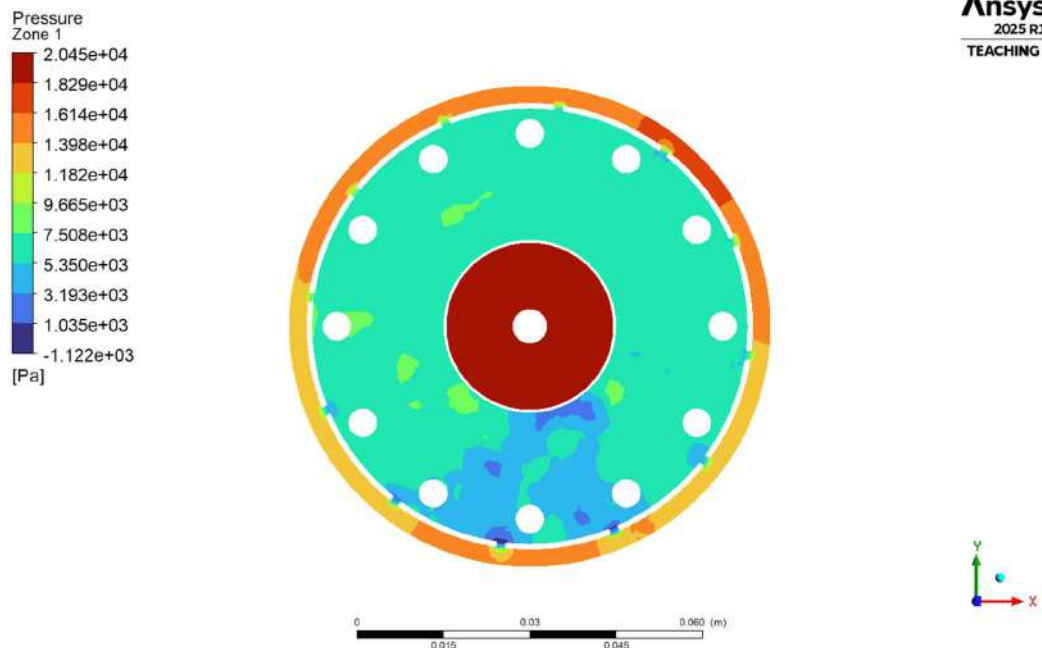


Fig.4.1.1.3(b) zone 1 Pressure Contour

c. Zone 2 Pressure Distribution :

The pressure scale ranges from around -1.81×10^4 Pa (dark blue – very low pressure) to roughly 2.88×10^4 Pa (red – very high pressure). Red and dark orange colours indicate the high-pressure zone at the centre – when very hot winds begin to mash at its top due to stagnation or compression. Surrounding this central region is a blue ring, representing a sudden drop in pressure created when the fluid begins to jet outwards from the centre. Farther from the centre, the pressure ramps up again (0.2–3). The outer regions are coloured green/yellow to show where moderate pressure is distributed through the main flow region. The tiny white circular holes grouped along the outer periphery presumably are air or fuel injection ports, which disrupt the adjacent flow and create minute pressure differences in their surrounding neighbourhood. More orange regions in the lower part of the chamber are visible, which may indicate increasingly localized pressure increases enabled by flow interaction, including mixing or recirculation. The outer boundary ring is yellow-orange, suggesting that pressure is relatively stable along the wall. In summary, the contour provides an overview of the transition from high pressure at the core to low pressure surrounding it and then stabilizing towards its outer boundary.

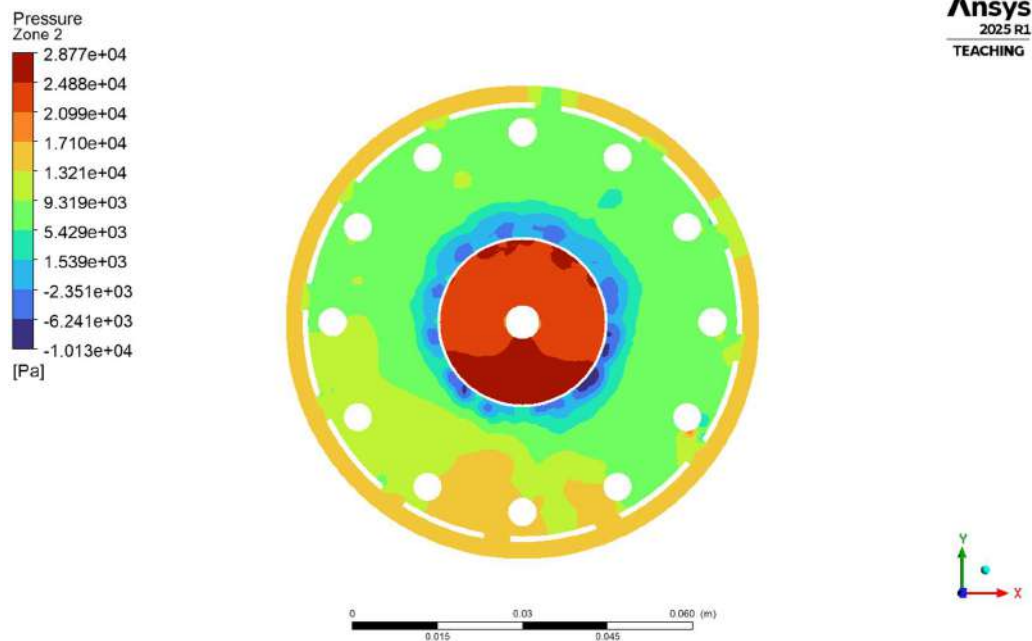


Fig.4.1.1.3(c)Zone 2 Pressure Contour

d. Outlet Pressure Contour:

This is the pressure contour of an annular flow domain from the simulation in ANSYS. The colour scale shows pressure in Pascals, ranging roughly from -1.25×10^3 Pa (dark blue – lowest pressure) to $\sim 2.68 \times 10^3$ Pa (red – highest pressure). From the contour, it is apparent that higher pressure regions are predominantly localized near the internal circular boundary, particularly in the bottom part where red to orange colours show. This indicates a local increase in pressure resulting from flow impingement, recirculation or stagnation on the interior wall. The pressure becomes lower with the flow moving outward, as coloured yellow and green then light or dark blue in the figures here. This means that the fluid gains velocity as it passes through the annular space, and a subsequent decrease in pressure is observed. The first noticeable observation is that the pressure distribution around the circumference is not entirely uniform, with some irregular patches of colours seen, which might lead to turbulence, vortex formation or uneven mixing of flow. The inner and outer values of contour get a maximum value near the wall and become lower as go outside, displaying some regions for higher pressure (P1) and lower pressure (P2).

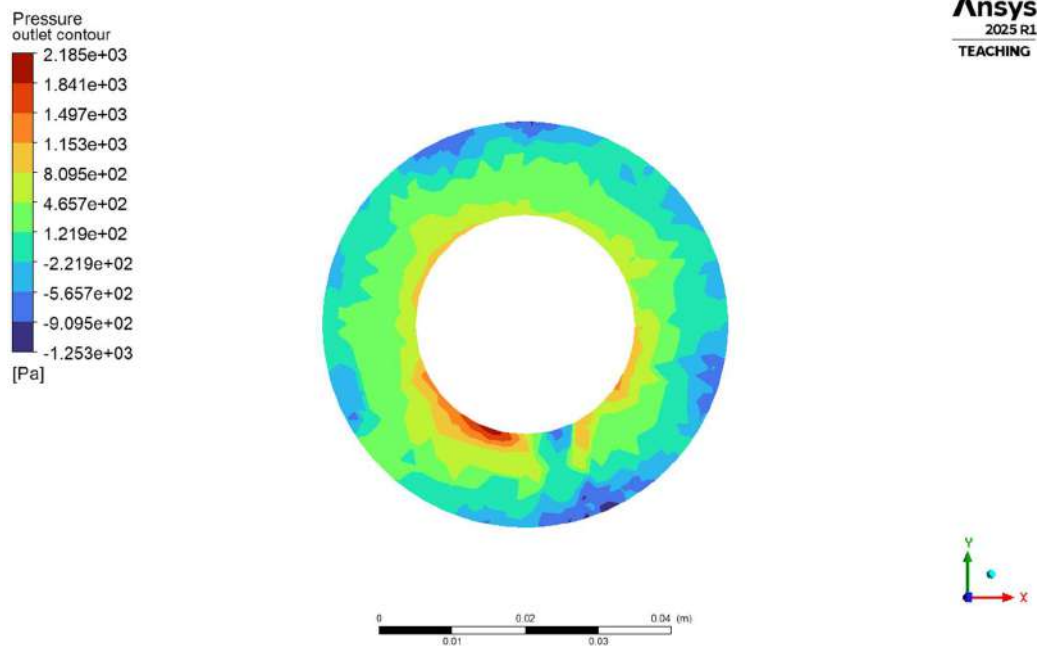


Fig.4.1.1.3(d) Outlet Pressure Contour.

e. Cross-Section Pressure Distribution :

Pressure contour distribution of a cross-section internal flow chamber is illustrated in the figure below, obtained from an ANSYS simulation. The scale of pressure is shown on the left, where values span from roughly 2.69×10^3 Pa (dark blue – low pressure) to about 4.77×10^4 Pa (red –high pressure). In fact, the area with the greatest hydrodynamic pressure is localized over the left-side inlet (in red and orange), which corresponds to where the flow entering into a chamber from an inlet point. As we can see, as the flow goes through inlet and starts to go towards central passages pressure start to drop gradually from blue part in regions (inlet)to yellow-greenish region which represent moderate low/high fluid pressure levels as it spreads towards rest of the system. The pressure decreases even further in both the upper and lower chambers, where it is primarily represented by light blue and dark blue colours indicating that the flow accelerates and widens in those areas. Here, the pressure stays subdued in the right-side outlet zone as it reveals the discharge area through which every single report of fluid flows out of the box. The patterns are similar in upper and lower parts of the geometry, indicating a fairly symmetrical flow distribution between both channels. In general, the contour depicts the pressure drop from inlet to outlet due to flow expansion, frictional losses and velocity changes inside the internal passages of the chamber.

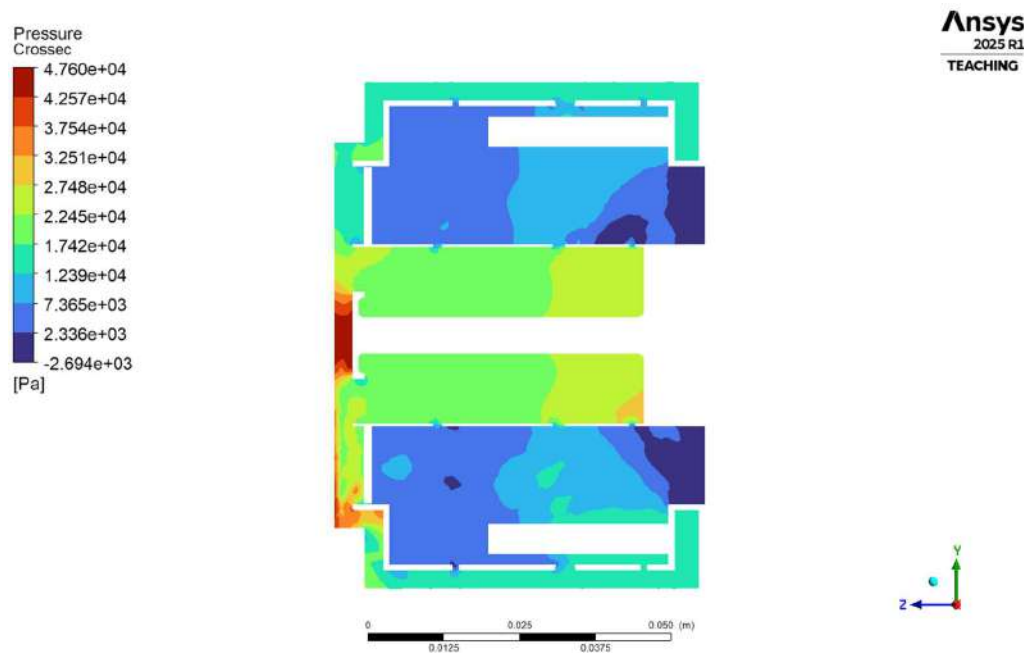


Fig.4.1.1.3(e) Cross-Section Pressure Contour.

4.1.2 Velocity Contour Comparison

4.1.2.1 Velocity Contour 8 Blade

a. Inlet Velocity Contour:

The colour scale shows velocity magnitude in m/s relative to approximate values from about 0 m/s (dark blue – very low velocity) to roughly 1.38×10^2 m/s (red – very high velocity). The central region and the outer limit are coloured dark blue, reflecting very low velocity caused by stagnation or solid boundary conditions that restrict fluid movement. From the centre outward, even lower areas appear to be slightly green yellow at the edges, indicating that this is where viscosity begins crossing its sweet spot because it accelerates fluid across the passage. As shown in the radial compartments or channels, regions of high velocity have been represented as yellow, orange and red colour zones which demonstrate flow movement through these passages. These high-velocity zones are found where the flow becomes directed and accelerated by the geometry of the channels. However, the distribution is not completely uniform between all the sections which indicates altering flow acceleration or turbulence or mixing effects locally. Again, close to the outer wall, the velocity decreases again due to the no-slip condition being applied at the boundary of a cylinder causing a boundary layer effect. Overall, such contour shows the velocity behavior with respect to flow within the circular system; as it portrays how fluid accelerates along radial channels and near solid boundaries decelerates.

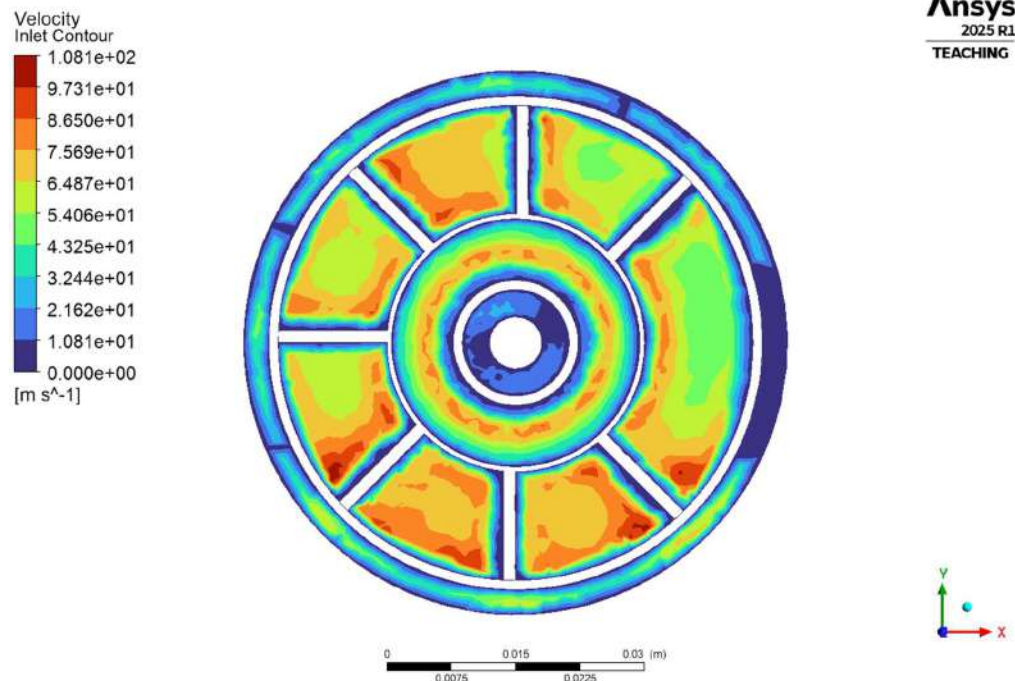


Fig.4.1.2.1(a) Inlet velocity Contour

b. Zone 1 Velocity Contour :

Velocity contour distribution of ANSYS simulation in circular chamber with multiple peripheral holes. The scale for speeds varies from 0 m/s (dark blue, very low velocities) to 1.31×10^2 m/s (red, very high velocities). The blue core in the middle conservative region suggests a low velocity zone near the dead on the central shaft which can also be due to stagnation or slow flow circulation. As we move away from the centre, speed increases sharply and forms high speed regions (yellow, orange to red) indicating acceleration of flow in the surrounding circular passage. These high-speed regions seem to take place in various patches around the cavity, indicating that there are geometric restrictions on the flow due to the network of holes in their span that could act as an injection or a flow by-passing system. The small circular holes with white rounds on the outer circumferential surface suggest that they could have been used as injection or cooling ports which might create a localized area into mixing flow. Heading toward the outer edge, the contour reverts to blue, corresponding to a decrease in velocity because of the no-slip wall condition, where the fluid decelerates. In general, the contour describes that the velocity is increasing when the flow goes far from the centre and localized high-velocity zones are appearing opposing spherical cross-section in a circuit around chamber before decreases again opposite outer wall which means circular system have complex distribution of flow.

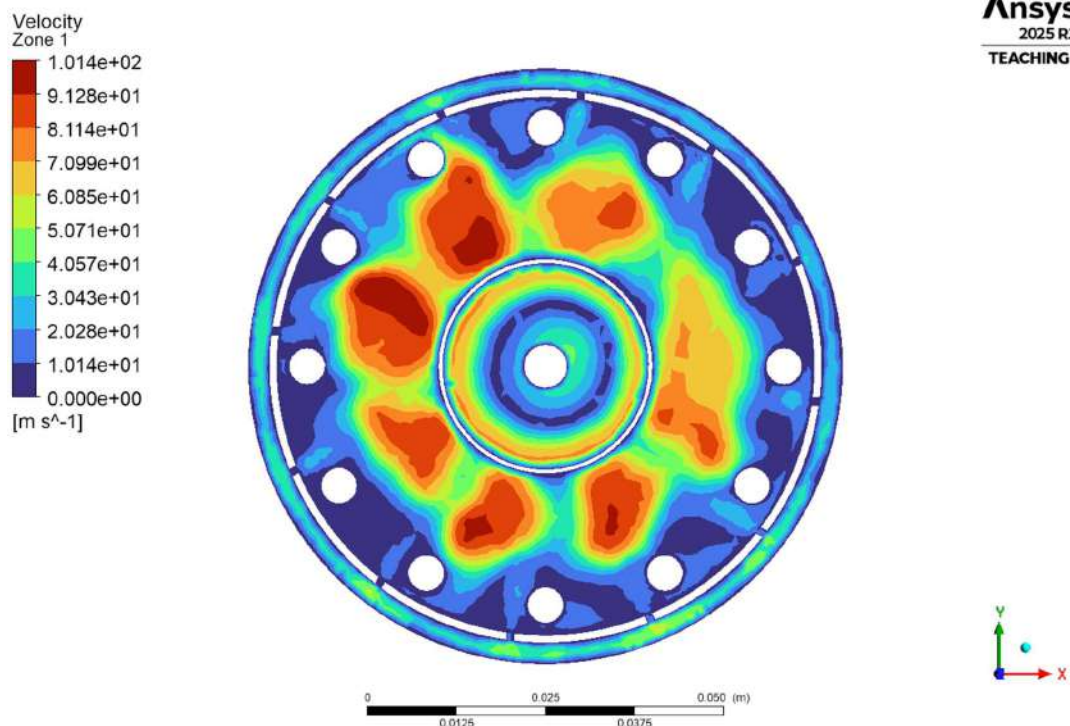


Fig.4.1.2.1(b) Zone 1 velocity Contour

c. Zone 2 Velocity Contour:

The velocity distribution contours value from an ANSYS simulation of a circular chamber with multiple peripheral holes. The velocity scale ranges between 0 m/s (only seen in dark blue – very low velocity) to $\sim 1.15 \times 10^2$ m/s (red – very high velocity). Dark blue colours are visible near the centre of the chamber and might be interpreted as low velocity zone surrounding the central shaft due to stagnation flow or slow recirculating region in this area. At the same time as we are moving outward from the centre, our velocity is greater than before and indicated by increasing green, yellow, orange, red colours displaying rapid acceleration of the fluid around the surrounding circular channel flowing out of the double helix. The mechanism of variation in the high-speed area might be due to turbulence dispersion or heterogenous geometry due to its non-uniform patch around Chamber. The many small white circular holes lining the outer circumference can be assumed as air injection or cooling ports which bring in additional flow into the chamber and create localized jets. The jets appear as greenish streaks directed toward the holes that they emanate from, where the injected flow comes in contact and mixes with the original flow. The velocity after reaching to the outer boundary wall is reduced again and is dark blue, which at the solid steady state, on no slip wall condition due to fluid viscosity with solid it reduces its strength. In this contour, we can see that the system velocity increases when moving away from the centre, creates local high-speed zones due to injection and mixing effects before decreasing again near the outer wall.

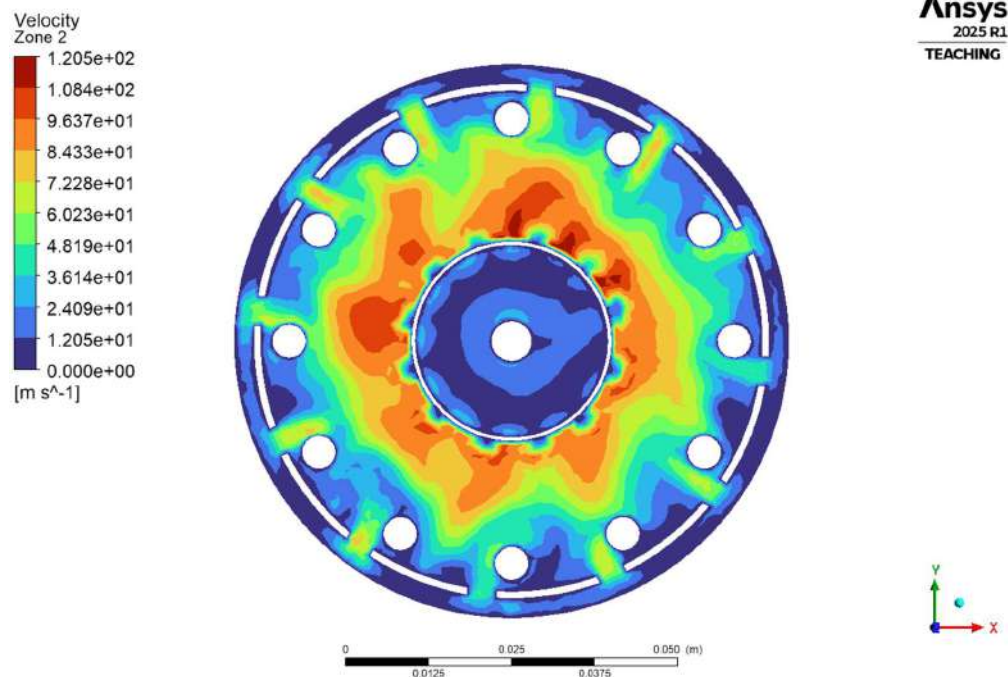


Fig.4.1.2.1(c) Zone 2 velocity Contour

d. Outlet Velocity Contour:

The velocity contours of an annular flow region obtained from ANSYS Colour scale for velocity: from -0 m/s (dark blue – very low velocity) to $\sim 1.99 \times 10^2$ m/s (red – very high velocity). Near the walls, due to no-slip boundary condition, the fluid comes to a stop at the solid surface and thus the outer circular boundaries would appear on dark blue representing low velocity. As we move away from the walls towards the inner annular region, the velocity increases and is coloured in green and yellow colours (flows at moderate speed). In some parts, particularly at the upper part of annulus orange and red areas are having high velocity, where flow accelerates strongly. The presence of such localized areas of high-speed flow indicates an uneven distribution, which may be indicative of vortices, turbulence or geometric effects in the annular flowing passage. In general, the contour indicates that the maximum velocity is in the middle of two cylinders from annular flow path, where both sides of inner and outer wall has decreasing velocity that specifies a good complex of velocity distribution behavior inside an annular chamber.

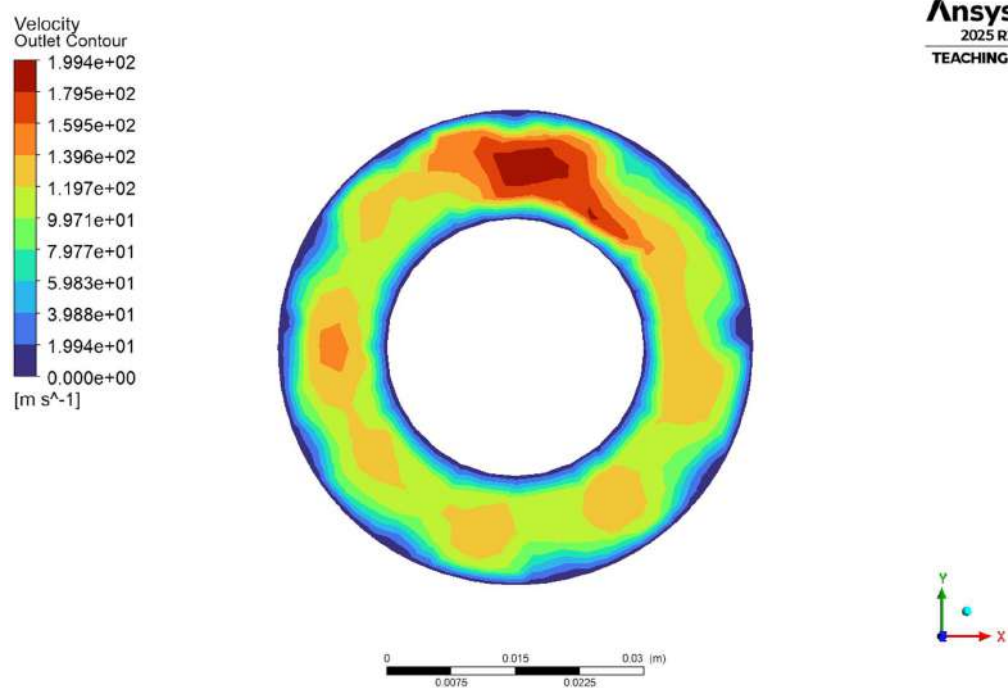


Fig.4.1.2.1(d) Outlet velocity Contour

e. Cross-Section Velocity Distribution:

The illustration shows the velocity contour distribution obtained from a 3D cross-sectional internal flow chamber simulation in ANSYS. The velocity scale ranges from 0 m/s (dark blue – very low velocity) to $\sim 2.02 \times 10^2$ m/s (red – very high velocity). The right-side outlet sections, coloured yellow, orange and red in the contour plot indicate that these are the highest velocity regions — hence the fact that the fluid accelerates strongly as it leaves out of the chamber. As the flow travels inside through the central passage, it loses speed and is presented in green and light blue colour, depicting moderate flow speeds. The upper and lower chambers have mostly low velocity regions shown to be dark blue, where slower moving fluid results from the expansion of flow area or potential recirculation zones. Areas of low velocity may also indicate where vortices and flow mixing occurs within the chamber. This similarity in both top and bottom indicates a fairly even flow being passed through the two passages. Overall, this contour shows that velocity profile increases toward the outlet as fluid passes through the passage; we see lower velocities are found inside the larger chamber regions indicating complex flow behaviour and distribution.

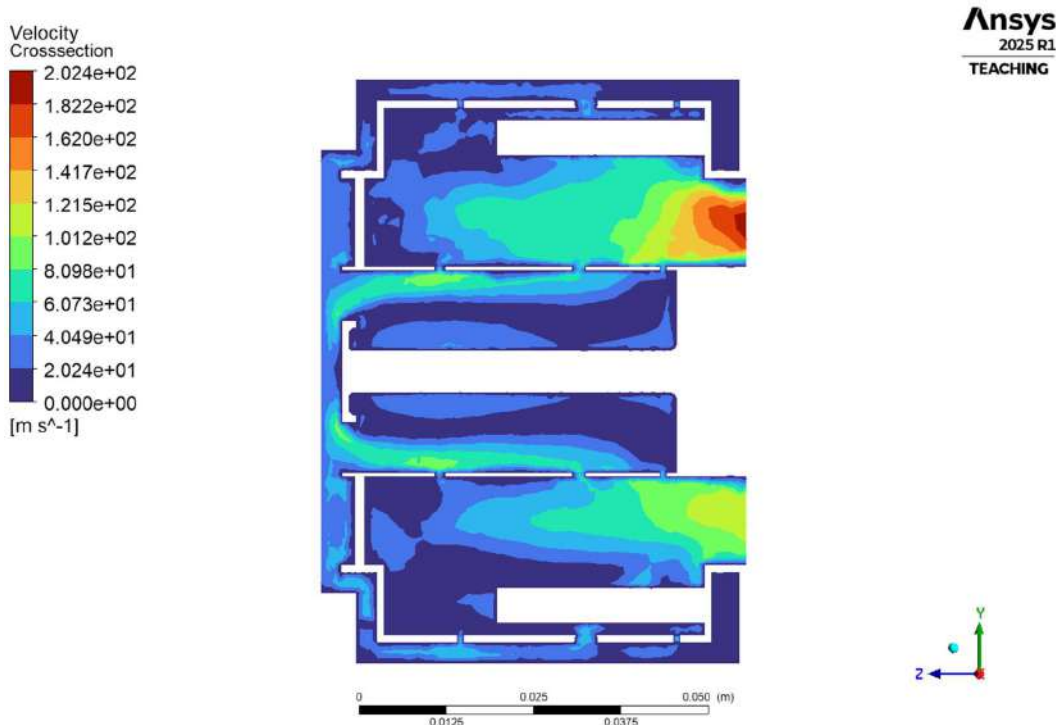


Fig.4.1.2.1(e) Cross-Section velocity Contour

f. Velocity Vector Field:

This is a plot of the velocity vector projection from an ANSYS simulated cross-sectional flow chamber. The velocity magnitude varies from 0 m/s (dark blue – very low velocity) to about 2.03×10^2 m/s (red – very high velocity). The vectors show the direction and magnitude of flow in the chamber. Through analysing the contour, we can tell that the highest velocity happens on the outlet regions (right side) and represented in yellow, orange and red vectors showing that our fluid has a high acceleration as it comes out of the chamber. As the flow shifts from the central passages into the larger upper and lower chambers, velocity slows — appearing almost entirely as greens and light blues that indicate moderate flow rates. Within the larger sections of these chambers, there are several small vectors with darker colours that represent low-velocity regions (indicative of recirculation or mixing areas) where fluid is moving slower. It is observed that the upper and lower degassing sections have very similar flow distributions, suggesting a relatively even share of fluid through these passages. So, the arrows in a vector plot effectively represent both direction movement of the flow from internal passages out to outlet as well as areas of high acceleration near exit and smaller circulation within chamber.

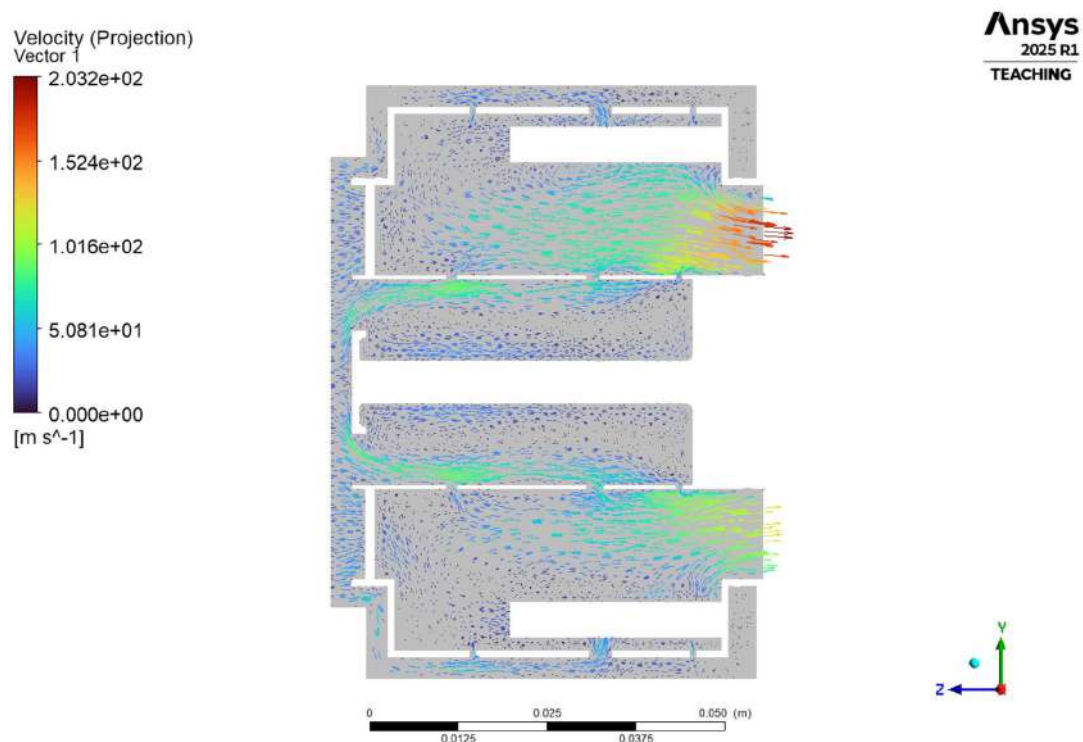


Fig.4.1.2.1(e) Cross-Section velocity vector Contour

4.1.2.2 Velocity Contour 12 Blade

a. Inlet Velocity Contour:

The airflow area distribution in the circular chamber after entering multiple radial passages, as illustrated by velocity contour in ANSYS simulation. The colour bars represent velocity magnitude between 0 m/s (dark blue – very low velocity) and approximately 1.15×10^2 m/s (red – very high velocity). The contour indicates that in the middle section around the shaft, where there is low velocity (the blue region), a dimensional flow restriction is imposed and stagnation occurs at the solid hub. As the air flows through the radially oriented passages, its velocity is gradually increased by virtue of reduced cross-sectional flow area and guidance from an arrangement of vanes. The areas in red and orange within the passages represent high velocity zones, where the air accelerates rapidly as it travels through these channels. So, the velocity distribution looks quite uniform between the passages which is in Favor for fluctuating flow around the chamber, such as combustor or related turbine applications. This illustrates how the nature of use of the vanes guides, accelerates and distributes symmetrically [the air] in regard to the contour.

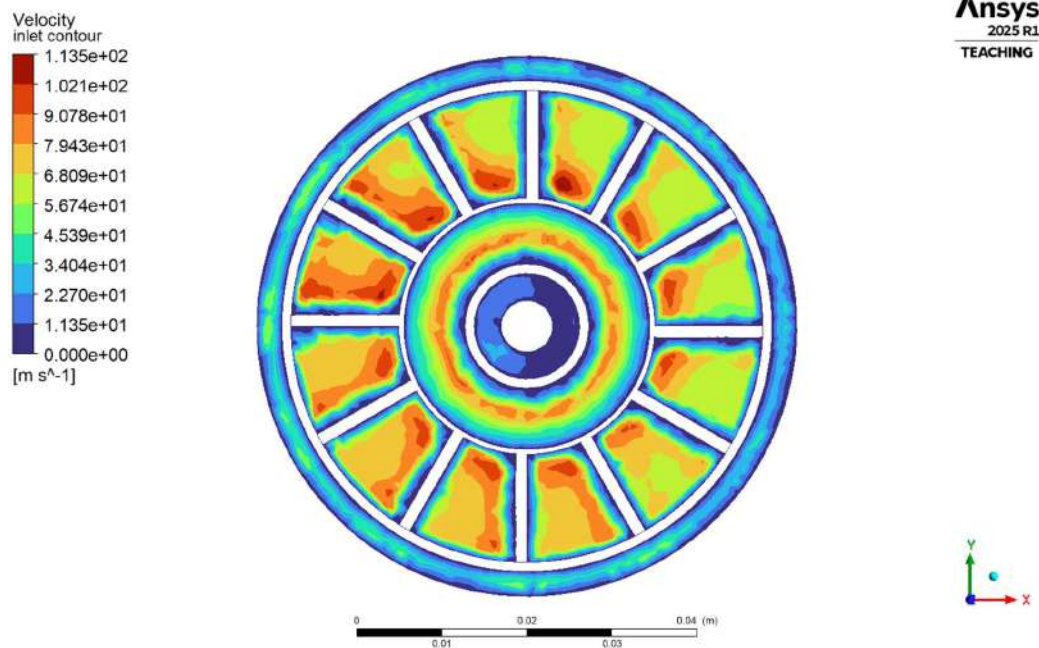


Fig.4.1.2.2(a) Inlet velocity Contour

b. Zone 1 Velocity Contour:

For velocity, the scale goes from 0 m/s (dark blue – very low velocity) to $\approx 1.03 \times 10^2$ m/s (red – very high velocity). The shaft and central area have low velocity (in blue and light blue colours), indicating that the flow is confined, resulting in lower airflow near the hub. Outwards, there is a ring-shaped region of high speed (yellow to red) showing that the air accelerates as it moves through the annular flow path. The red zones surrounding the circular ring identify high-speed, localized areas created by interaction between the bulk flow and the surrounding holes and geometry of the chamber. The velocity further drops in the two-dimensional region at left and around the holes (blue area) due to flow expansion and wall effects. This overall contour indicates that there is a circumferential velocity distribution in the annulus with alternating high and low velocity fields which can enhance mixing as well as turbulence inside the chamber, both of which are very crucial for combustion and stability of flow behavior.

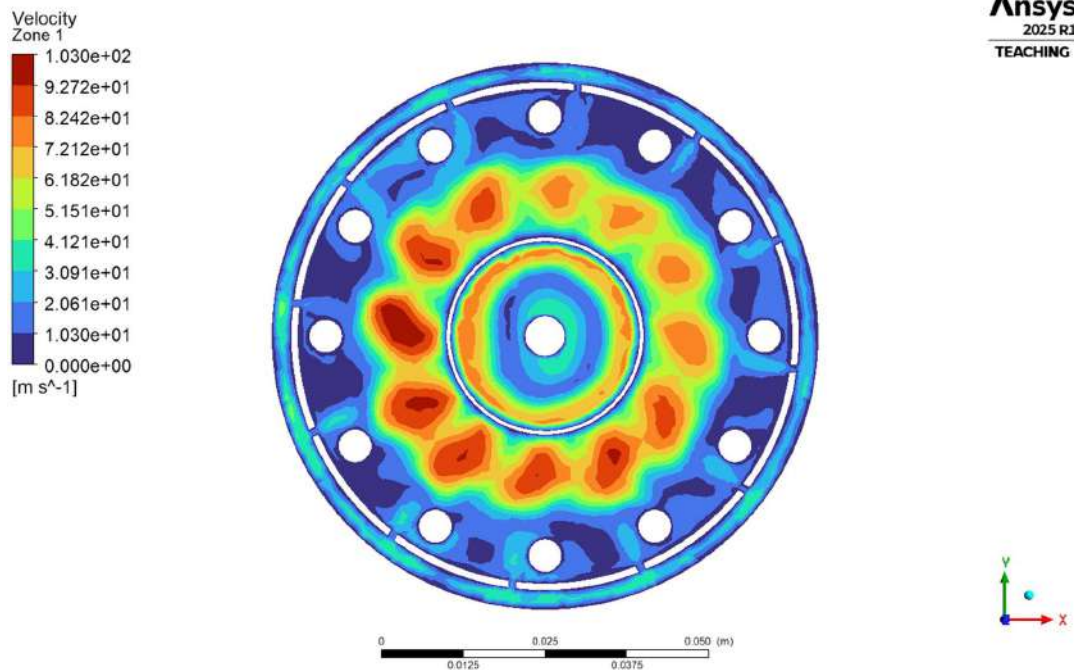


Fig.4.1.2.2(b) Zone 1 velocity Contour

c. Zone 2 Velocity Contour:

The velocity scale is in the range of 0 m/s (dark blue - very low speed) to an approximate 1.50×10^2 m/s (red - very high velocity). In the central hub area seen on the contour plot, dark blue represents low velocity as airflow is blocked off directly by the rigid shaft and a stagnation point forms. As you move outwards from centre, velocity increases steadily into what is termed moderate speed range (green and yellow together) running around the doughnut shape, indicating that as it spreads radially airflow accelerates. The small red patches along the inner circular border represent rare, localized high velocity areas formed when after having rotated for such a long time the stream paths are finally coming into contact with sides of chamber. Near the outside walls and at peripheral holes as shown by blue shading velocity increases once more, indicating that air becomes looser in its interactions with a chamber boundary. Overall, the contour shows that airflow is coming from low velocity at centre to higher speeds in the mid-region in a gradual shift, and then once again to low velocity near wall boundaries: this means good radial distribution of active gases inside chamber.

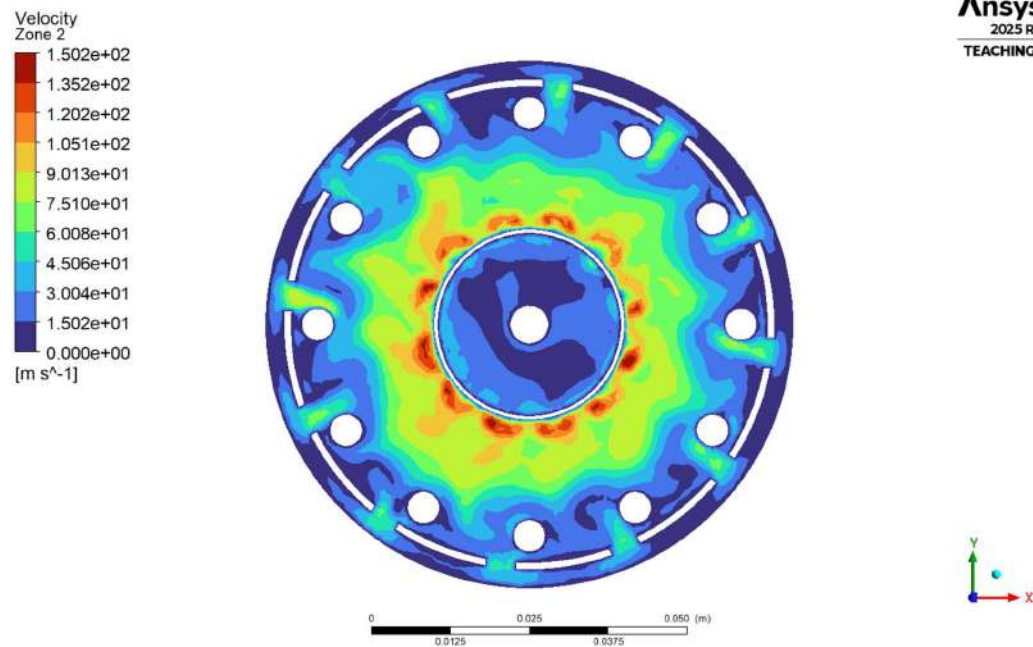


Fig.4.1.2.2(c) Zone 2 velocity Contour

d. Outlet Velocity Contour:

From this contour, the scale the dark blue represents very low velocity region; the upper limit being roughly 199×10^2 m/s (red, high velocity). From the contour, the highest velocity region is found in the upper left corner of annular outlet; this area is represented by red, a signal that air at that point is greatly accelerated in its outward motion from chamber. On the whole annular area, mostly yellow and orange are observed, representing moderate to high speeds of fluid flow with at best only one small error or less than half a mistake per hour While leaving chamber; decreased velocity closer to wall reaching at pressure zone in inner-outer perimeter When seen in plain view, the high-velocity zones around the annular ring have different amounts of flow intensity as well as all emanating from a common source in part because chamber geometry conditions how air flows into and exits it. According to the figure, on average air leaves the chamber with relatively high velocity and moderate circumferential variation. This can affect downstream mixing combustion performance.

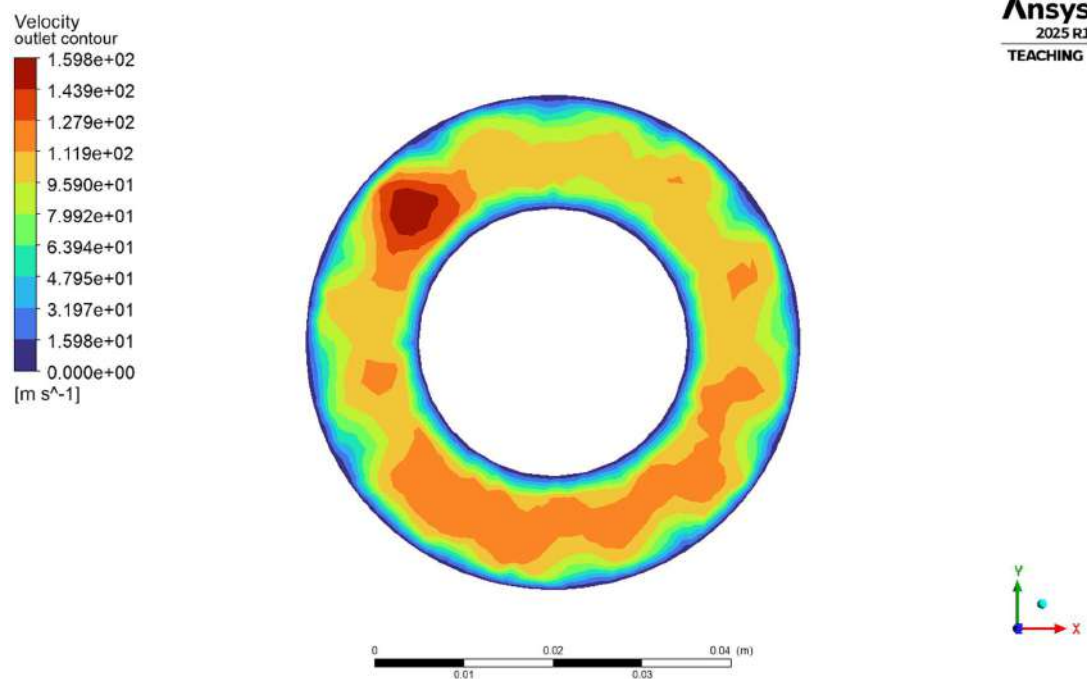


Fig.4.1.2.2(d) Outlet velocity Contour

e. Cross-Section Velocity Distribution:

The contour shows us that high velocity areas are located near the out let passages on the right, as at these points the stream contracts and its speed increases. These locations are depicted using orange and red, indicating that raw air is being sent out of air water .The velocity is lower in the centre of the cavity (represented as green-blue shades), suggesting that formation of recirculating zones and slower moving fluid regions also occur here. These recirculation areas assist in the elimination of dead zones and stabilization of the flow within the chamber. As one moves toward the cavity walls, velocity drops off even more because of laminar boundary layer effects and friction at the wall. Overall, the contour reveals that airflow enters the chamber, passes through internal paths, and finally accelerates toward the outlet.

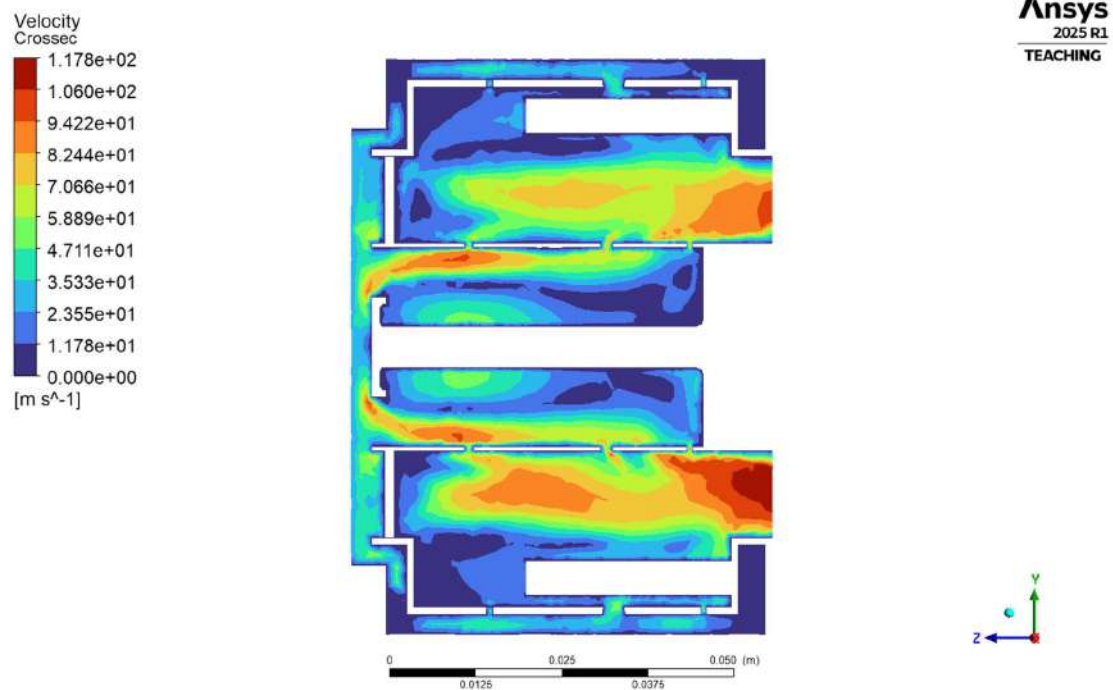


Fig.4.1.2.2(e) Cross-Section velocity Contour

f. Velocity Vector Field (Bottom-Right):

Velocity Vector Projection is an arrow (by plane diagram) which by scaling and directing to show the fluid speed or direction at each point in a region. The Colour Scale shown here indicates the velocity in the simulation range from 0 m/s (dark blue--the lowest possible speed) to approximately 1.68 times (red--high end of the scale). The vectors below detail how the airflow flows through the internal cavities of the chamber. According to the figure, we can see that the flow enters from the left-hand side and is confined by narrow channels leading to a right-hand outlet where due to acceleration of this rolling stream it becomes faster. Denser and longer vectors near the outlet show higher-speed flow, while shorter ones in wider chamber regions represent slower moving air. Middle sections of the chamber have curved flow patterns mixed with minor recirculation areas which serve to enhance mixing and more uniform distribution of airflow in general. Nearby walls of the chamber also drop speed due to wall friction and boundary formation layers. Between them and in the end the vector projection clearly shows the flow path, channel acceleration and that there is re-circulation in chamber before it enters reactor section.

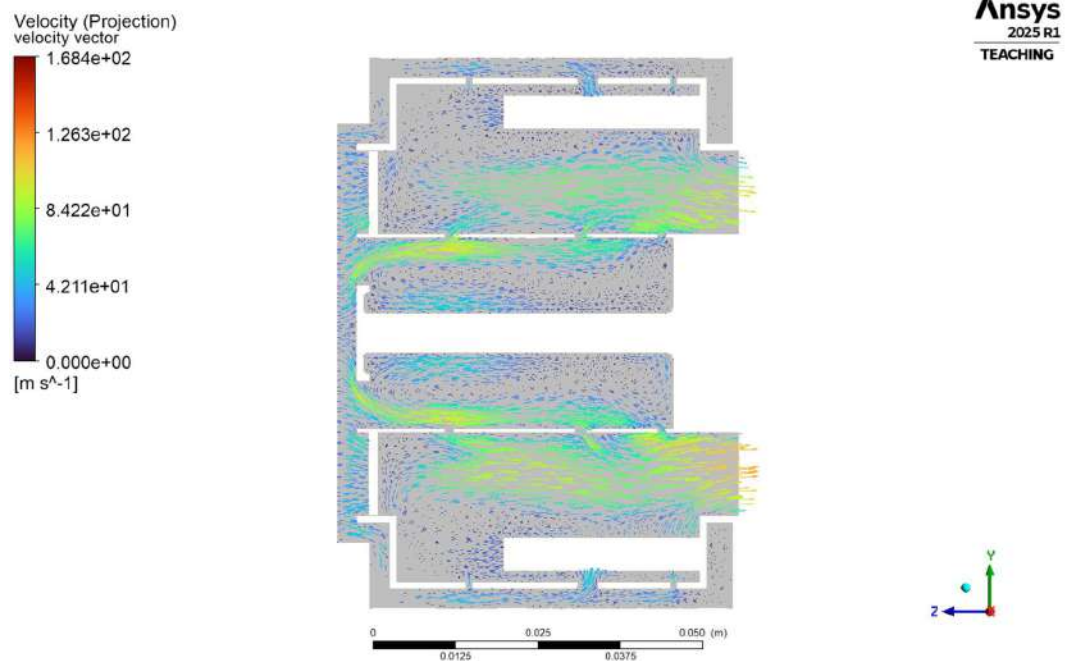


Fig.4.1.2.2(f) Cross-Section velocity vector Contour

4.1.2.3 Velocity Contour 16 Blade

a. Inlet Velocity Contour:

From the contour, it is seen that the central hub area has relatively low velocity: this is reflected in blue tones, which indicate cooler. As the air course outwards into radial passages formed by additional vanes (and therefore without a solid shaft in between them), velocity increases in steps of direction while gently converging with distance. Yellow and red zones appear in the vane passages, indicating higher velocities still: it is evident that incoming air speeds up as this mass passes through narrow channels. Most of the outside circular region is at moderate speeds (Green to light blue), suggesting that airflow spreads and disperses around the chamber after passing these channels. The red scatter in sections of vane passages shows small familial differences in airflow intensity among diverse sections of vanes, because of geometric effects and local flow interactions. Ultimately, the contour shows the structure of a plate being tormented by wind: however, as an added party Favor we can see that air comes in from centre and is pushed radially outwards, gaining speed within these channels where the flow distribution loop ends up being very uniform.

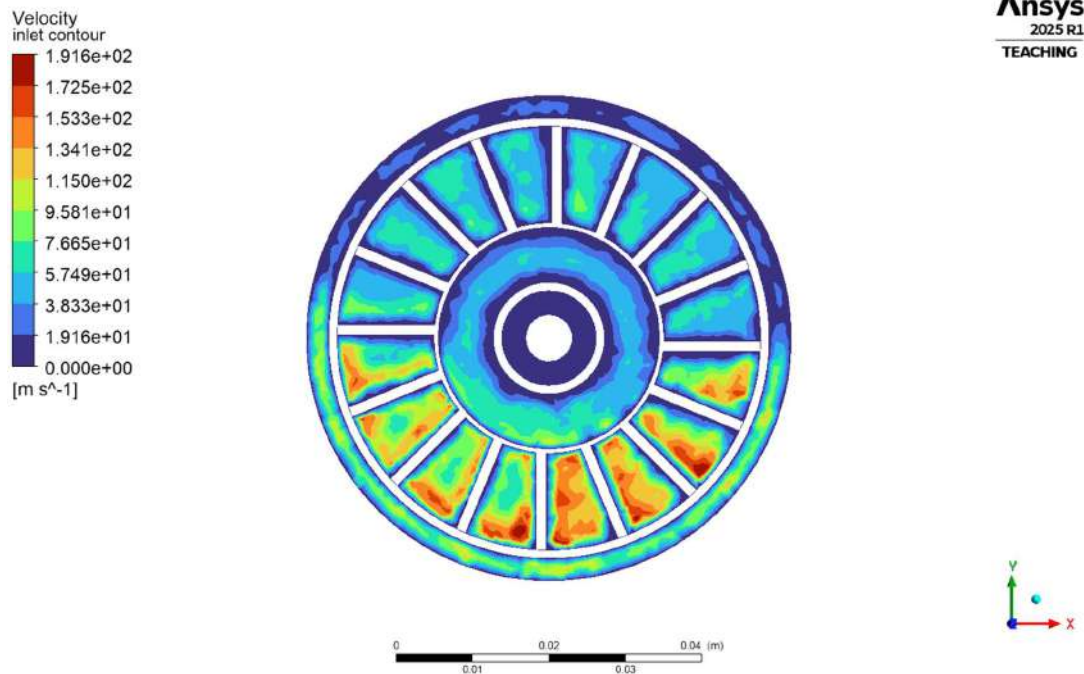


Fig.4.1.2.3(a) Inlet velocity Contour

b. Zone 1 Velocity Distribution:

Velocity information can go from 0 m/s (dark blue--extremely slow velocity) to around) There might be a range of low velocities (blue to green bands) in large internal spaces. From the plot, it can be seen that even the general parts of housing are predominated by low to moderate velocities of internal airflow. The central hub also has low velocities, because physical restrictions (solid shaft) are causing direct flow of air and it creating a static area. At the bottom of the chamber (yellow to red) is a region of high-velocity air that indicates the flow rate at this point has increased greatly. Apart from the speed at peripheral holes slightly rises and respectively rises to form a local flow path, also due to boundary layer effects and wall friction the rate of flow near the chamber walls drops. The contour shows that overall, the velocity distribution is not uniform, with some areas being concentrated, as for example in a high-velocity zone surrounded by lower areas. This can increase internal circulation and help mix air inside the housing.

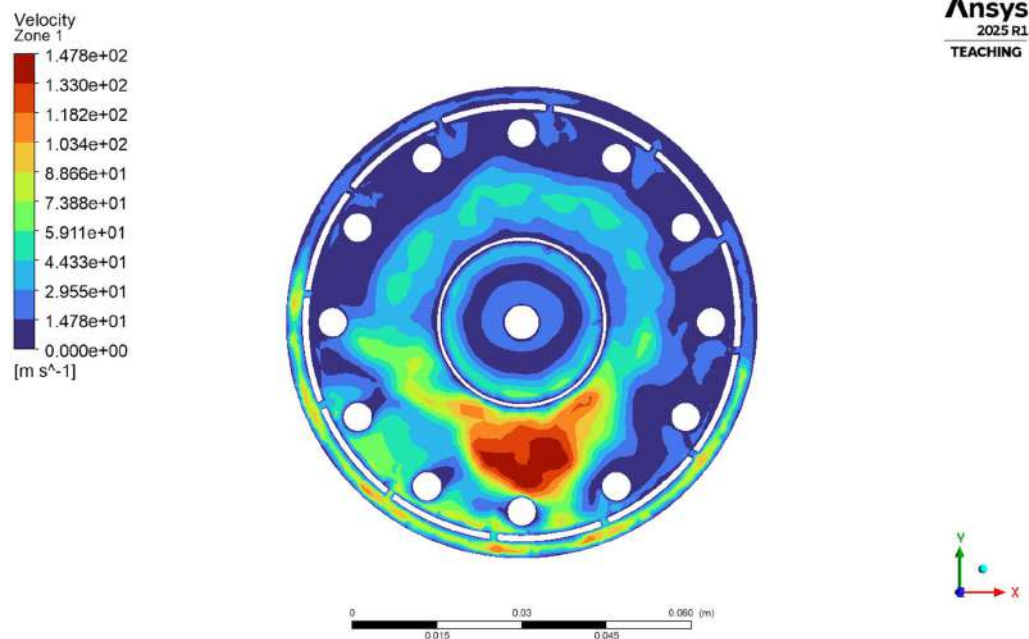


Fig.4.1.2.3(b) Zone 1 velocity Contour

c. Zone 2 Velocity Distribution:

The colossal at the left indicates the value of velocity in units of m/s, where dark blue is very low velocity regions (near 0 m/s) and red corresponds to highest amalgam fluid regions (approximately above 120 m/s). From the contour distribution you can see that most of flows enter from outer region of swirler and it is guided by vanes or flow passages generating a vortex as flowing toward to central outlet. Areas of high velocity (yellow to red zones) are clearly defined adjacent to some vane exits and toward the bottom of the inner circular chamber; these are regions that exhibit flow acceleration, forced by geometric constriction and directional change from swirler blades. The blue and lighter-green regions, on the other hand, indicate relatively lower velocities, which correspond to recirculation zones or regions where flow is losing momentum due to turbulence and mixing. The semicircular holes all around the outer ring seem like secondary air passages or structural elements, affecting local velocity and introducing their own mixing ends. A strong velocity gradient moving inward from the edge to the centre suggests that an efficient swirler developed both angular momentum and radial pressure gradients, which are crucial in flame stabilization, as evident in combustion applications like gas turbine combustors and industrial burners. In general, the contour plot shows how swirler geometry is controlling airflow patterns generating regions of high velocity, low velocity and recirculation with increasing fuel–air mixing, flame stabilization, and combustion efficiency in combustor.

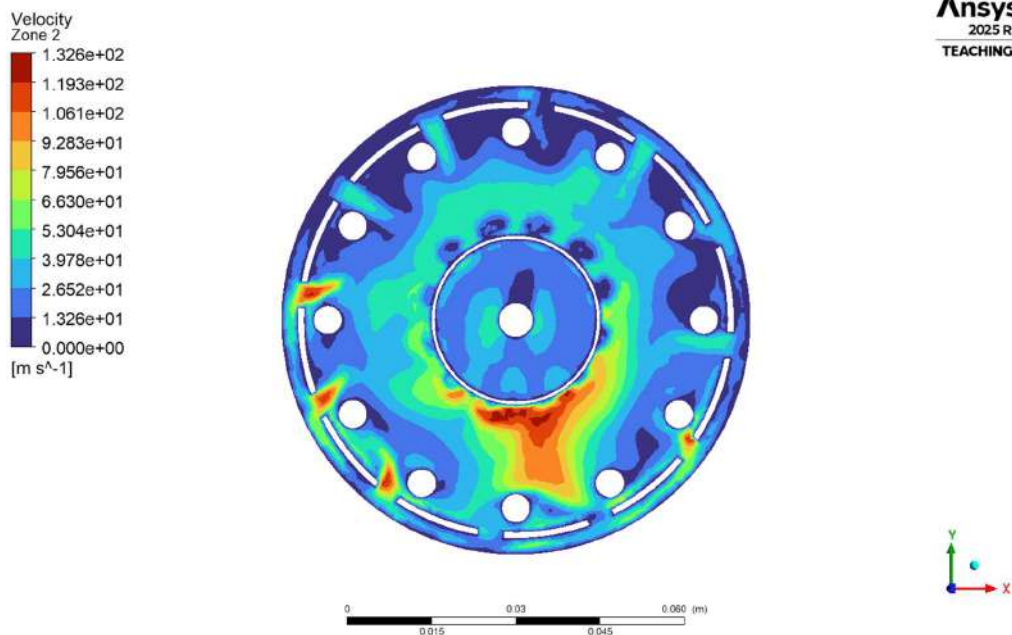


Fig.4.1.2.3(c) Zone 2 velocity Contour

d. Outlet Velocity Contour:

The velocity scale goes from 0 m/s (dark blue – very low velocity) to about 1.83×10^2 m/s (red – very high velocity). Because of the contour, the average velocities are moderate (green to light blue) through most of the annular region, suggesting quite smooth airflow as it leaves the chamber. Notice that we see a separate high-velocity region (orange to red) at the lower part of the outlet ring, which indicates strong acceleration in. Such high-speed zone is probably resulted on local geometric effects originated due to the flow direction that pushes it toward that portion of the outlet and towards upstream. The velocity is lower near the inner and outer walls of the annular passage (blue shades) owing to boundary layer formation and wall friction effects. Non-uniform distribution of high velocity regions throughout the annulus implies that even though it appears through the satisfactory discharge packet analysis, flow coming out from the chamber is non-uniform over packed flow and could dictate downstream mixing performance as well as overall flow performance.

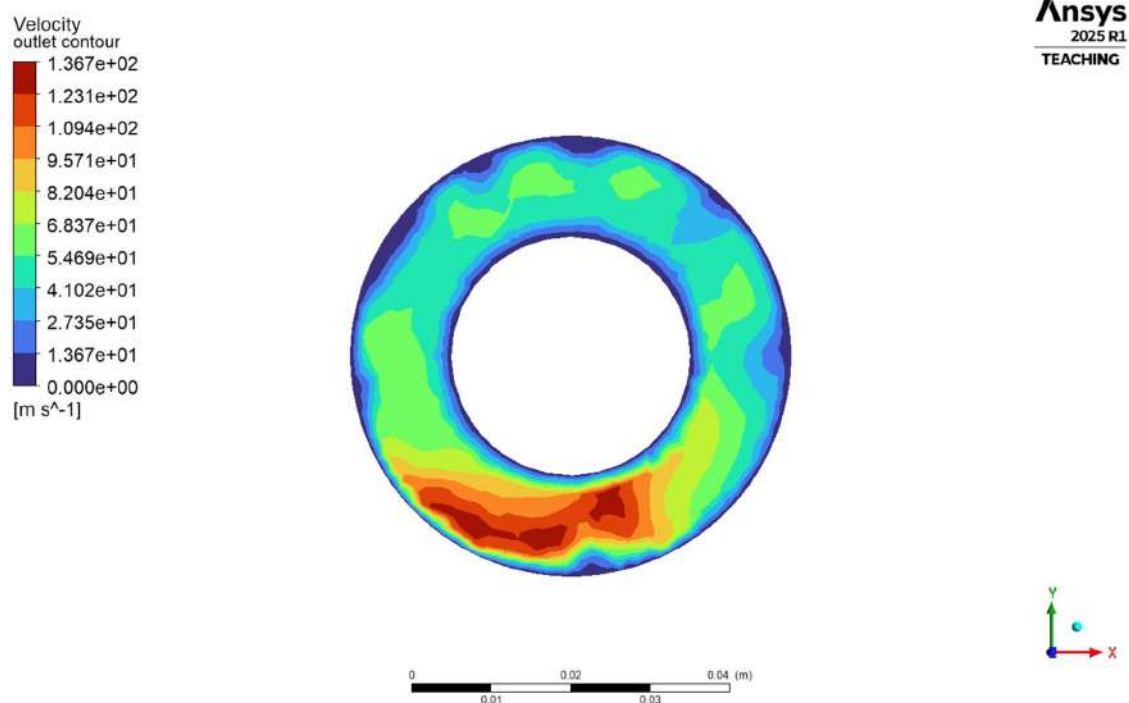


Fig.4.1.2.3(d) Outlet velocity Contour

d. Cross-Section Velocity Distribution :

The velocity scale goes from 0 m/s (dark blue – very low velocity) to approximately 1.80×10^2 m/s (red – very high velocity). The upper chamber region has a low flow velocity, shown as dark blue and light blue colours, which represent static air and recirculation zones within the broad inside cavity. The air is forced through the narrower flow channel, which result in significant increase in air velocity due to pressure drop and also leads to fluid acceleration as they are flowing towards the outlet. That's shown in the orange and yellow and red areas of lower chamber, where fluid is fast-moving before it leaves through that passage. Outside the core flow, effects of boundary layer and wall mechanic can be seen as the velocity gradually decreases near walls and is represented by blue regions along the boundaries. In all, the contour shows that with larger chamber spaces, the flow slows down due to which internal circulation zones are formed and then it accelerates in outlet channel creating high-velocity localized regions which further affects total flow distribution inside the system.

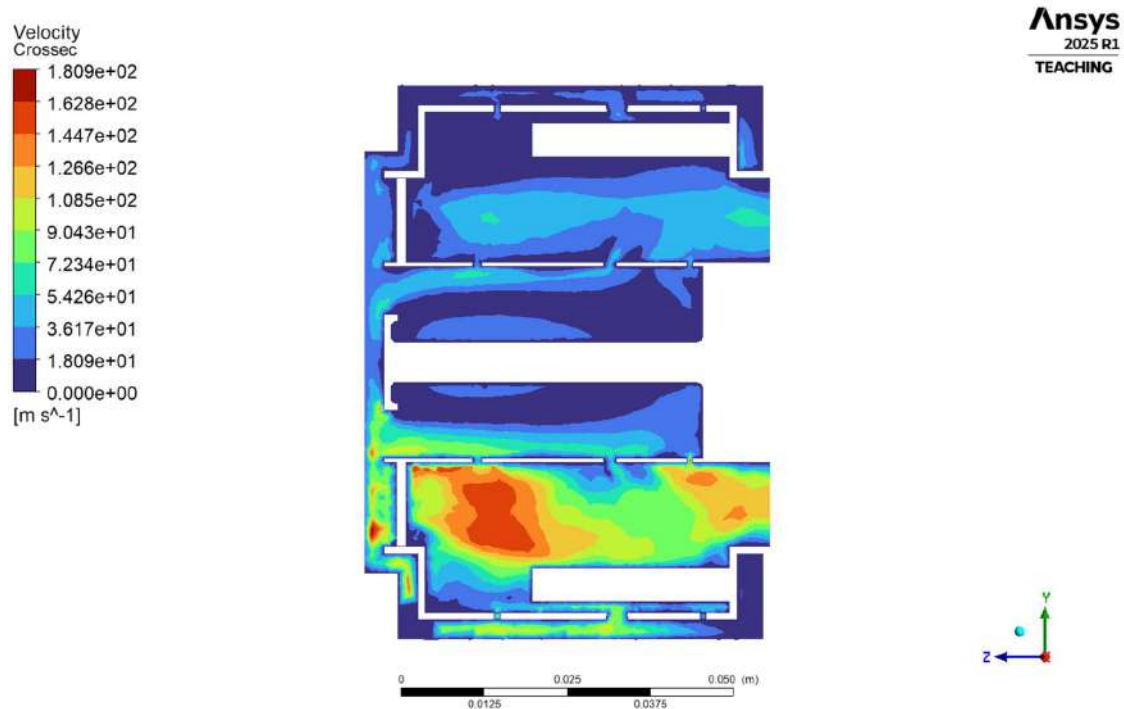


Fig.4.1.2.1(e) Cross-Section velocity Contour

e. Velocity Vector Field:

These vectors indicate the path of airflow as it travels through the chamber's internal channels. In the corresponding figure, we can see that within most of the chamber region, low velocity vectors (blue shades) appear to be present, which means that airflows will be slower in the larger internal cavities. In the lower passage, the air flows to its exit; with each vector, it becomes denser (passing more closely together) and a little longer, indicating an increase in flow velocity, as well as a clear flow direction to the outlet. In a few areas of the vectors, the lines widen slightly and move to an angle, which indicates recirculation as well as mixed shelves where fluid circulates before moving again. Next to walls, boundary layer formation and frictional effects reduce the vectors. "A projection of the vector shows the direction and magnitude of air movement at any given point, allowing us to see how airflow travels through the chamber, slows in wider sections, creates internal circulation patterns and continues moving along connecting channels towards outlets.

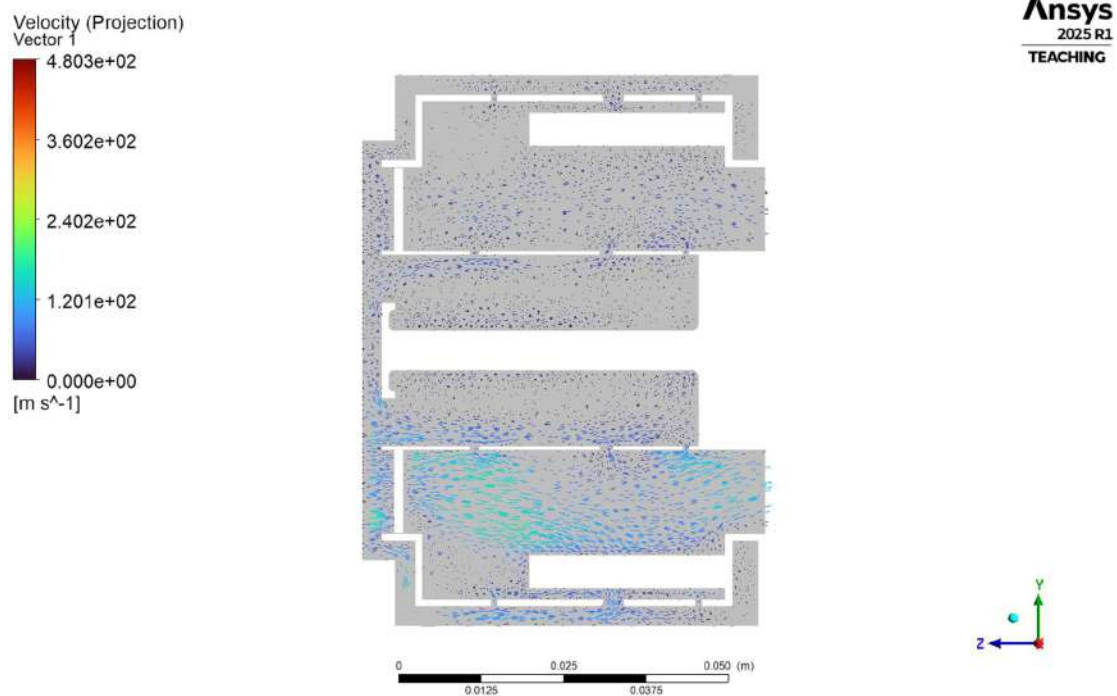


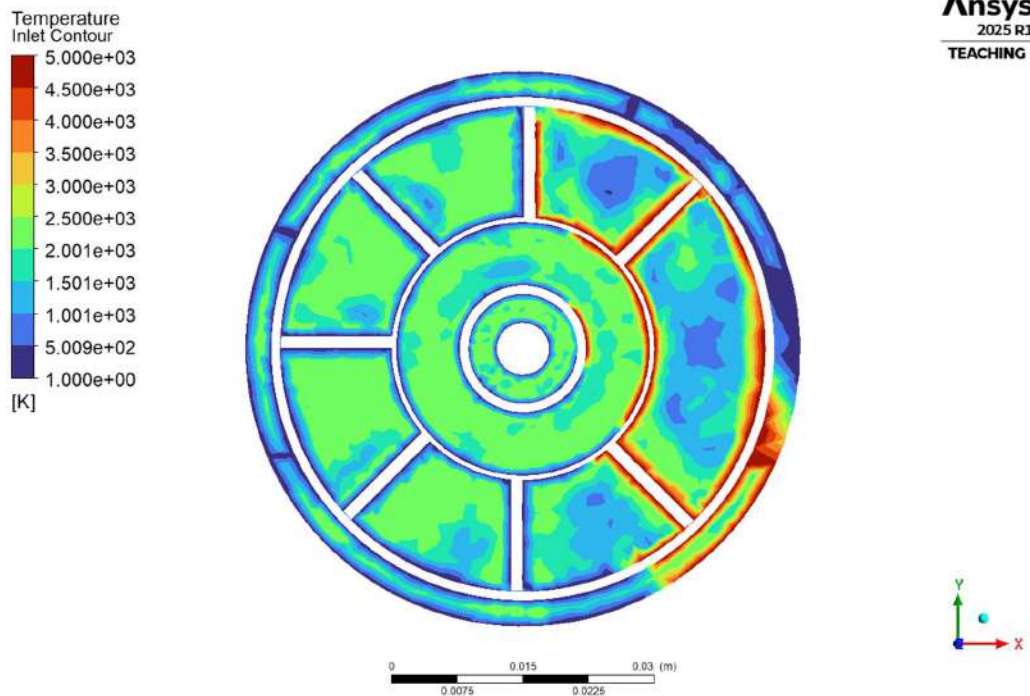
Fig.4.1.2.1(f) Cross-Section velocity vector Contour

4.1.3 Temperature Contour Comparison

4.1.3.1 Temperature Contour 8 Blade

a. Inlet Temperature Contour :

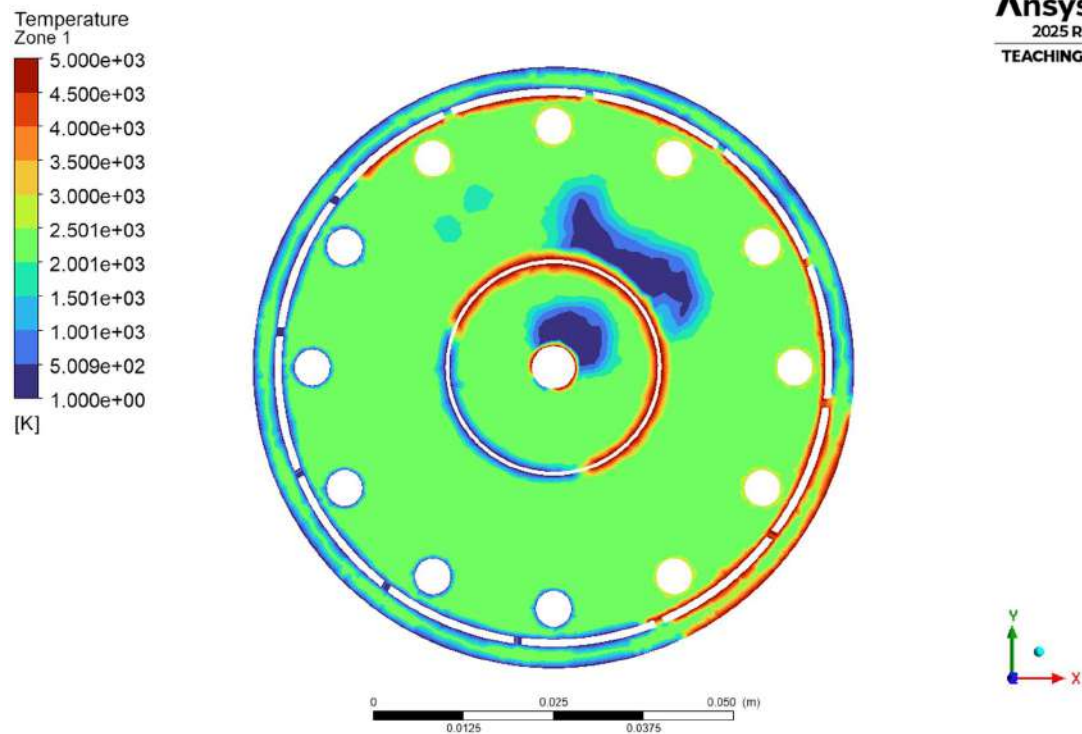
The scale for temperature ranges from ~300 K (dark blue – low temperature) to ~5000 K (red – very high temperature). From the contour, most of the chamber region has moderate temperature (green to light blue) which means the inlet flow is relatively uniform, not just in the passage(s) which lead into them. Due to low heat transfer around the shaft and a limited flow movement, the central hub region remains relatively stable in temperature. But notice the higher temperature area (yellow to red) along the right of the outer annular region — this indicates some localized heating associated with concentrating flow or wall interaction in a small portion of the chamber. Temperature differences across the outer boundary show wall heat transfer and flow circulation effects. These patterns give evidence that the temperature remains relatively uniform throughout the inlet passages with only a few hot spots localized at certain areas, affecting not just thermal behavior but also observations post-mixing within our chamber.



4.1.3.1(a) Inlet Temperature Contour

b. Zone 1 Temperature Distribution :

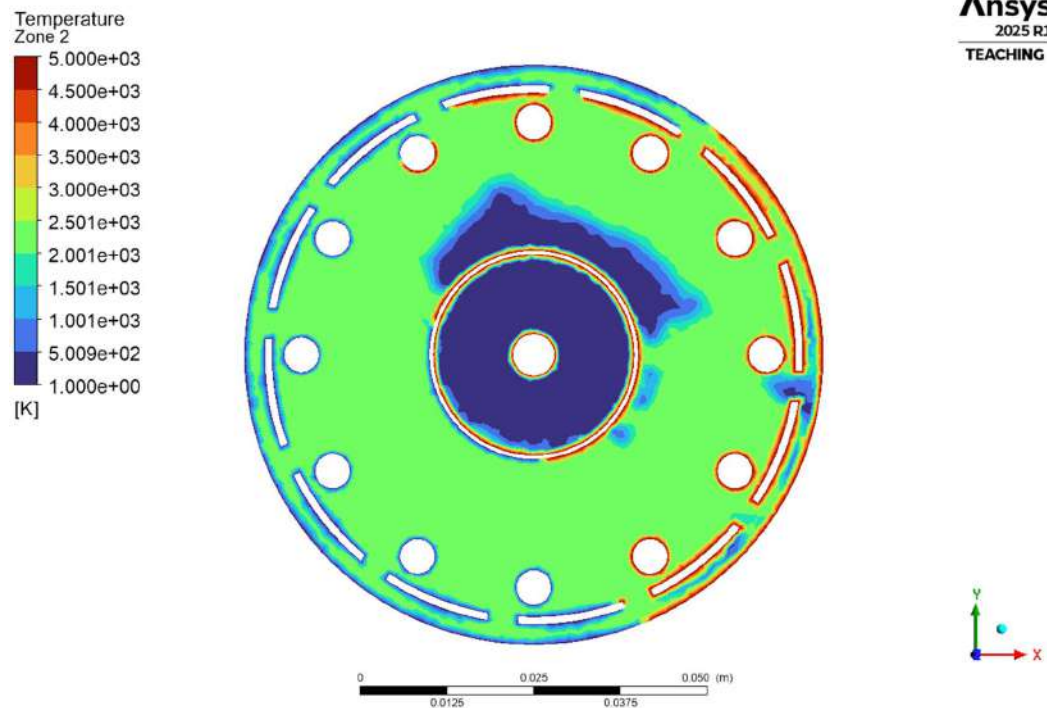
From the contour, most of the chamber region is dominated by moderate temperatures (green shades), implying a relatively homogenous thermal distribution within the entire chamber. In the middle hub region, less flow interaction and lower heat exchange next to the shaft lead to cooler temperature zones (blue colour). A localized colder area (dark blue) can also be seen in the upper central region which may result from recirculating flow or entrainment of cooler oncoming air. Higher temperature zones (yellow to red) are also observed along the outer annular boundary, indicating that heat is trapped next to the wall following flow interaction and potential thermal accumulation in the external duct. In general, the contour shows a relatively uniform temperature field with some local hot and cold spots, which can be associated to the mixing behavior as well as heat transfer within the chamber.



4.1.3.1(b) Zone 1 Temperature Contour

c. Zone 2 Temperature Distribution :

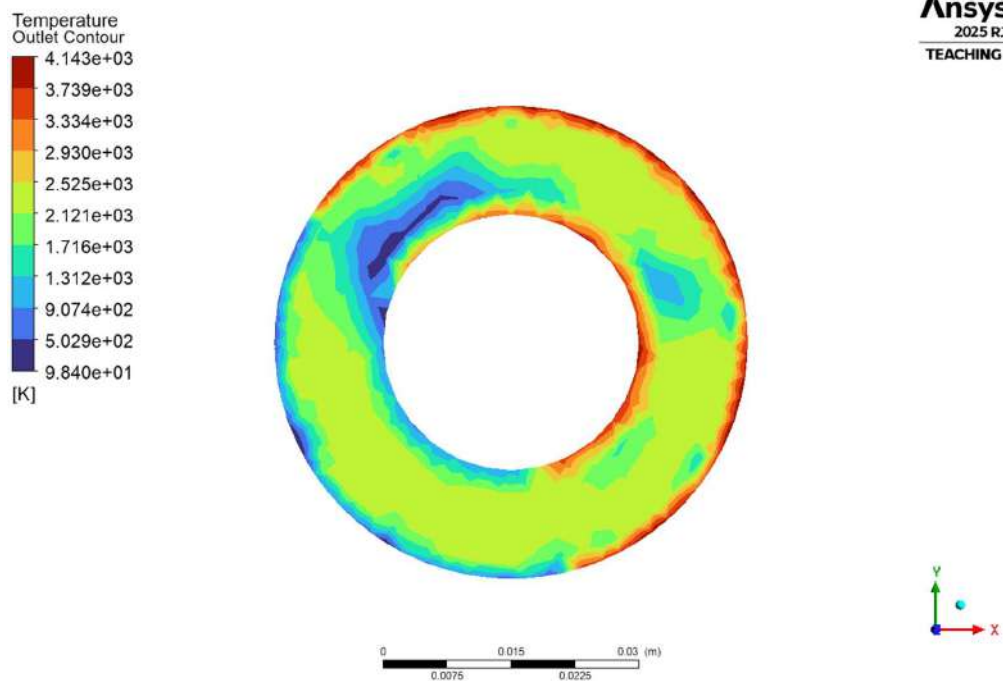
The scale for temperature runs from around 300 K (dark blue – low temperature) to about 5000 K (red – very high temperature). Most of the chamber area displays moderate temperatures (green shades) on the contour, meaning that heat is fairly distributed through it. In the central hub region, there is a high zone of low-temperature (dark blue), which is formed when the solid shaft and limited airflow in that area lead to less heat transfer and mixing. In this context, the cooler region that appears to span from the centre through to the upper portion of the chamber indicates circulating or sluggish fluid which potentially transports air at lower temperature. Stronger flow interaction and heat accumulation in proximity to the chamber walls can yield higher temperature regions (yellow-to-red) along both the outer annular boundary as well as adjacent to the slots. Generally, the contour shows a non-uniform heating of system with cooler central zone and outer warmer region that corresponds to the mixing pattern and thermal behaviour inside of chamber.



4.1.3.1(c) Zone 1 Temperature Contour

d. Outlet Temperature Contour :

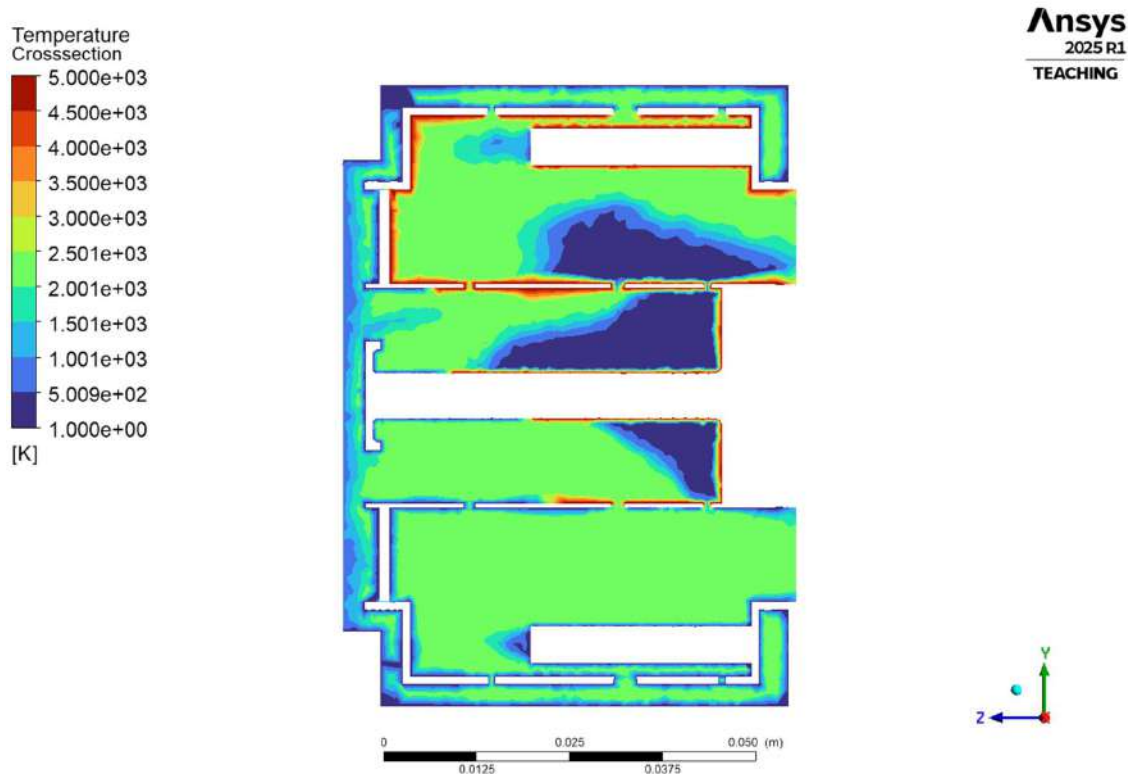
The temperature scale extends from around 300 K (dark blue – low temperature) to about 4.16×10^3 K (red – very high temperature). Based on the contour, over most of the annular outlet area, moderate temperatures (green to yellow colours) can be seen; this shows that the temperature value of the outflow is relatively consistent across the outlet region. But also, some higher temperature areas (colouring from orange to red) can be seen over sections of the outer contour of the neighbouring ring, indicating flow interaction and thermal accumulation at zones near to the outer wall. Observe a cooler region (blue shades) along the inner annular flow, which might reflect downwelling or mixing of relatively cold material due to recirculation. The gradients in colour around the annulus indicate that it is not perfectly even uniformly circumferentially, yet this can be considered somewhat evenly escorted out from the chamber when performed. In summary, the contour suggests that the flow leaves out of the chamber at moderate temperature levels with localized hot and cold spots that occur due to mixing.



4.1.3.1(d) Outlet Temperature Contour

e. Cross-Section Temperature Distribution :

From this contour, moderate temperatures (green shades) the upper chamber region suggests that heat is productively distributed in that part of the flow path. In comparison, the middle passage exhibits lower temperature regions (blue shades) that may be produced by colder trailing air or recirculation zones where fluid velocity is low. As the flow passes through these narrow connecting channels between the chambers, hotter regions (yellow to red) are observed consistent with greater thermal interaction and heat transfer as the fluid accelerates through these constricted areas. The lower chamber displays mostly moderate temperatures, indicating that the flow enables mixing and redistributes thermal energy as it progresses through the system. Thermal diffusion results in slight temperature gradients near the walls from heat transfer between the fluid and chamber boundaries. In general, along the contour of the chamber sections, temperature changes—high and low temperatures—in narrow space are observed in the passages; cooler zones are visible in an area with broad recirculation. The outer section is a result of flow movement, mixing, migration, and heat transfer together within the system.

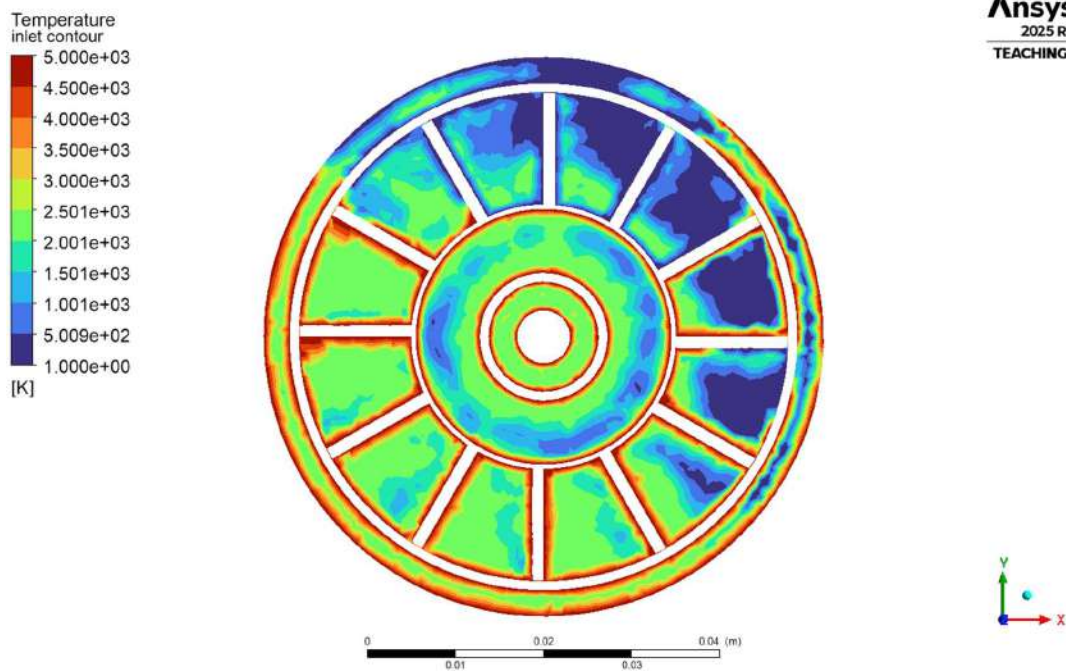


4.1.3.1(d) *Cross-Section Temperature Distribution Contour*

4.1.3.2 Temperature contour 12 blade

a. Inlet Temperature Contour :

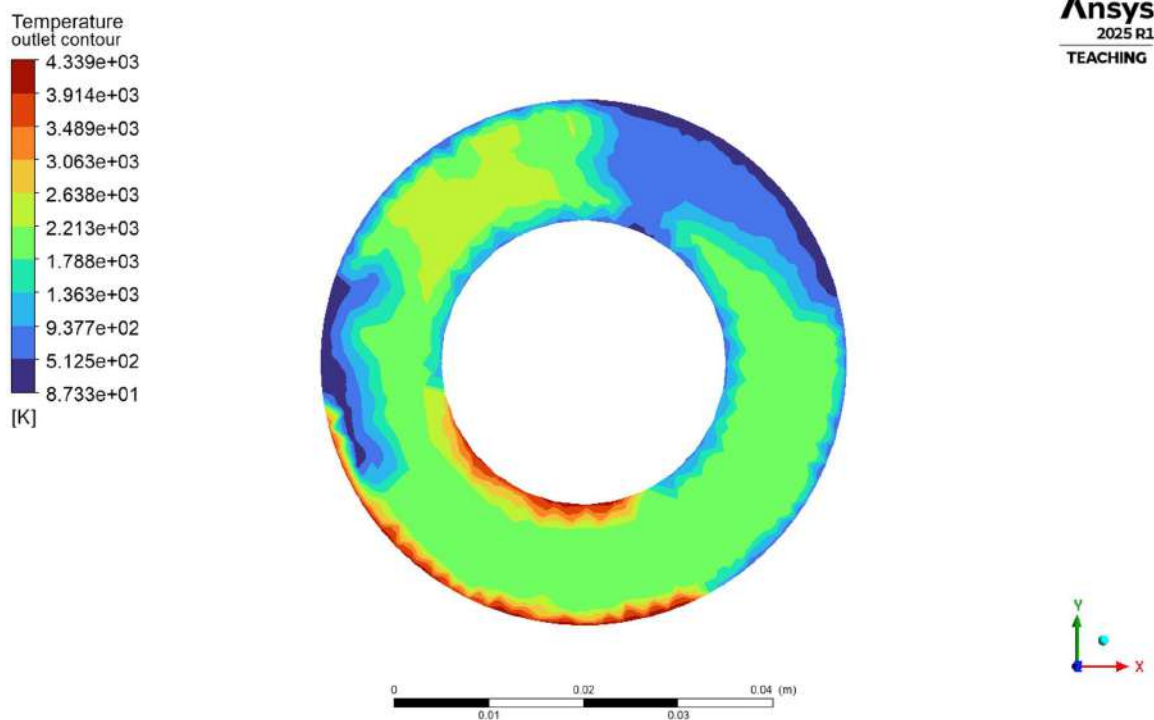
The outer annular contour indicates temperature concentration (orange to red) around the chamber walls, as flow inhibits heat transfer due to interaction and thermal buildup. The radial vane passages are primarily identified as moderate temperature channels (greenness to blue) implying that incoming flow is distributed through the vanes with homogeneous thermal conditions. Some pump vane sectors show cooler regions (dark blue), which means that lower-temperature fluid is entering or that recirculation reduces heat transfer. Temperature levels are scattered within the central hub region, indicating that the flow around the shaft is quite stable with little thermal overlap. In general, the contour shows that inlet temperature is well distributed within the chamber with a few hot spots in some areas close to the outer boundary of the combustion chamber and cold pockets observed in certain passages of vanes indicating effects of flow distribution and mixing characteristics inside the chamber.



4.1.3.2(a) Inlet Temperature Contour

d. Outlet Temperature Contour:

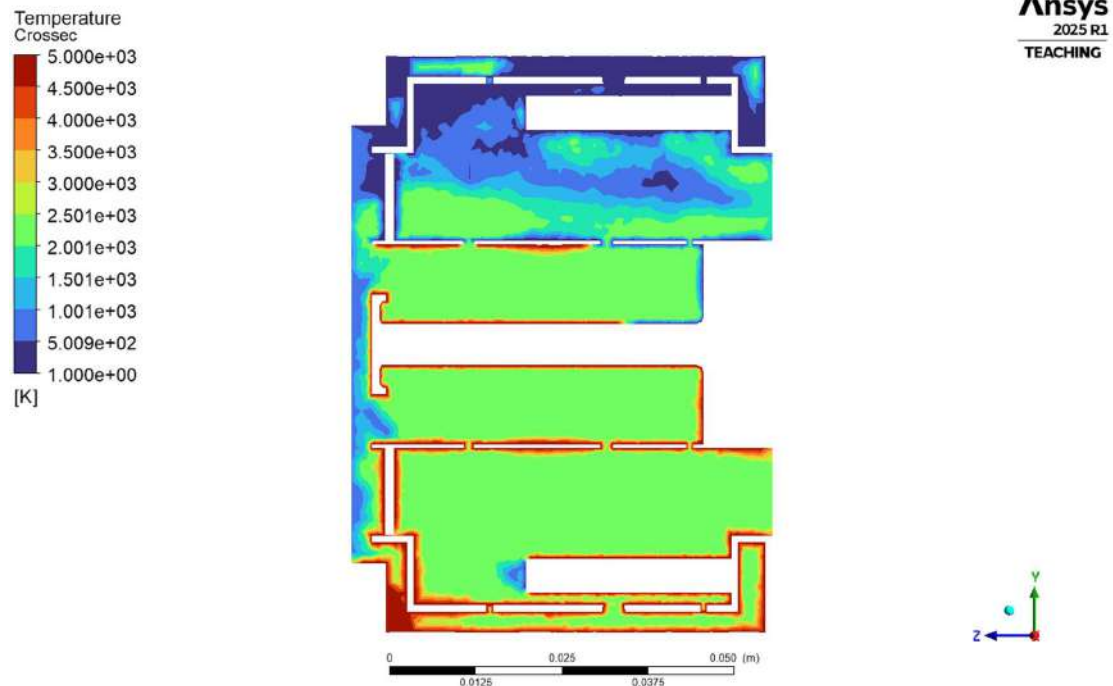
The temperature scale covers approximately from 300 K (dark blue – low temperature) to $\approx 4.23 \times 10^3$ K (red – very high temperature). In the annular section of the exit, hotter fluid is concentrated downwards as it exits the chamber (using a colour code where yellow to red indicates higher temperatures) From the outlet ring, it is evident that its higher part shows lower temperatures (blue colours), indicating that a cooler air flow or mixed one reaches there. The annular region is largely covered in green to light blue colours, which are indicative of moderate temperatures and a partial mixing of hot and cold fluid prior to the exit. Inhomogeneity in the flow pattern, mixing effects, and heat conduction to chamber walls produce temperature differences throughout the annulus. In general, the contour suggests that the outlet stream has a non-homogeneous temperature distribution and exhibits localized 'hot' areas followed by colder zones in its flow plane representative of thermal mixing and flow behaviours established within the chamber before leaving it.



4.1.3.2(d) Outlet Temperature Contour

e. Cross-Section Temperature Distribution:

The upper chamber is cooler and these places are at lower temperature and in fact might be reflux areas, since we can lead to low heat collection. Moving through the middle passages, however, we see the temperature (in shades of green) mellows out significantly to make for more even mixing and a build-up of heat within the chamber. But as more fluid flows through those restricted passages (yellow heads of arrows) and out of the lower chamber, regions with increased temperature (yellow to red) can be seen in the converged region and along narrow connecting channels. The hotter area around the walls is also due to thermal interaction and heat accumulation at the chamber limits. In general, this contour shows the temperature distribution along the flow trajectory that is, cool regions observed in the upper chamber and high temperatures preferably in lower passages indicating effects originating due to flow mixing recirculation and wall heat transfer phenomenon dynamics within the system.

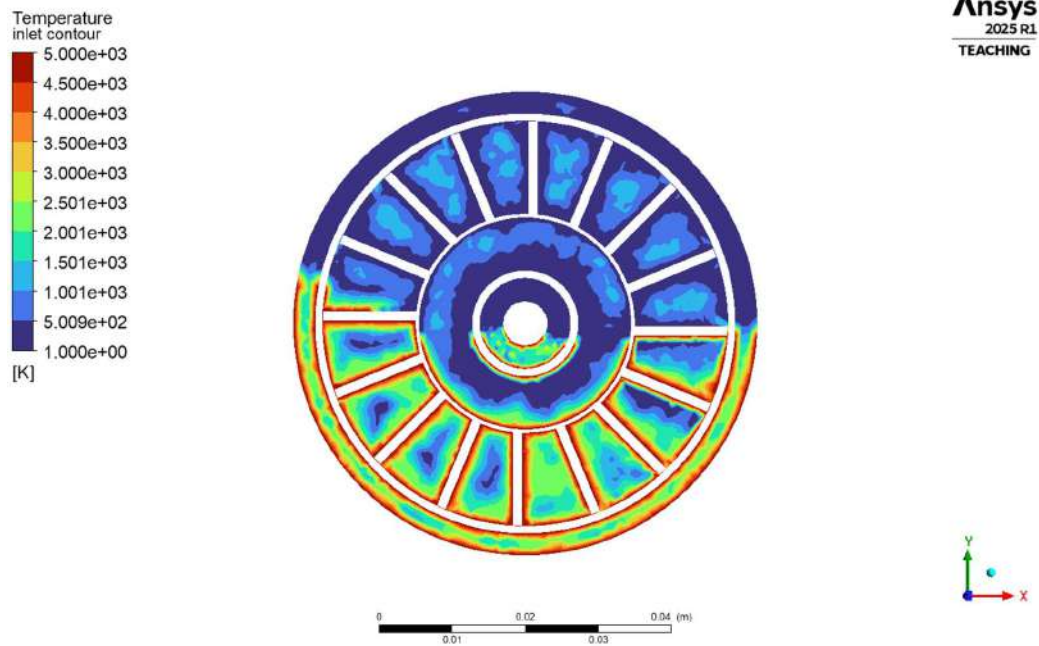


4.1.3.2(e) *Cross-Section Temperature Distribution*

4.1.3.3 Temperature contour of 16 blade

a. Inlet Temperature Distribution :

Temperature scale between 300 K (dark blue - low temperature) and 8.20×10^3 K (red - very high temperature). From the contour, majority of upper-side vane passages have lower temperatures (in blue shades), and indicate relatively cooler incoming flow. On the contrary, lower vane passages present moderate to higher temperatures (green to yellow) indicating that there are stronger thermal mixing and higher heat in this region of chamber. Please increase your compatibility with browsers and follow the good performance on other ones, i.e. along the outer annular boundary where thin bands of higher temperature (orange to red) are evident because of heat build-up as well as interaction between flowing fluid and chamber wall. Moderate temperature levels are present within the central hub region, indicating slight variations around the shaft area. Such a contour reveals that the temperature at the inlet is non-uniform and relatively cooler in upper passages while being warmer in lower parts which clearly states the effect of flow distribution and thermal mixing happening within the chamber.



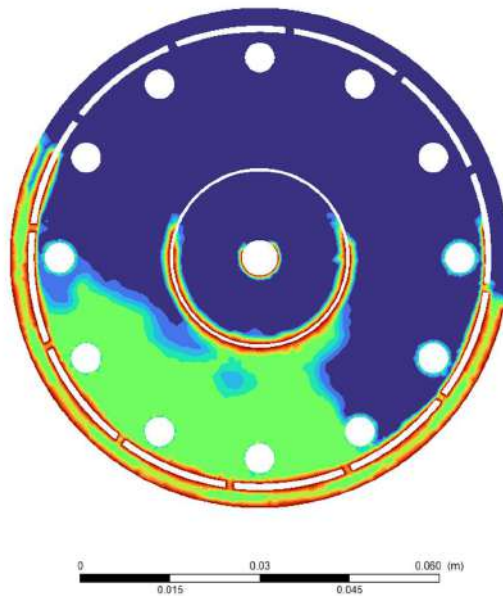
4.1.3.3(a) Inlet Temperature Distribution

Temperature
Zone 1



5.000e+03
4.500e+03
4.000e+03
3.500e+03
3.000e+03
2.501e+03
2.001e+03
1.501e+03
1.001e+03
5.009e+02
1.000e+00

[K]



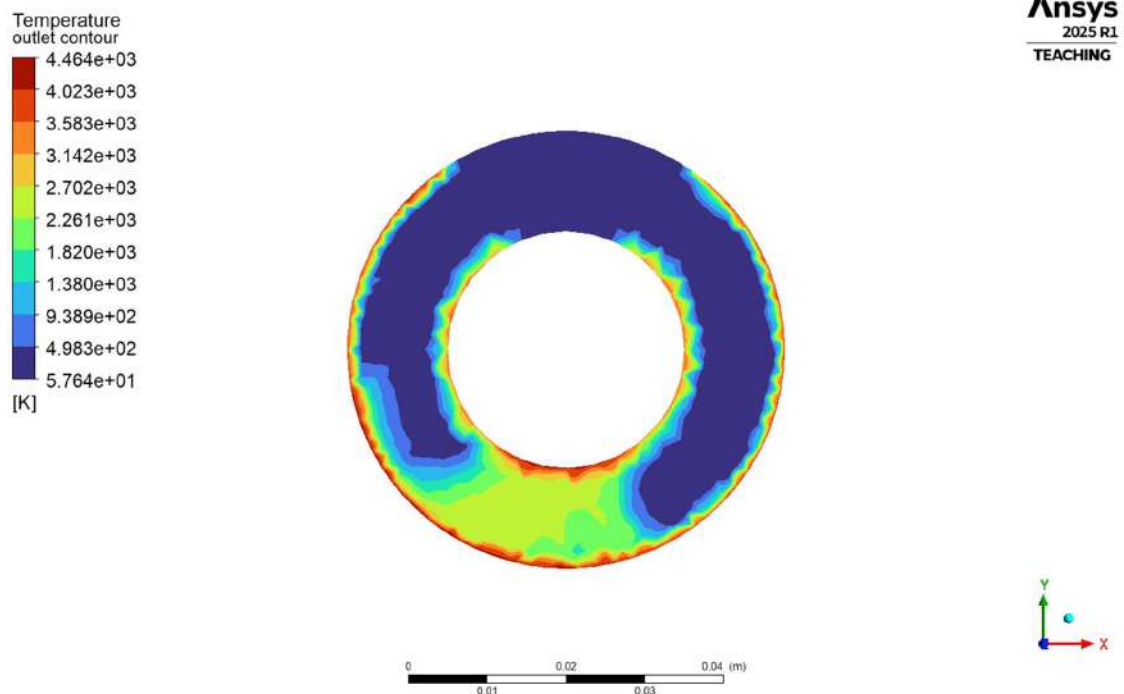
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d. Outlet Temperature Distribution:

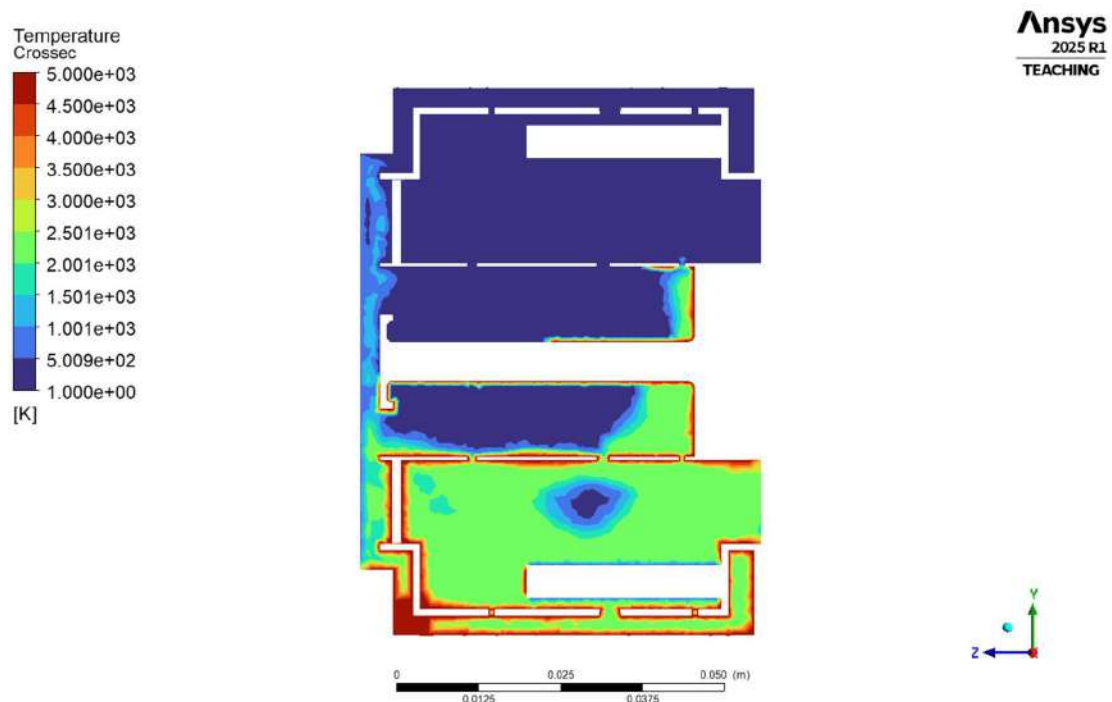
From the contour, more of the upper part of the annular outlet is at low temperatures (dark blue), suggesting relatively cool fluid exits in that area. Meanwhile, the bottom of outlet ring is brighter (green to yellow and red) which indicating that high-temperature fluid is concentrated in this area before it out. Flow circulation and mixing inside the chamber results in uneven thermal distribution around the annulus, which is reflected by temperature variation. Near the inner and outer edges of the annular channel (the gap between the fluid flow) there are small gradients in temperature due to heat transfer as well as interaction with the walls of the chamber. In general, the contour indicates that the outlet flow has non-uniform temperature distribution with localized hot and cool zones in its different sections reflecting fluid mixing and heat transfer inside the chamber before the flow exits.



4.1.3.3(d) Outlet Temperature Distribution

e. Cross-Section Temperature Distribution:

From the contour, it can be seen that low temperatures (dark blue) dominate the upper chamber region, meaning cooling fluid occupy this region and heat transfer there is relatively limited. During the middle passage, however, temperature rises to moderate ranges (green hues) as flow travels through chamber and mixing starts. Stronger thermal contact and heat accumulation of fluid are realised in restricted passages, which manifest as high-temperature zones (yellow to red) around the walls of narrow channels in the lower chamber section. Regions of lower temperature (blue area) in the middle of the lower chamber indicate a recirculation zone where cooler fluid has been trapped and mixes with adjacent flow. The overall contour reveals that the temperature rises from top to bottom along the length, which shows the effect of flow circulation, mixing, and heat transfer with walls.



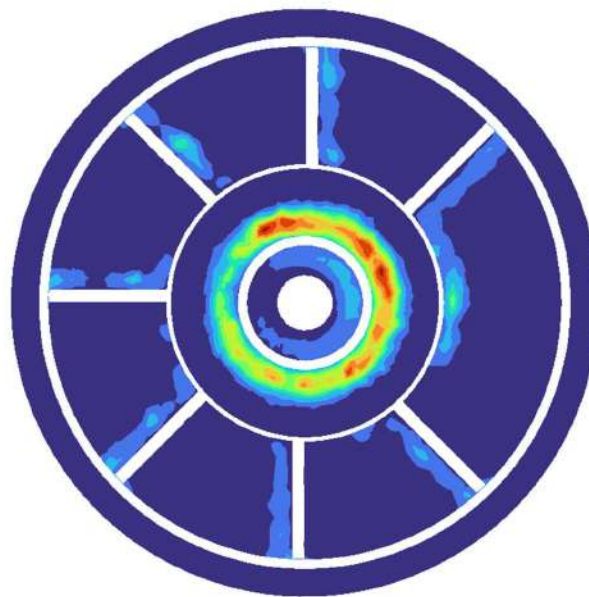
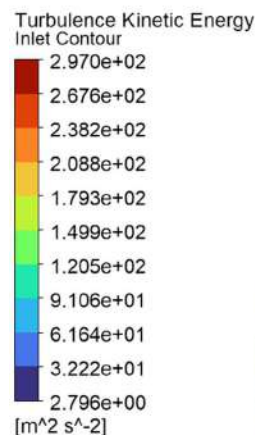
4.1.3.3(e) *Cross-Section Temperature Distribution*

4.1.4 Turbulence Kinetic Energy

4.1.4.1 TKE Contour 8 Blade

a. Inlet Turbulence Kinetic Energy :

Most of the sections outside the chamber and in the vane passages exhibit insignificantly low levels of turbulence, as shown by dark blue portions, indicating relatively smooth inflow. But a clear ring of higher turbulence (yellow to red) is around the central hub region. This suggests that enzymes are strongly mixed near the inner annular region where fluid flows into the central structure and is deflected [35]. This is mainly due to velocity gradients, flow separation and interaction of the radial passages with rotating or central element which leads to turbulence for this region. Flow moves outward through the vane passages and as turbulence decreases the outer sections tend to be blue. In general, the contour indicates that turbulence is primarily localized in an inner annular region of the chamber while the remaining areas of the chamber are relatively quiescent, suggesting mixing and energy dissipation be primarily occurring closer to the centre of the chamber.

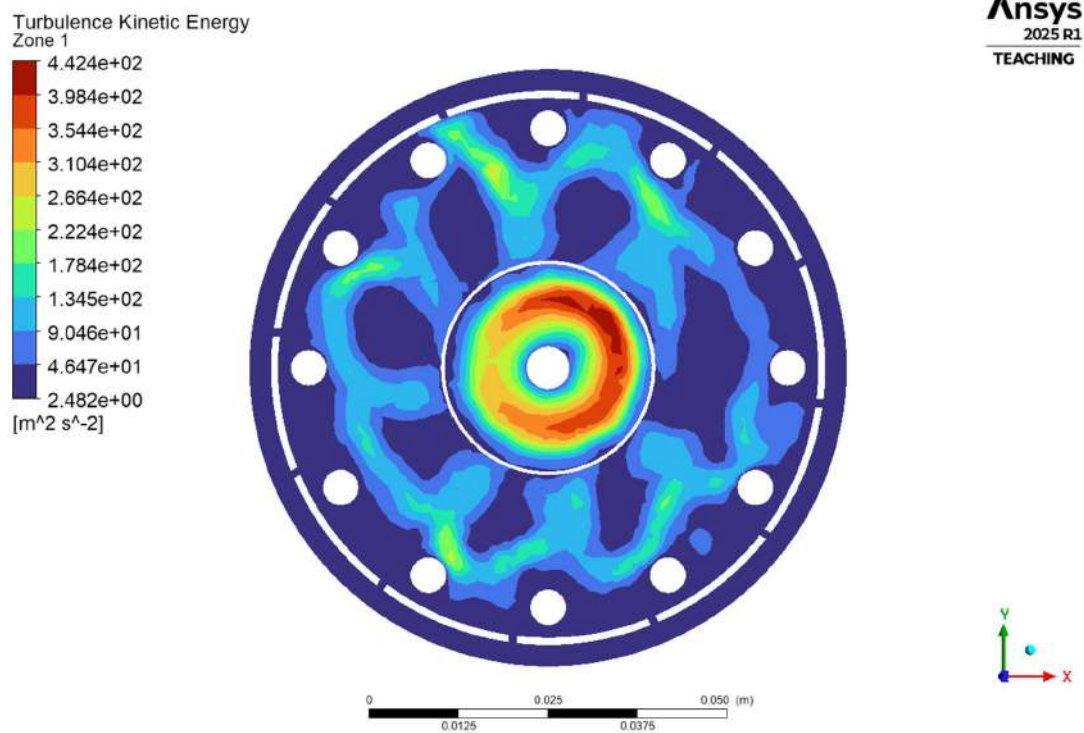


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4.1.4.1(a) Inlet Turbulence Kinetic Energy

b. Turbulence Kinetic Energy in Zone 1:

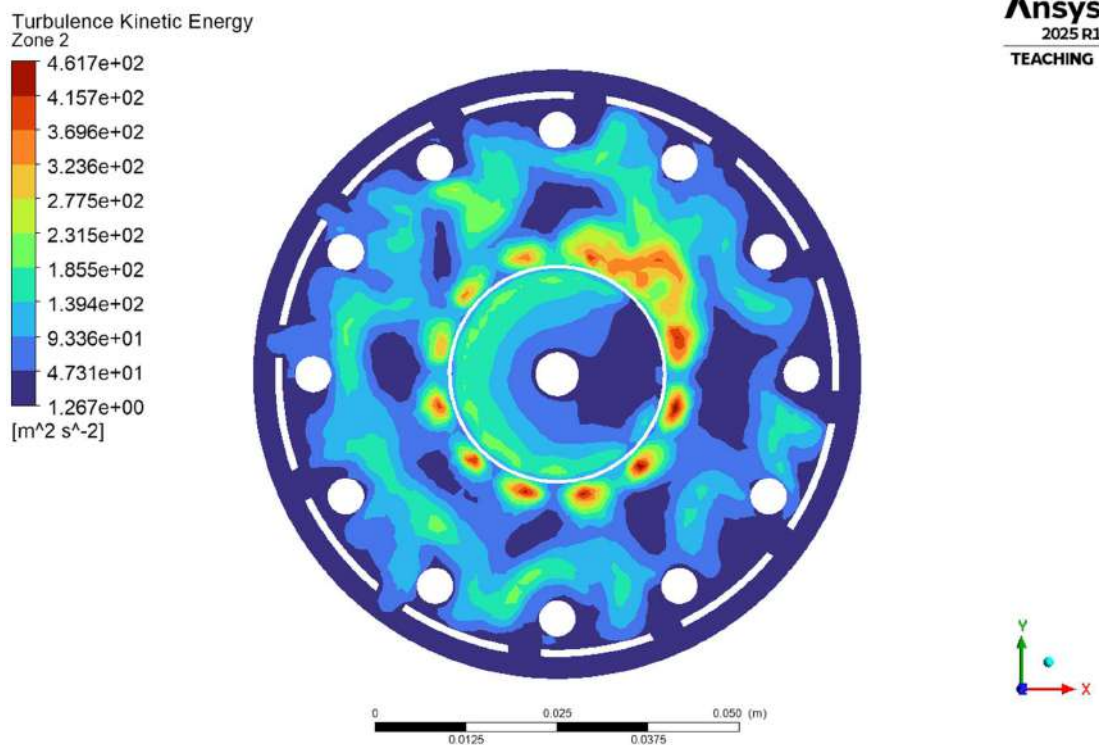
Based on the contour, the most turbulent area is clustered around the inner annular zone near to central hub due to show in red and orange colours. This shows strong mixing and lots of fluid interaction as the airflow redirects and interacts with the closed geometry inside the chamber. Zoned around the centre of the chamber are swirling areas, which emit turbulence and can be seen as the reddish type patterns; then moving outward into less turbulent formations you have greenish to light blue areas. These are the regions of moderate turbulence due to the flow circulation and interaction with the bounding perforations. Due to this reduction in velocity fluctuations, the flow near the walls becomes more stable, leading to mostly low turbulence levels (dark blue) on the outer boundary of chamber. This contour, as a whole, shows that the turbulence is most intense in the vicinity of the innermost area and decreases toward the outer chamber indicating that most of mixing and energy loss occur at close to inner annular space but there is relatively more smooth flow in outer region.



4.1.4.1(b) Turbulence Kinetic Energy in Zone 1

c. Zone 2 Turbulent Kinetic Energy :

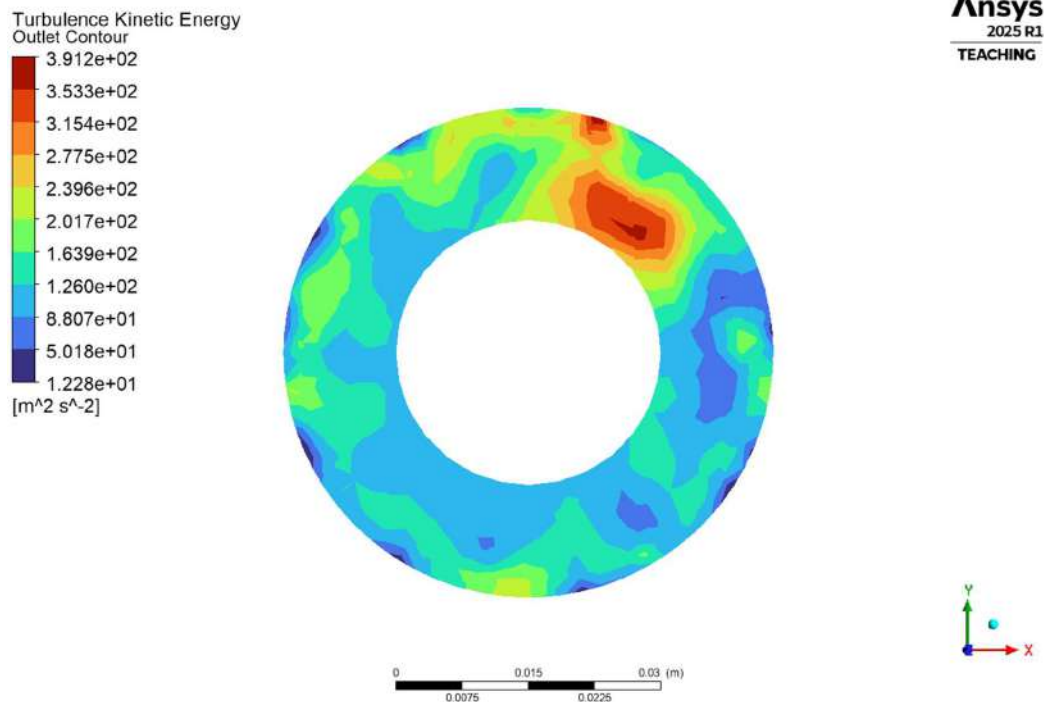
Turbulence kinetic energy contour open from ANSYS simulation output of the turbulence in the circular chamber with several peripheral holes the colour scale goes from dark blue for the lowest turbulence levels to red for high turbulence kinetic energy. This contour shows that most of the outer region of the chamber, far from the inlet and outlet openings (approaches: blue to green shades) experiences low turbulence (low angle left). On the inner annular region close to the central hub, multiple localized high-turbulence spots (yellow to red) can be seen. These areas represent strong flow interaction and demonstrate where the fluid reverses direction, interacting with chamber geometry. The main sources of turbulence in these regions are due to velocity gradients, flow separation, and interaction between the incoming flow and recirculating fluid. Turbulence decreases progressively as the flow acquires an outward direction towards the peripheral holes and chamber walls, indicated in online Supplementary Movie 1(17) by the passage from green to blue hues. Generally, the contour suggests that turbulence is more intense around the inner circular zone and regular flow occurs in outer chamber which signals that a larger portion of mixing processes and energy dissipation happens near to the central part of chamber.



4.1.4.1(c) Zone 2 Turbulent Kinetic Energy

d. Outlet Turbulence Kinetic Energy :

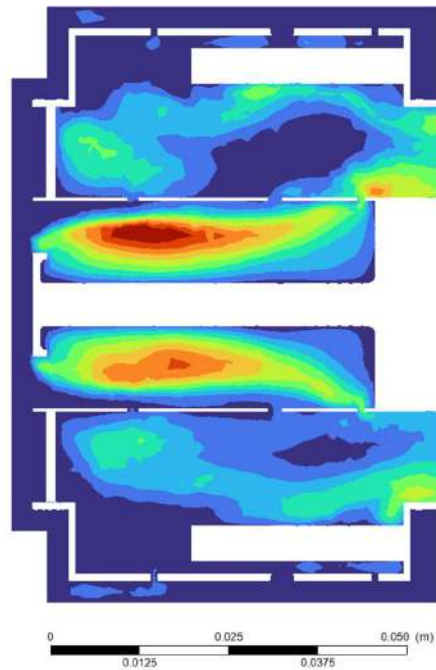
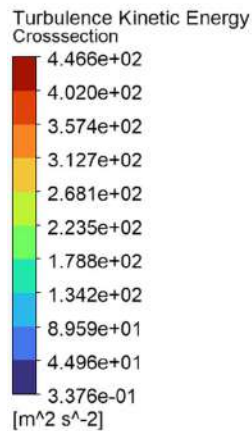
Turbulence Kinetic Energy Outlet Contour from ANSYS Simulation The colour scale varies from low levels of turbulence (dark blue) to high levels of turbulent kinetic energy (red). From the contour, a large portion within the annular outlet region shows moderate turbulence levels (represented by shades of blue to green) suggesting that an even flow is achieved near the outlet. Nonetheless, we can see a small local region where turbulence has increased (yellow to red) in the upper-right region of the annular ring which indicates stronger velocity fluctuations and mixing in that part. It can be due to uneven flow distribution or merging of various streams inside the chamber prior to reaching the outlet which results in a higher turbulence. There are slight differences in turbulence due to the interaction of boundary layers with the wall near through those inner and outer walls of annular passage. In summary, the contour suggests that the outlet flow has generally moderate turbulence with some high-turbulence spots, indicating the degree of mixing and manner of flow which occurs within a combustion chamber prior to exiting.



4.1.4.1(d) Outlet Turbulence Kinetic Energy

e. Cross-Section Turbulence Kinetic Energy :

The upper chamber portion is mostly low to moderate turbulence (blue to green colours) from the contour meaning not much mixing occurred there, implying quite a stable flow in that region. There is also a high turbulence region (yellow to red) in the middle connecting passage that clearly shows strong velocity fluctuations and intense mixing due to flow acceleration through the narrow channel. Moving along, below, the bottom chamber section sees another region with moderate to high turbulence (green to orange) indicating further mixing as the flow branches out in the deeper cavity. The near wall jump in velocity is because of boundary layer formation and lower velocity fluctuations, therefore the reason for darker blue if you are further from the chamber walls. In conclusion, the contour indicates that turbulence is primarily formed in the constricted areas where the flow rate increases following a change of direction and exhibits lower levels of turbulence within the larger chamber regions.



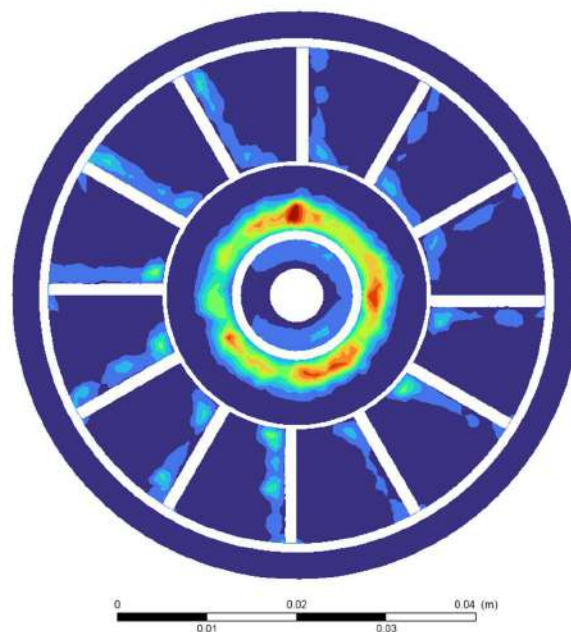
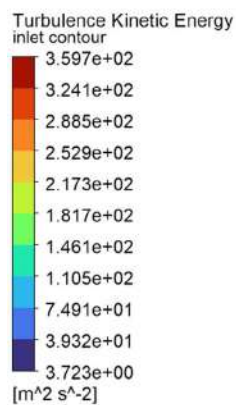
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4.1.4.1(e) Cross-Section Turbulence Kinetic Energy

4.1.4.2 TKE contour 12 blade

a. Inlet Turbulence Kinetic Energy :

From the contour, we can see that most of the outer chamber and vane passes has low turbulence (dark blue), in which the ingoing flow is smooth & stable. However, noticeable is a second annular region of increased turbulence (marked yellow-red) shown around the circular central hub due to strong velocity fluctuations and mixing from asymmetry in the airflow interacting with the inner geometry causing redirections. Intra-vane moderate turbulence (light blue to green) is observed in small localized patches resulting from the flow interactions with the radial walls and no recirculation due to back flows. Overall, as illustrated by the contour, we can observe that turbulence is concentrated on the central annular region in a vortex flow manner, while the outer vane passages are less affected by or have more stable flow showing that most of mixing and energy dissipation occurred at regions closer to the inner part of the chamber where fluid enters into the system.



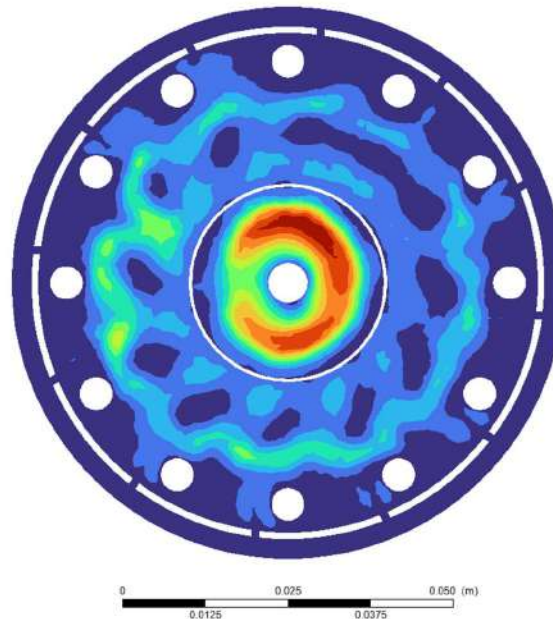
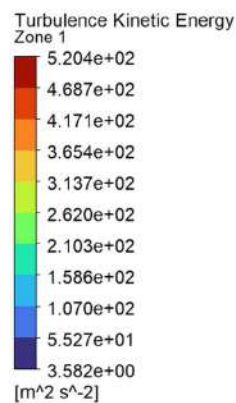
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4.1.4.2(a) Inlet Turbulence Kinetic Energy

b. Zone 1 Turbulence Kinetic Energy

From the contour, the highest turbulence is concentrated around the inner annular region near the central hub, where the red and orange colours appear. This indicates strong velocity fluctuations and intense mixing as the airflow interacts with the inner geometry and changes direction around the central component. Moving outward from the centre, the turbulence gradually decreases and appears as green to light blue regions, which represent moderate turbulence caused by flow circulation and interaction with the chamber structure and peripheral holes. Near the outer boundary of the chamber, the turbulence levels become very low (dark blue) because the flow becomes more stable and velocity fluctuations reduce close to the walls. Overall, the contour shows that turbulence is strongest near the central region and gradually reduces toward the outer chamber, indicating that most of the mixing and energy dissipation occurs close to the inner annular zone.



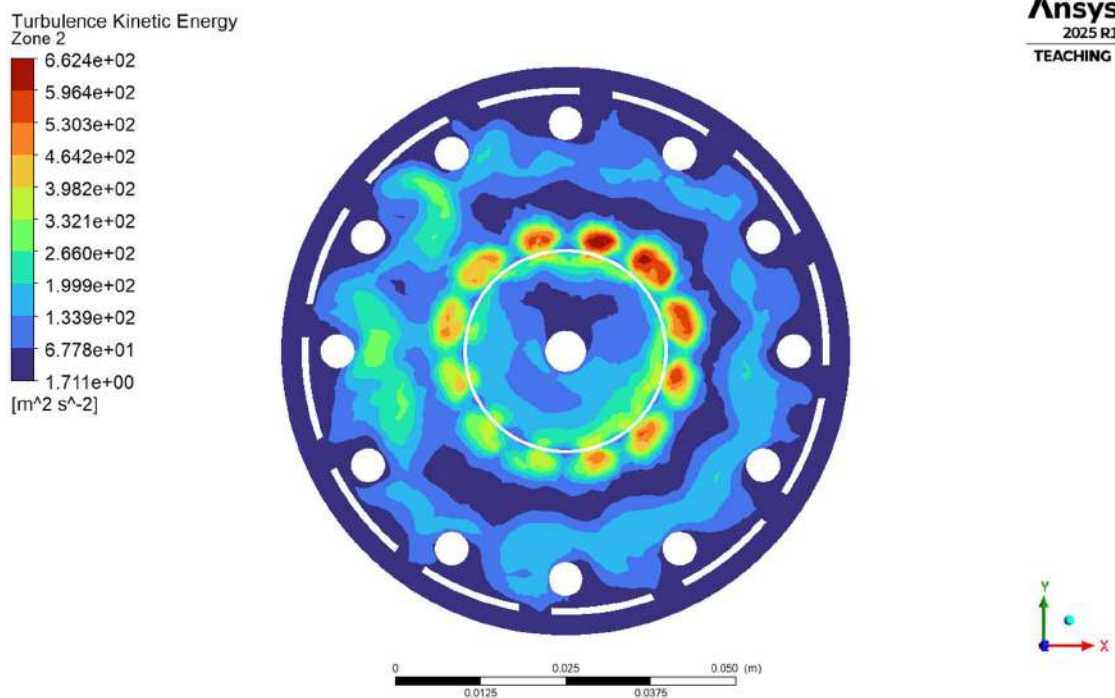
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4.1.4.2(b) Zone 1 Turbulence Kinetic Energy

c. Zone 2 Turbulence Kinetic Energy

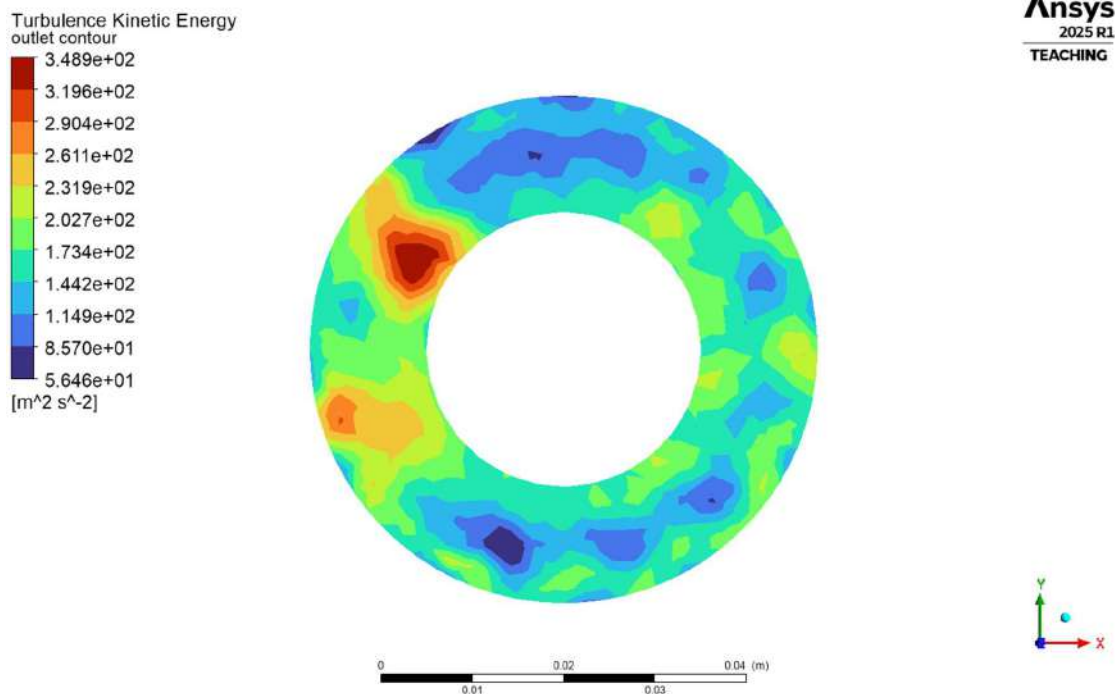
From the contour, several high-turbulence areas (yellow to red) are clearly seen from the contour surrounding the inner annular area next to the central hub, where there are significant velocity fluctuation and strong mixing occurring as airflow passes through the inner circular boundary. These high-turbulence zones arise from flow separation, velocity gradients and the mutual influence of the rotating flow and the geometry of the chamber. As we move away from the circular core, we see decreasing turbulence, indicated by pale green to light blue areas indicating moderate turbulence generated by the internal flow and its interaction with the chamber walls and peripheral holes. For the outer half of the chamber, turbulence approaches zero (dark blue), which denotes a relatively steady flow with little to no fluctuations in velocity near walls. In general, the contour depicts that turbulence is predominantly confined to the inner annular zone whilst the outer chamber flow appears to be relatively laminar, substantiating that most of mixing and energy dissipation are taking place near the centre of a chamber.



4.1.4.2(c): Zone 2 Turbulence Kinetic Energy

d. Outlet Turbulence Kinetic Energy :

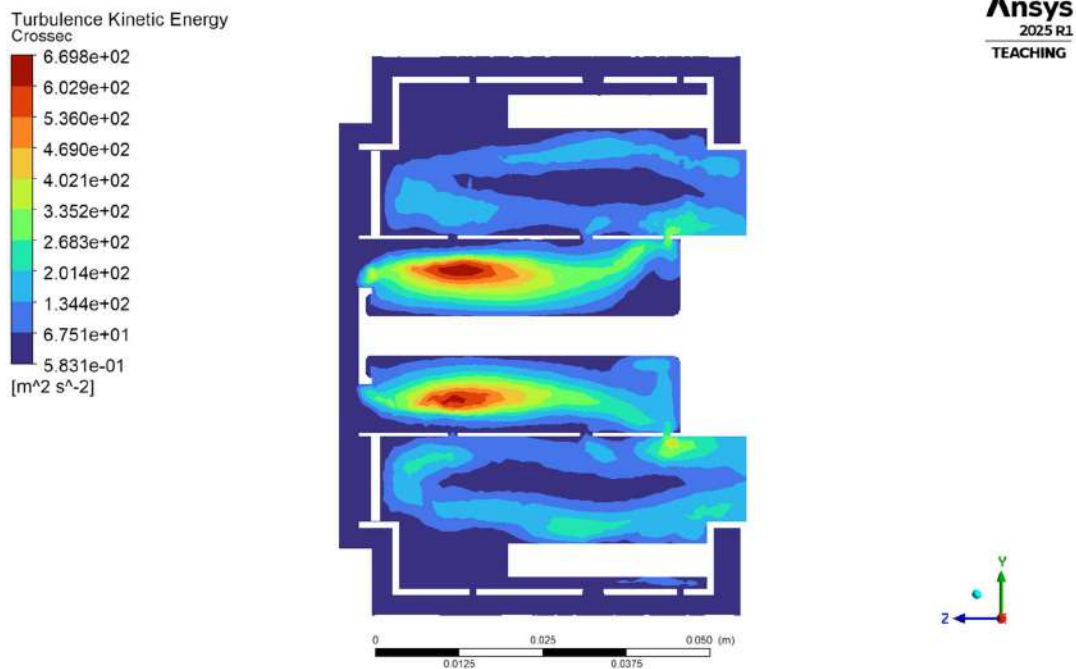
There is an obvious primary high turbulence region (yellow to red) for the upper-left tabs trim, indicating high fluctuation velocity mixing in that region. Localized high turbulence is likely due to non-uniform flow distribution in the chamber or individual flow streams interacting before leaving the chamber. Thick markers of turbulence are seen around that outlet, and other regions with moderate turbine levels, though a few small dark blue blobs — calmer flow zones relative to the rest. In general, the contour shows that most of the outlet flow have a moderate turbulence with some localized areas of high-turbulence, which has spread and characterized into the behavior of mixing and flows developed inside the chamber before leaving fluid



4.1.4.2(d) Outlet Turbulence Kinetic Energy

e. Cross-Section Turbulence Kinetic Energy :

The upper chamber regions appear largely blue to light green (low to moderate turbulence) from the view contour suggesting that there is highly stable flow with minimal mixing in that area. When the flow goes through the narrow connecting channel in the middle, a high turbulence region (yellow to red) can be detected, which is caused by significant velocity fluctuations and mixing due to flow acceleration and sudden change of geometry. The lower chamber section of the same simulation reveals another moderate to high turbulence area as the fluid fans outwards into a larger cavity and meets up with adjacent walls. Turbulence is less near the edges of each chamber and shows as dark blue regions due to boundary layer effects, and lower fluctuations in the velocity closer to walls. In general, the contour shows that the areas of turbulence generation are in narrow passages where flow acceleration and/or change of direction takes place, whereas the larger chamber regions will have relatively smooth nozzles and steady-state conditions.

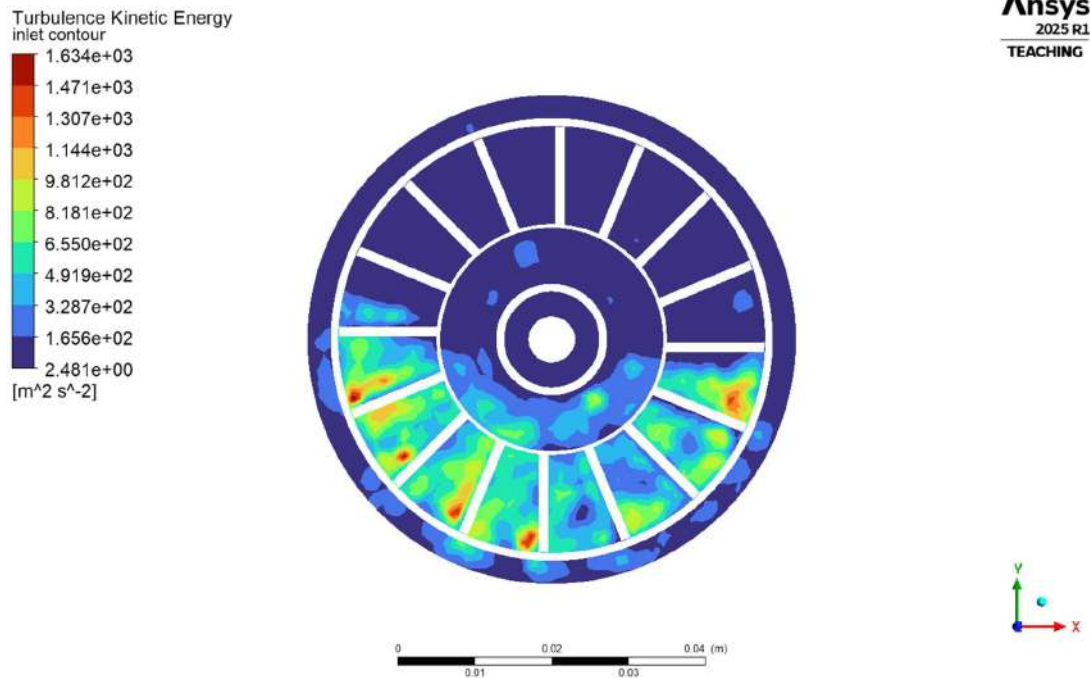


4.1.4.2(f) Cross-Section Turbulence Kinetic Energy

4.1.4.3 TKE contour 16 blade

a. Inlet Turbulence Kinetic Energy :

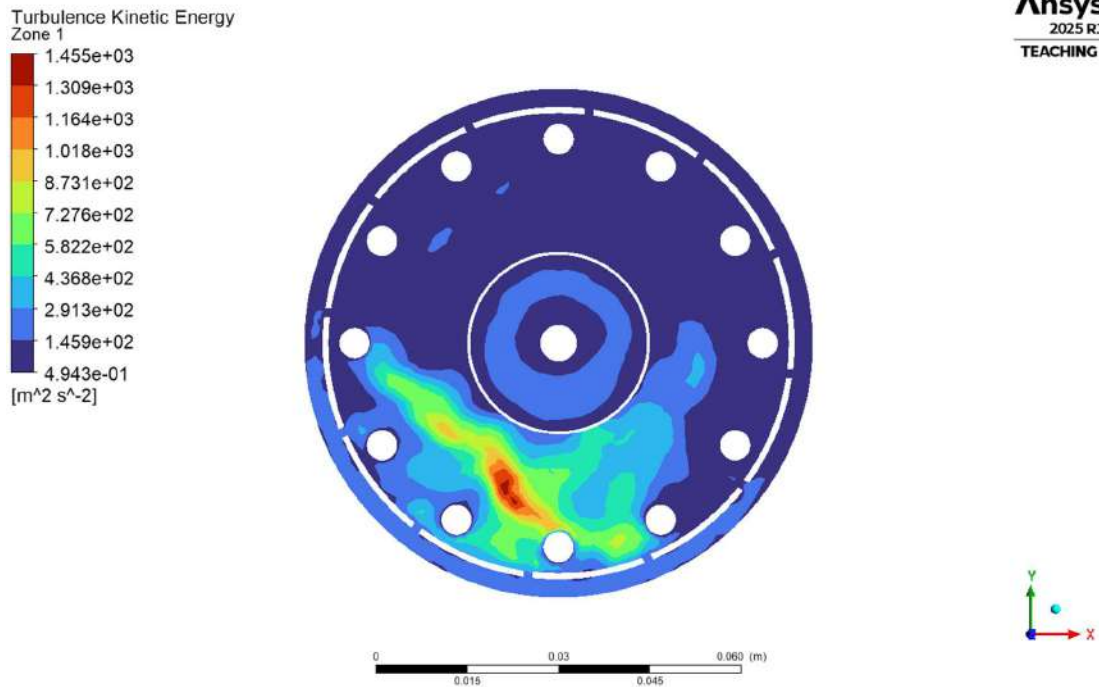
The upper half of the chamber exhibits largely laminar flow as evidenced from the dark blue contour, which signifies very low turbulence in this region of incoming flow. In contrast, the lower vane passages have moderate to higher turbulence levels (green to yellow with small red patches), indicating stronger velocity fluctuations and mixing as air is presented into those channels. Localized high-turbulence areas are attributed to the interaction of flow with radial vane walls, as well as alterations in the direction of flow and small recirculation regions formed through passages. At the vicinity of the central hub, where turbulence is low compared to when the magma tube escapes from stable behaviour around and within a concentric inner region. Generally, the contour shows that at the inlet, a non-uniform distribution of turbulence is observed: There are areas with greater turbulence in the lower vein hollows, while the top region presents rather smooth flow.



4.1.4.3(a) Inlet Turbulence Kinetic Energy

b. Zone 1 Turbulence Kinetic Energy

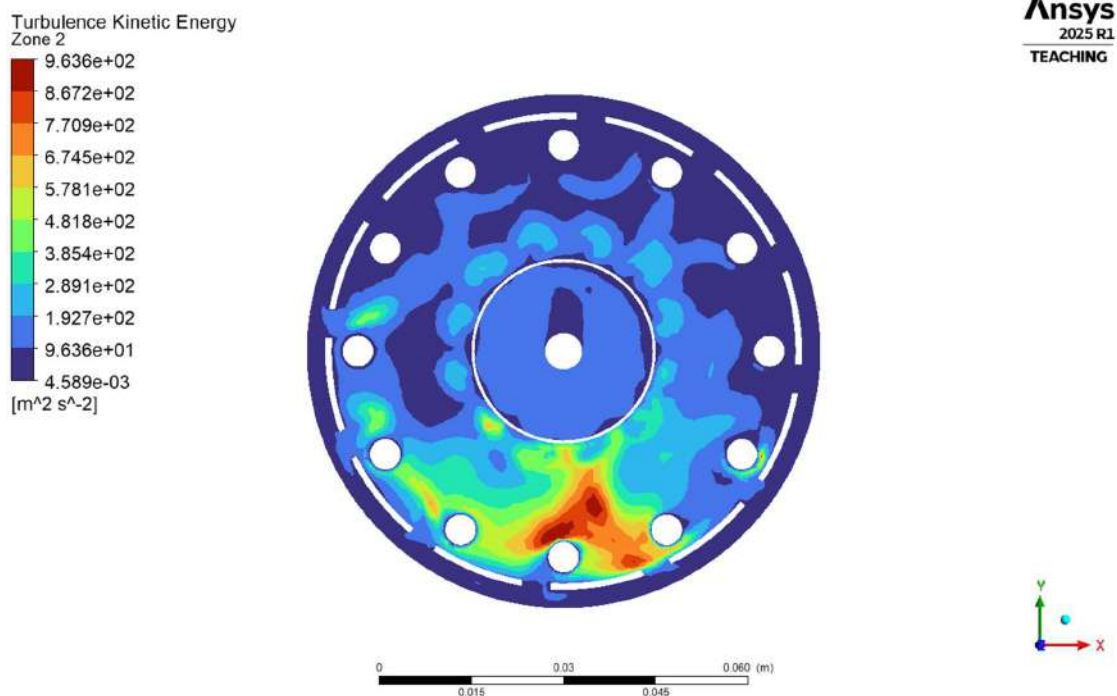
From the contour, it can be seen that most of the chamber area comes with very little turbulence (dark blue), meaning that the flow is quite going stably and there is no significant fluctuation in velocity. But in the lower-left part of the chamber there is a unique region of high turbulence (green to red) which indicates very strong mixing and velocity changes. The increased turbulence intensity in the surface not only indicates interaction of flow with peripheral holes, but also shows that small recirculation zones appear due to sudden change of flow direction. The turbulence around the central hub region is from low to moderate, suggesting that flow behaviour near the inner structure is relatively smoother. In general, that contour shows an uneven dispersion of turbulence across the chamber with utterly chapters of lower flow outlets being laminar while possibility pockets near it show stability turbulence due to intersection or interaction.



4.1.4.3(b) Zone 1 Turbulence Kinetic Energy

c. Zone 2 Turbulence Kinetic Energy:

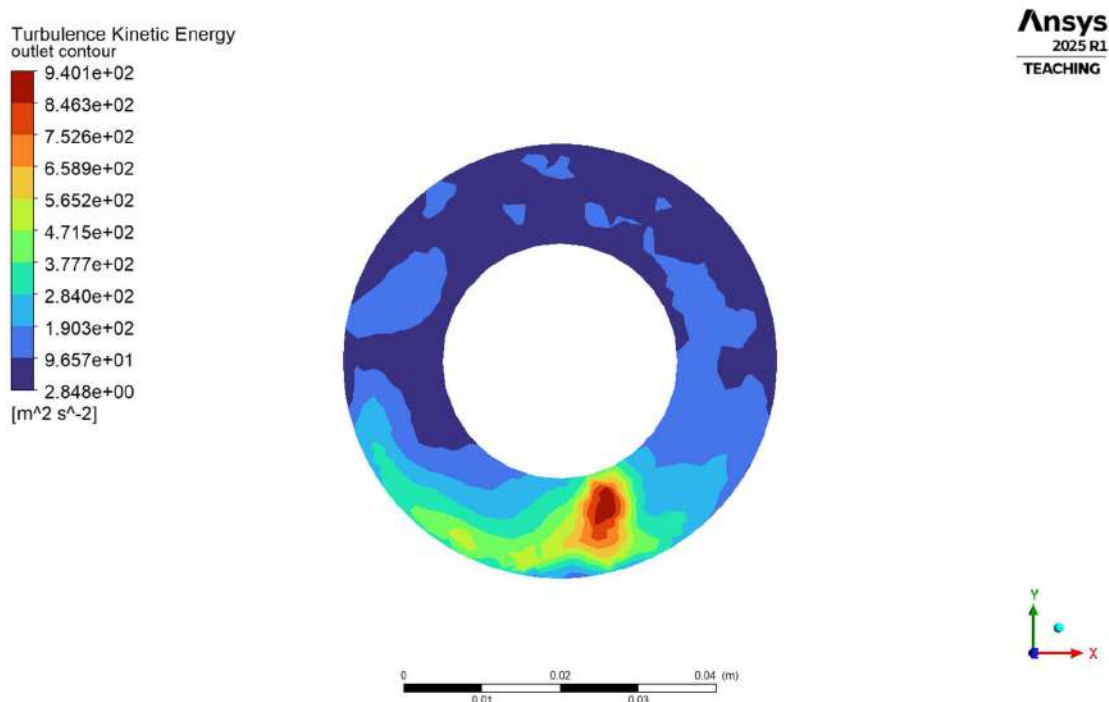
This turbulence kinetic energy contour was obtained to simulate the distribution of turbulence within the ANSYS circular chamber with peripheral holes. Legend from low (dark blue) to high (red) turbulence levels of turbulent kinetic energy. Most of the chamber region, shown in contour, is dominated by low to moderate turbulence (blue to light green) through substantial portions of the chamber itself -- this would imply that one has relatively stable flow in many areas. However, there is a separate high-turbulence zone (yellow to red) in the lower part of the chamber where large velocity fluctuations and mixing take place. The event of high turbulence is possibly attributed due to flow interaction with the peripheral holes and sudden changes in direction of flow thereby creating recirculation zones increasing the turbulence intensity. The turbulence near the central hub is relatively low, where the flow behavior appears to be much smooth around the inner structure. Generally, the contour shows that the turbulence lack uniformity throughout the chamber and exhibits high turbulence zones in both lower chamber regions, whereas other parts of the cavity are under quite stable flow conditions.



4.1.4.3(c) Zone 2 Turbulence Kinetic Energy

d. Outlet Turbulence Kinetic Energy :

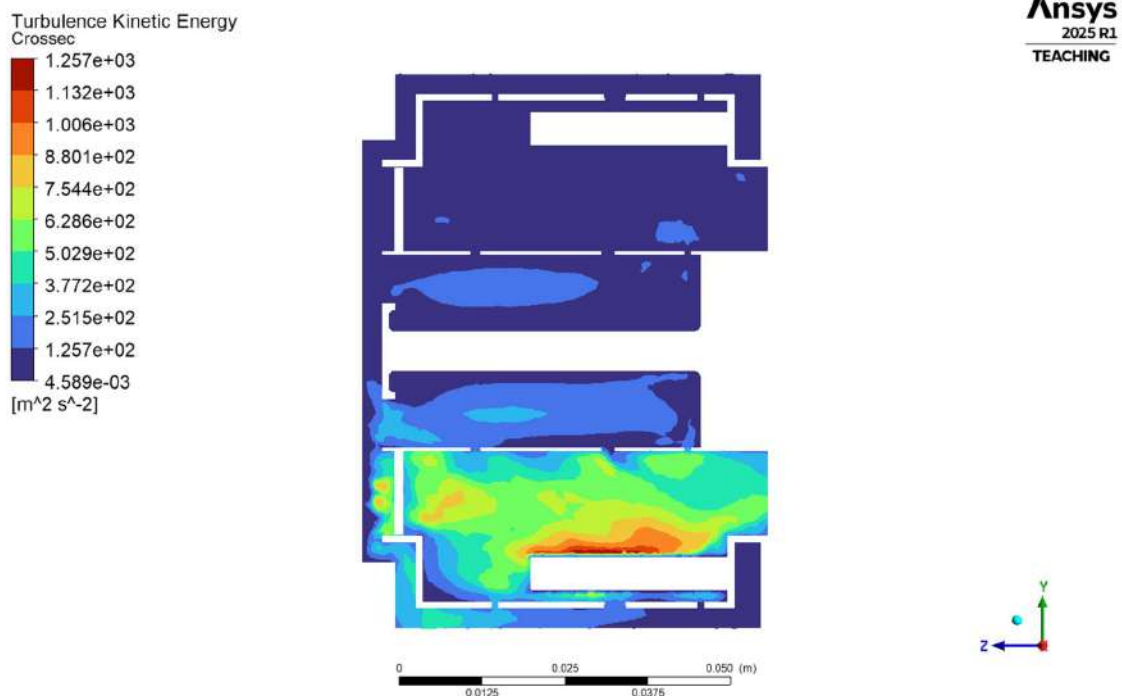
Turbulence kinetic energy outlet contour from ANSYS simulation showing how this component fills the annular chamber where it exits. This colour scale spans from low values of turbulence intensity (dark blue) to high levels of turbulent kinetic energy (red). It can be seen in the contour that the majority of the outlet region is controlled by low turbulence levels (dark blue to light blue), meaning that before leaving the chamber, the flow becomes relatively stable. But a separate region of high turbulence (yellow to red) becomes visible in the lower part of the annular outlet, which indicates strong velocity fluctuations and mixing being intensely developed there. The uneven flow distribution and interaction among various flow streams in the chamber are likely results, contributing to this localized turbulence before reaching the outlet. The surrounding areas show moderate turbulence (green colours) which diminish more and more toward the inner and outer boundaries, respectively, due to boundary layer effects (low order of Bossiness approximation) and decaying velocity fluctuations in the walls vicinity. In general, such contour indicate the outlet flow is predominantly steady and with small regions of turbulence, which is a signature of mixing and internal motion prior to exiting from the chamber.



4.1.4.3(d) Outlet Turbulence Kinetic Energy

e. Cross-Section Turbulence Kinetic Energy :

Turbulence kinetic energy cross-section contour depicts how turbulence is extended inside the chamber internal flow passages. Dark blue indicates low turbulence levels; red, the very high kinetic energy of turbulence. The contour indicates predominately very low turbulence (dark blue) in the upper chamber regions, again suggesting that where flow separation does occur it is largely stable with only minor velocity fluctuations. However, as the flow travels down the middle and lower passages, we observe that this upheaval becomes increasingly turbulent, indicated by green through to yellow areas, with more mixing and interaction occurring within the chamber. Highest turbulence levels (orange to red) are found in the lower east passage close to outlet, where fluid jets through a narrow channel and interacts vigorously with walls of the chamber. Flow acceleration and sudden changes in geometry along with recirculation zones inside the passage contribute to these increases of turbulence. At flow layer walls, turbulence boundary layers. Essentially, the contour demonstrates that turbulence is primarily produced in the lower region of the chamber where flow acceleration towards the outlet occurs while upper sections exhibit relatively smoother and stabilized flow.



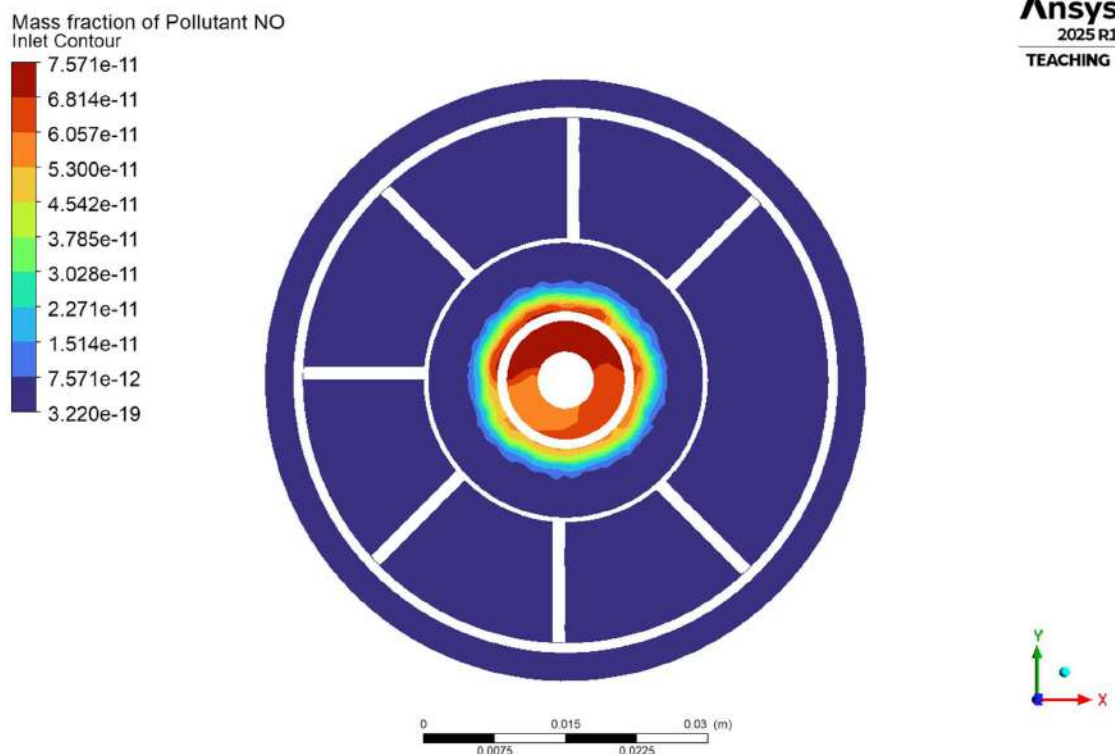
4.1.4.3(f) Cross-Section Turbulence Kinetic Energy

4.1.5 Mass Fraction of No_x

4.1.5.1 No_x Contour 8 Blade

a. Inlet Contour:

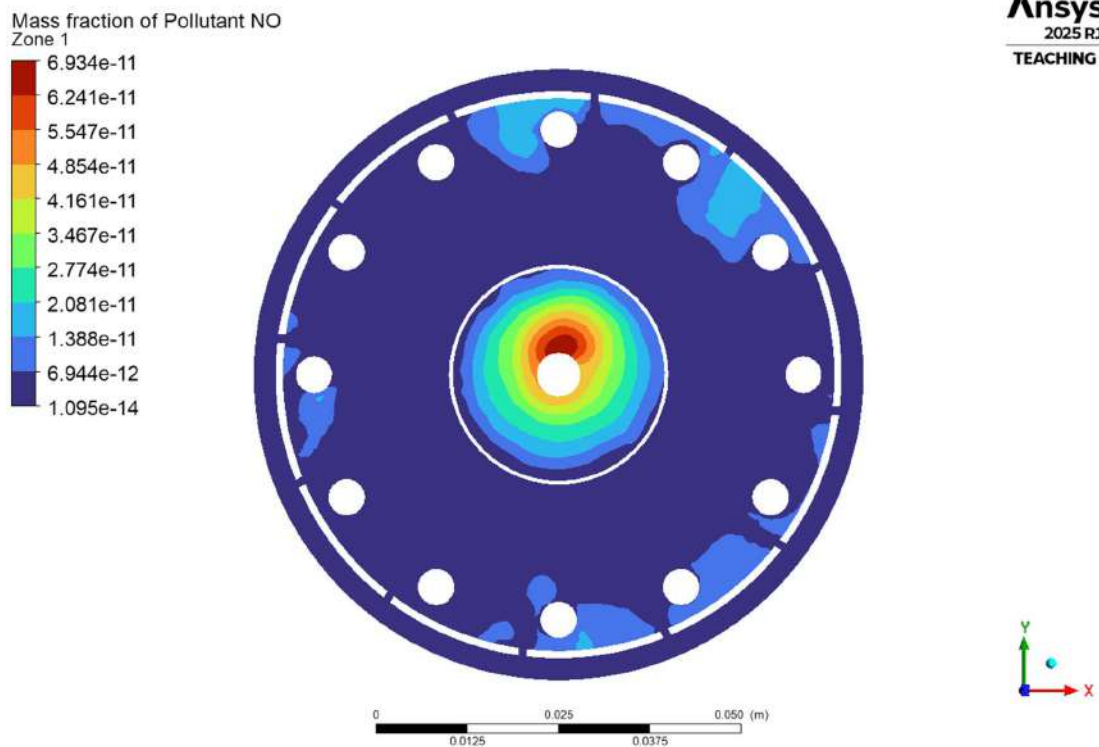
Mass fraction contour of pollutant NO at inlet . The colour scale indicates mass fractions from very low (dark blue) to high (red). The majority of the chamber and vane passages are shown as dark blue on the contour, which indicates almost insignificant levels of NO in those locations. In another remarkable observation, the NO concentration exhibits a surprising consensus with that in the crankshaft for vegetated cultures, clearly showing a circular region of higher NO concentration (in yellow to red) surrounding the inner annular region adjacent to the central hub. This also indicates that the generation or accumulation of NO mainly happens in the inner zone, where flow has strong interaction with the central structure and temperature/reaction strength is high. The outer regions are mostly blue because outside the vane passages, the NO concentration decreases rapidly as the flow moves outwards. In general terms, the contour implies that the NO pollutant concentration is strongly localized into a region nearby the inner core whereas outside this area an extremely low level persists, showing that in the outer sections of the flow domain there is little to no formation of pollutants.



4.1.5.1(a) Mass Fraction of No_x Inlet Contour

b. Zone 1 Contour :

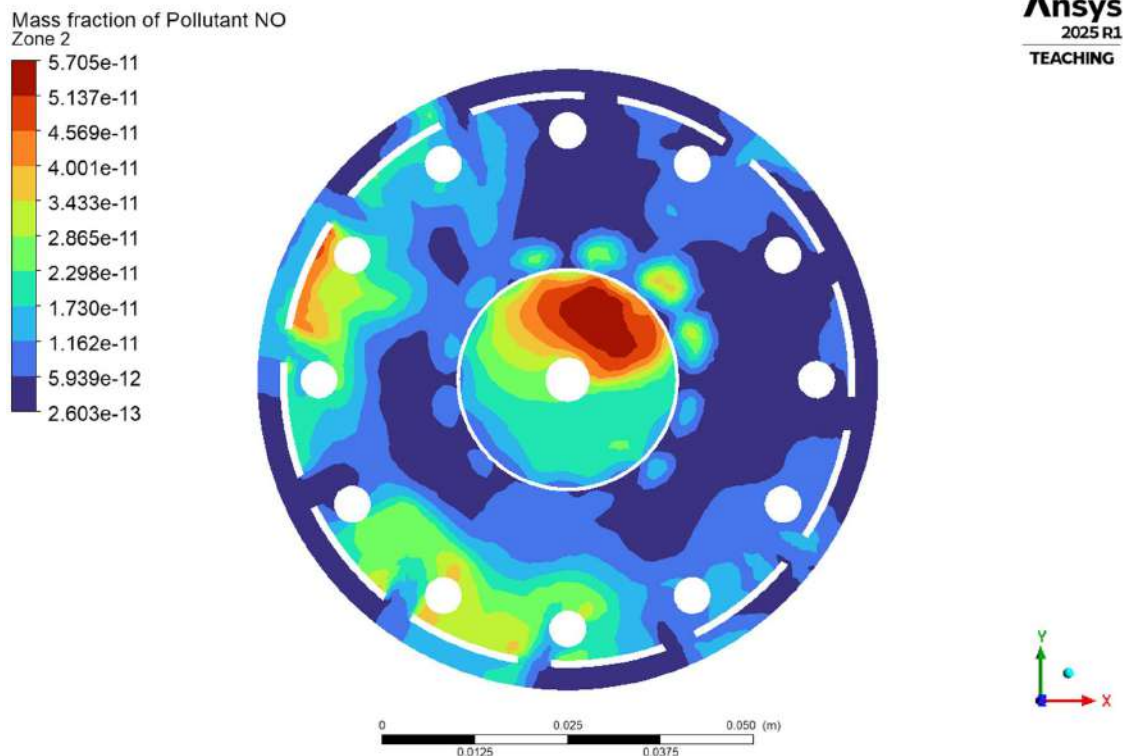
The vertical colour bars indicate NO mass fraction (low to high (dark blue → red)). For the low NO concentration areas (dark blue), this suggests that pollutant formation is limited to the outer portion of the chamber, shown from contour. However, there is a strong concentration zone (yellow to red) in the inner annular region accompanying the central hub where NO mass fraction is maximum. Such is the case since inner regions run at higher temperatures and with stronger chemical reactions that tend to yield nitrogen oxides. NO concentration drops off rapidly in the x-direction though, as the flow moves away from centre-line to outer chamber and peripheral holes, becoming green to blue shades. In summary, the contour suggests a localized formation of NO around the middle of the chamber with very low pollutant concentration in outer regions which shows that cleaning reaction and pollutants follows close to this central core of the system.



4.1.5.1(b) Mass Fraction of No_x Zone 1 Contour

c. Zone 2 Contour :

As we can observe, the influent NO mass fraction contour depicts how nitrogen oxide would specially spread out in following the air mixer inside of the circular chamber with several peripheral holes. Very low NO concentration (dark blue) to higher mass fraction values (red). Most of the chamber space is below detection level (blue to light green), which corresponds with limited formation of pollutants in many regions of the chamber, as seen from the contour. Nonetheless, a clear region with high concentration (yellow to red) can be observed at the adjacent inner annular area of the central hub that represents optimal NO formation regions. The reason for this is that reaction conditions of higher temperatures and intensity are in play in the inner area, which Favors nitrogen oxides formation. Moderate concentration regions (green to yellow) are scattered around parts of the chamber, although the pollutant expands slightly outward seen mixing and flow cycle. After all that, the contour shows that NO forms mostly in the central part of the chamber and much lower (less than order) in the outer region of it, showing how temperature, mixing and reactions evolve inside this system.

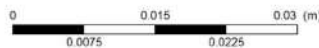
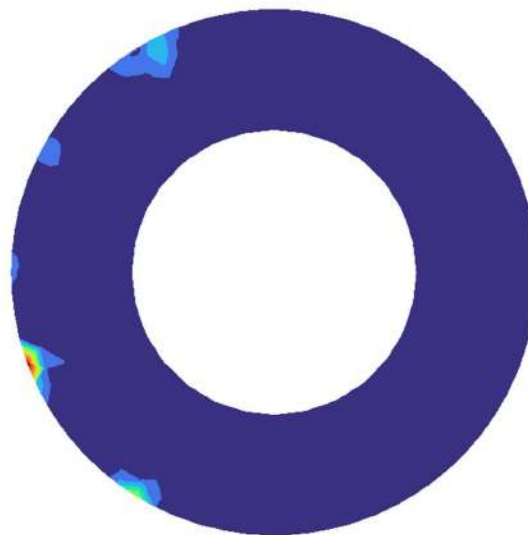
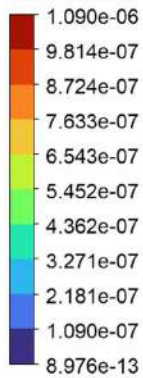


4.1.5.1(c) Mass Fraction of No_x Zone 2 Contour

d. Outlet Contour :

The colour scale reflects the concentration of NO from lowest (dark blue) to highest (green to red). The dark blue colour covering almost the entire annular outlet region from contour indicates that NO concentration at the outlet is very low and has an approximately uniform distribution. Only a few small localized patches of slightly higher concentration (light blue to green) can be seen along the perimeter, possibly due to some minor mixed flows or trailing reaction products dragged along with the outflowing fluid. The lack of intense red or yellow areas indicates that the majority of pollutant formation has already been diminished or dispersed by the time it reaches the outlet. This contour overall shows no (or minimal) NO emerging at the outlet, with a good distribution of this parameter within the chamber indicating that pollutant mixing and dilution had occurred before flow left the system.

Mass fraction of Pollutant NO
Outlet Contour



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4.1.5.1(d) Mass Fraction of No_x Outlet

e. Cross-Section Contour :

The mass fraction contour of pollutant NO in the cross-section, which represents the distribution of nitrogen oxide concentration inside the chamber passages. A dark blue colour fills most of the region, showing that the NO concentration is very low across the entire chamber. Green to red colours near the upper outlet passage show a localized patch of slightly higher concentration. It implies that NO formation is negligible and only occurs in a local area due to temperature and reaction influence, with all other regions of the chamber exhibiting very low pollutant concentrations.

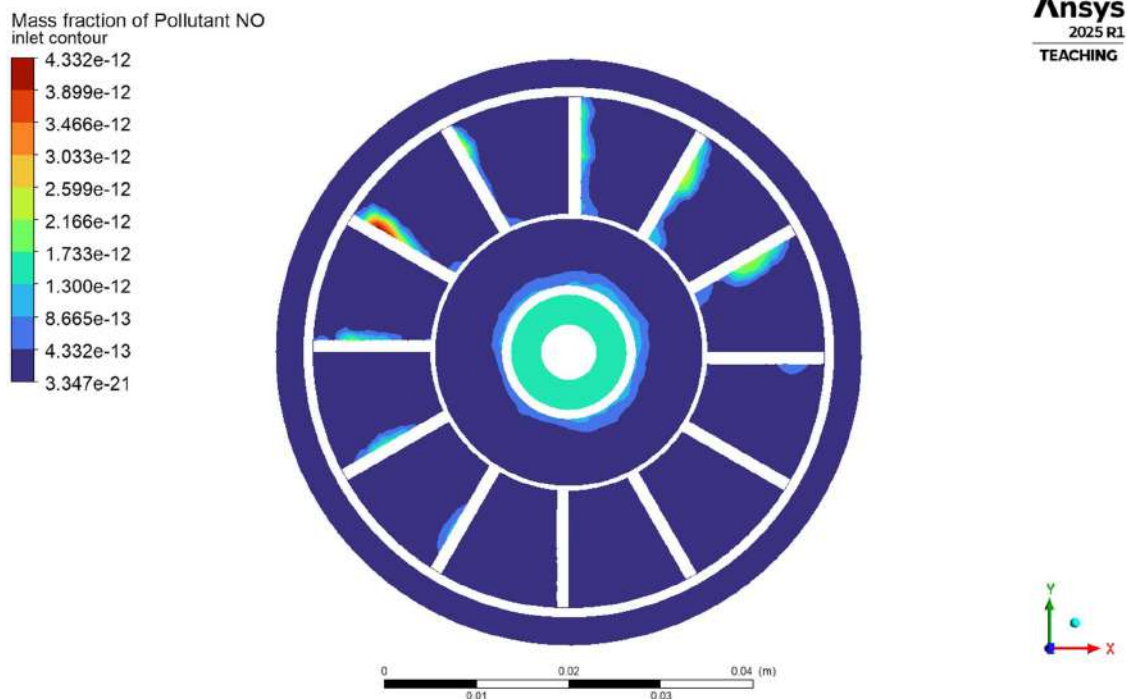


4.1.5.1(f) Mass Fraction of No_x Cross-Section

4.1.5.2 NO_x contour 12 blade

a. Inlet NO mass fraction

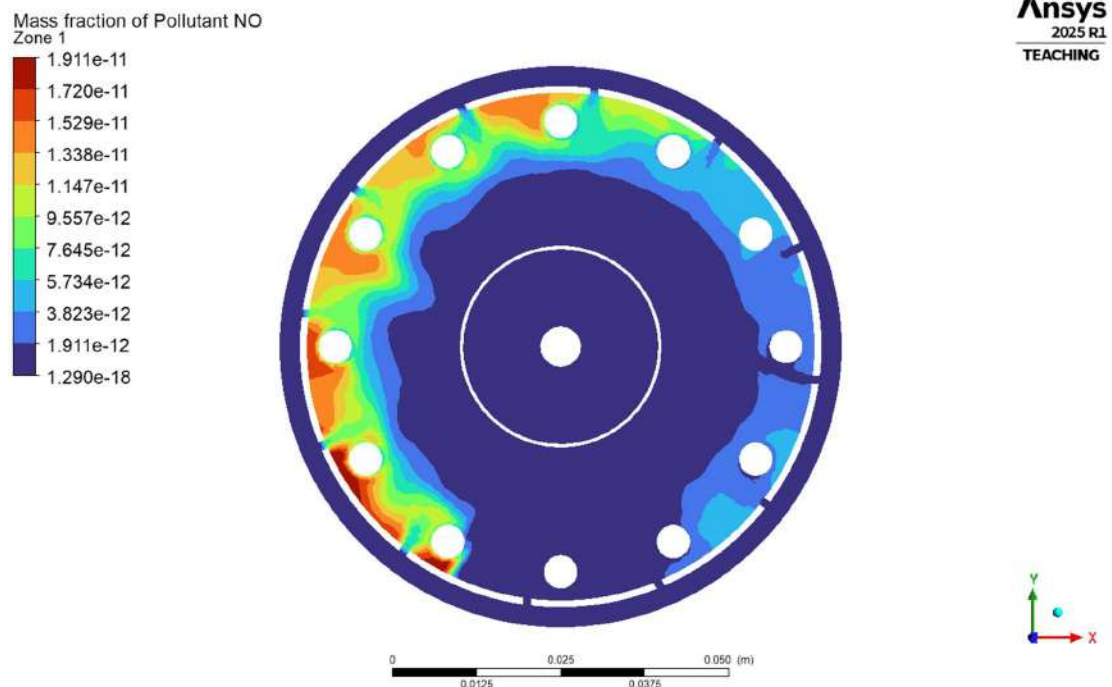
The contour of mass fraction for pollutant NO at the inlet reveals the concentration distribution of nitrogen oxide when entering into this chamber in flow. A large part of the region is in dark blue, which indicates very low NO concentration can be found in the incoming flow. A small increase in NO concentration (light blue to green) is observed close to the central hub and a few certain racket of vanes, indicative of minor formation due to local flow interaction. In general, the contour suggests that inlet NO levels are very low with only quite localized variations.



4.1.5.2(a) Inlet NO mass fraction

b. Zone 1 NO mass fraction

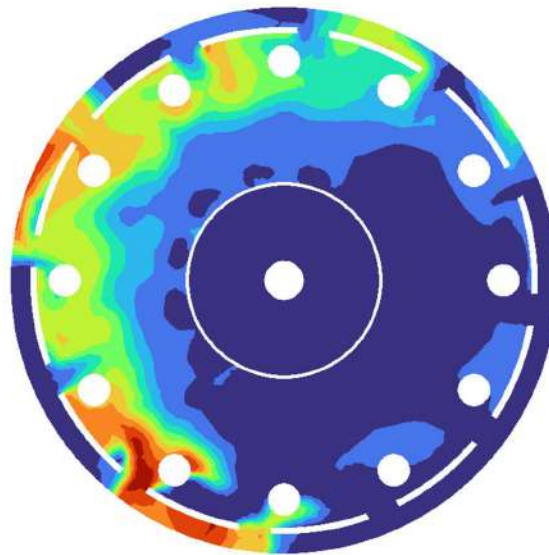
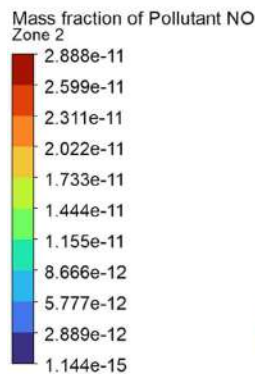
The mass fraction distribution of the pollutant NO (nitric oxide) inside the circular combustor visible with contour. The darker are the shades closer to red/orange, and in contrast, blue parts of the map indicate very low concentration of Free NO. The higher amount of NO formation is primarily taken place around the outer wall, and some border holes, which should be because MMAO enters high temperature under combustion condition with colourless-limits optimal reaction, it promotes formation of NO. The core region stays in dark blue, characterized by less production of NO owing to the comparatively lower temperature and lower intensity of reaction in that region. In general NO for constant- ϕ is not evenly distributed and is primarily concentrated around certain high-temperature flow regions.



4.1.5.2(b) Zone 1 NO mass fraction

c. Zone 2 NO mass fraction

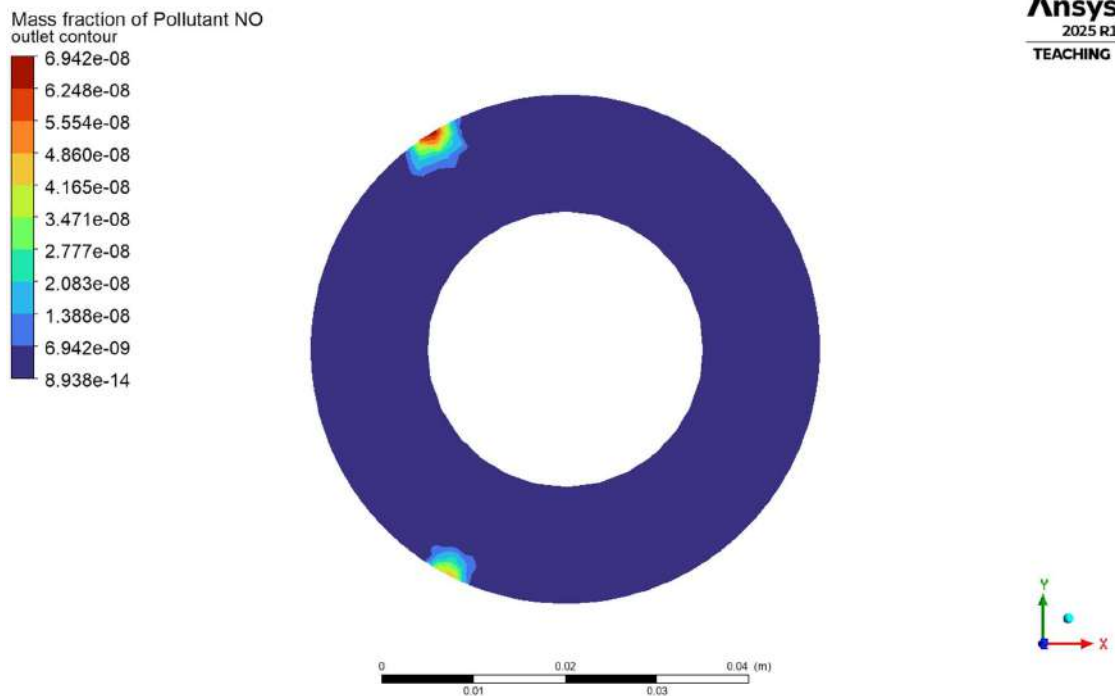
Mass fraction of the pollutant NO (nitric oxide) inside the circular combustor from ANSYS simulation A scale on the bottom indicates that regions shaded red and orange indicate high amounts of NO, while regions in blue represent very low levels of this gas. There is higher formation of NO closer to outer wall and around some peripheral holes which indicates that these areas experience higher temperature and stronger combustion, both of which are favourable conditions for the formation of NO. The central region keeps dark blue, related to less NO production thanks to lower temperature and reaction intensity compared with adjacent zones. Localization of the NO formation due to convective diffusion and NO rebuilding processes can be seen in the overall distribution, indicating that neither mechanism is uniform but are concentrated near specific high-temperature regions in the flow.



4.1.5.2(c) Zone 2 NO mass fraction

d. Outlet NO mass fraction

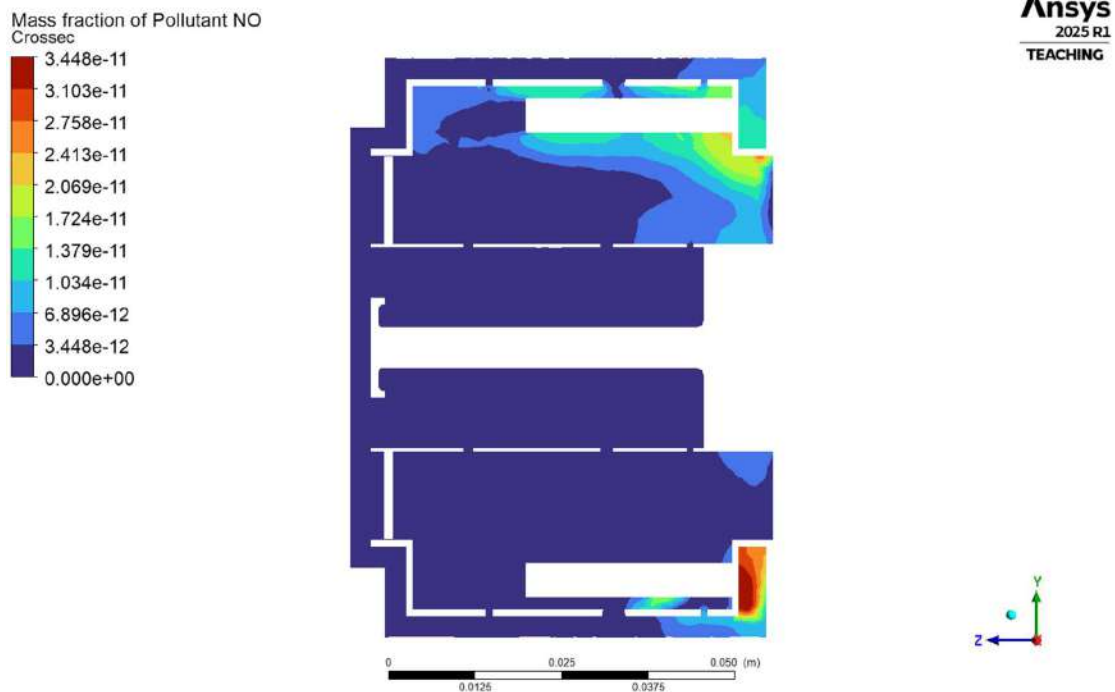
The contour displays mass fraction distribution of pollutant NO in a circular chamber. With the exception of a few areas where NO concentrations are high, the majority of the domain shows dark blue colouring which means that NO concentrations are extremely low across the entire chamber. Most of the outer boundary is thus going to remain green, as only a few small areas are coloured with yellow and red denoting slightly more localized regions of elevated NO formation near the edge. This localization may be related to temperature-peak or combustion stronger reaction near the wall region. Overall, the result suggests that NO generated in the chamber is comparatively minimal and spatially localized rather than distributed extensively throughout.



4.1.5.2(d) Outlet NO mass fraction

e. Cross-section NO mass fraction

The image shows the mass fraction of pollutant NO distributed in the combustor geometry from ANSYS. The large majority of the chamber is shown in dark blue, suggesting that the NO concentration is extremely low throughout most of the flow region. At higher altitudes, we observe slightly increased concentration of NO (green to yellow colours) in the upper right, which might correlate with regions where there are hot spots and enhanced combustion processes. Most NO is formed at the lower outlet region (red zone), indicating that gases leaving the chamber typically have higher NO, meaning combustion in the primary zone has initiated and high-temperature products accumulate near to outlet. Therefore, the distribution shows that NO formation occurs mainly in high-temperature regions or outlet zones instead of uniformly dispersed throughout the chamber.

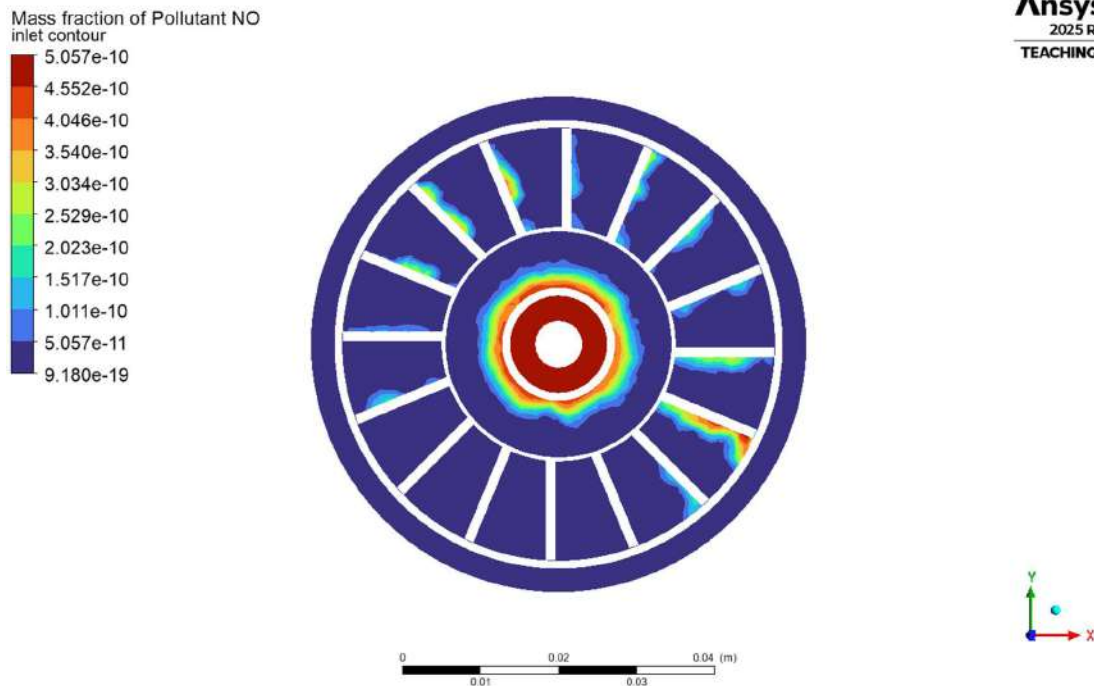


4.1.5.2(e) Cross-section NO mass fraction

4.1.5.3 NO_x contour 16 blade

a. Inlet NO mass fraction

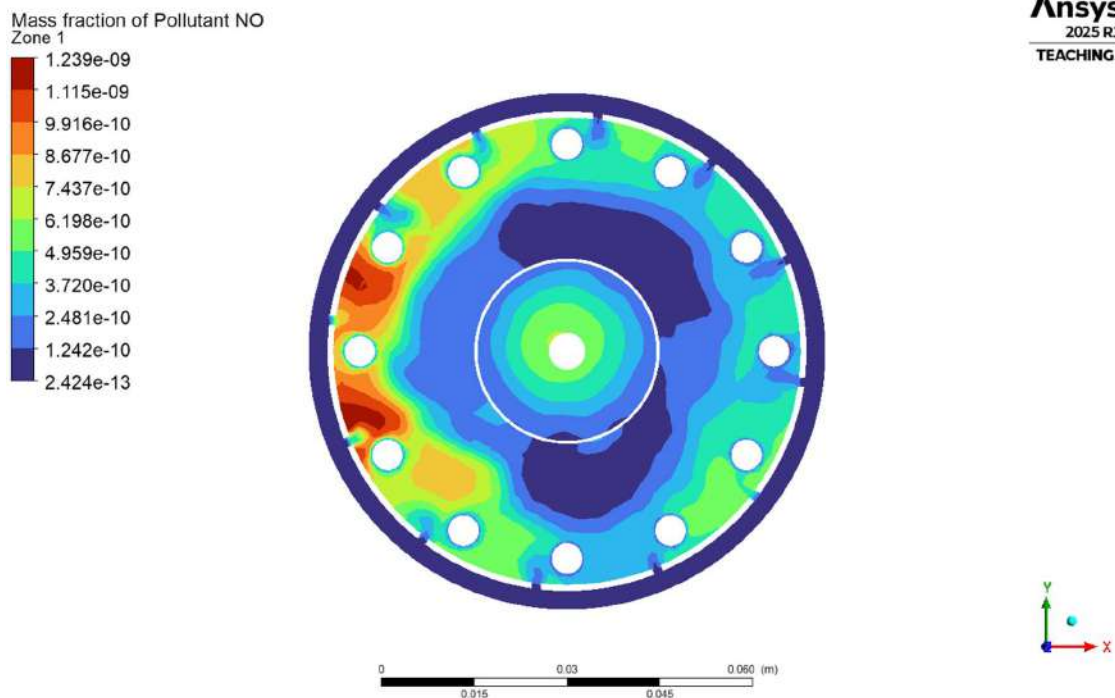
The contour presents the distribution of pollutant NO mass fraction at the combustor inlet section, imported from ANSYS simulation. The core part around the injector is shown in red and orange colours where the concentration of NO is very high because of the vigorous combustion and local temperature near the flame zone. As you can see, as the flow travels outward through the radial passages and into the outer chamber, the colour translates from a dark blue to bright green in stages which indicates that with mixing of cooler air and dilution effects there is progressively less concentration of NO. The dark blue colour that occupies most of the outer chamber reflects a significant reduction in NO formation outside of the main combustion zone. In general, the contours indicate that NO formation is at a peak close to the combustion core, and decreases as the gases flow outward in the combustor.



4.1.5.3(a) Inlet NO mass fraction

b. Zone 1 NO mass fraction

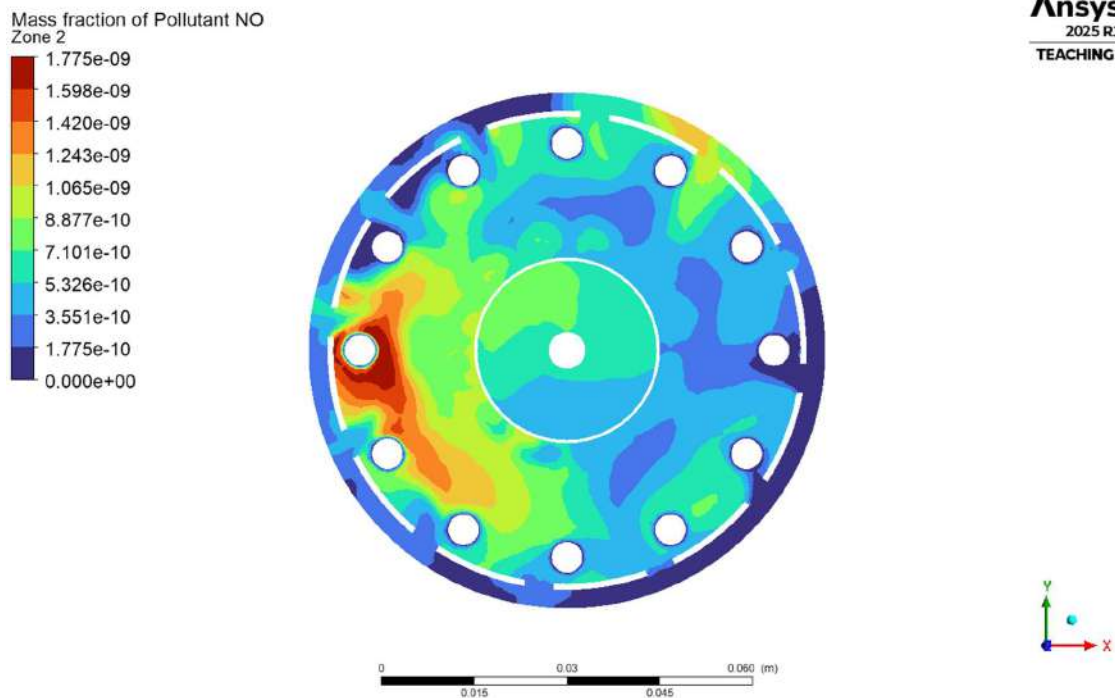
The contour showing mass fraction distribution of pollutant NO in combustor zone. The highest NO concentration (red and orange colours) is seen in the left outer-wall region, where temperature levels are higher and combustion reactions are more intense. It reflects moderate values (green and light blue) in the central region, where a bit of NO is formed around the core flow but not as much compared to outer high-temperature zones. The dark blue of large areas in the chamber shows very low amount of NO concentration due to cooling and mixing of air travelling inside the chamber. In general, it can be seen: the formation of NO is highly heterogeneous and primarily located in the region close to the outer wall with high combustion intensity from the contour.



4.1.5.3(b) Zone 1 NO mass fraction

c. Zone 2 NO mass fraction

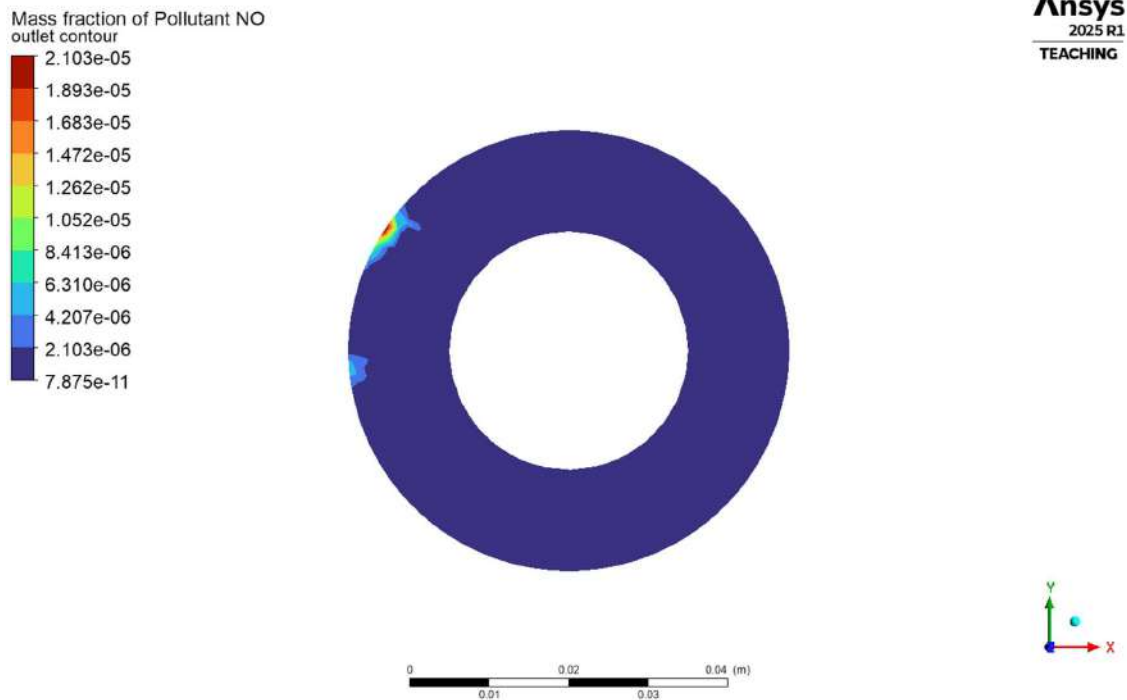
We can see that the highest NO concentration in the domain occurs near around the left side, close to the outer wall and one of air holes (in red region). This means that high temperature and strong combustion reactions occur locally in that region, which leads to more NO formation." Most of the chamber appears to be in green and blue colours, which represent moderate to low distributions of NO due to mixing and cooling effects from the incoming air. Thus, the distribution indicates NO formation is close to certain high-temperature areas rather than uniformly scattered across any chamber.



4.1.5.3(c) Zone 2 NO mass fraction

d. Outlet NO mass fraction

The most of the outlet region is dark blue which means you have a very low NO concentration in most part of your flow area. Above both images we see no green, yellow and red colours (high NO concentrations) except for a small area near the outer sides of each image that is making localized zones. Hot spots like these can form as a result of residual high-temperature gases that are delivered to the surface before they have fully mixed out in the chamber and have exited. In conclusion, the result indicates that NO emissions at the outlet is usually low and only a few small localized areas show higher concentrations.



4.1.5.3(d) Outlet NO mass fraction

e. Cross-section NO mass fraction

From the contour, the dark blue region is indicating that NO concentration is very low across most of the chamber. In contrast, slightly more prominent values can be observed in the outlet regions on the right-hand side where colours green and light blue areas appear, indicating some localized formation of NO due to higher temperature and combustion products accumulation around the exit flow path. Since the rest of the combustor is close to constant temperature (due to cooling) and spread with air, this allows for very little NO throughout so long as all combustion occurs in one chamber. In summary, the results indicate that NO generation is greatly limited and localized near small outlet regions instead of uniformly over the combustor.



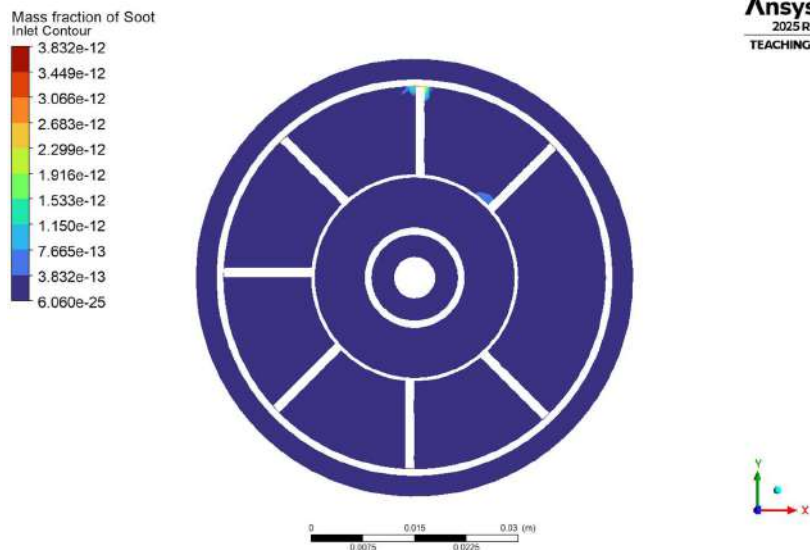
4.1.5.3(f) Cross-section NO mass fraction

4.1.6 Mass Fraction Of Soot

4.1.6.1 Soot Contour 8 Blade

a. Inlet Soot mass fraction

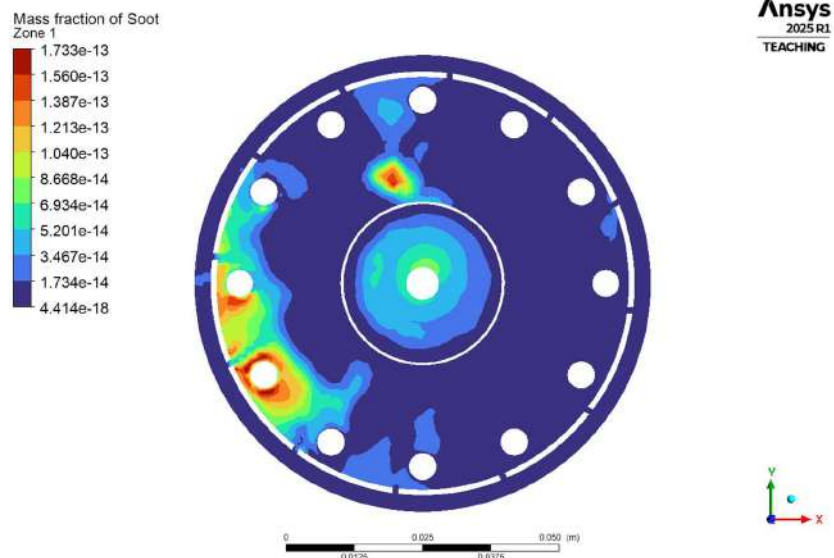
Soot concentration at the combustor inlet It was showing extremely low readings throughout the whole area, indicating that in the case of soot formation, it hasn't reached entry yet.



4.1.6.1(a) Inlet Soot mass fraction

b. Zone 1 Soot mass fraction

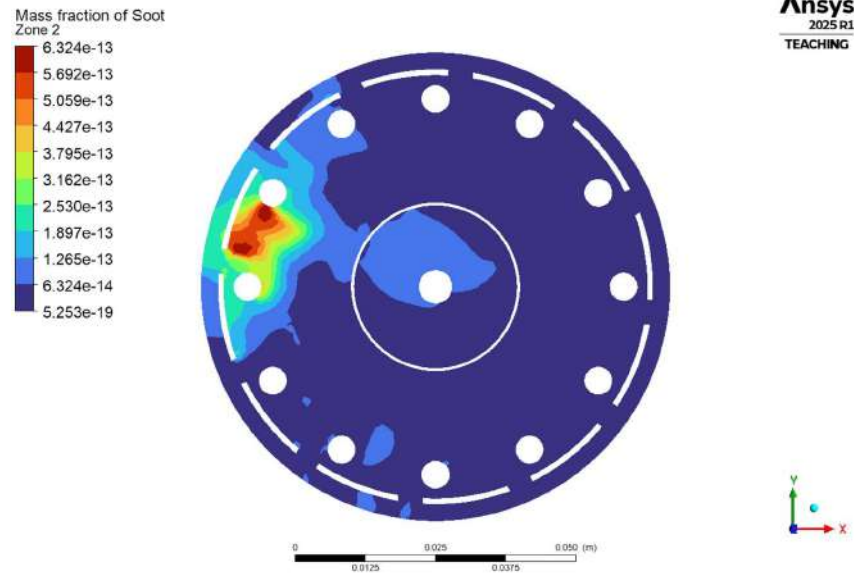
This contour corresponds to the first combustion zone, where fuel-air mixture reactions begin. Localized regions exhibit slightly elevated soot concentrations from incomplete combustion products at the combustion onset stage.



4.1.6.1(b) Zone 1 Soot mass fraction

c. Zone 2 Soot mass fraction

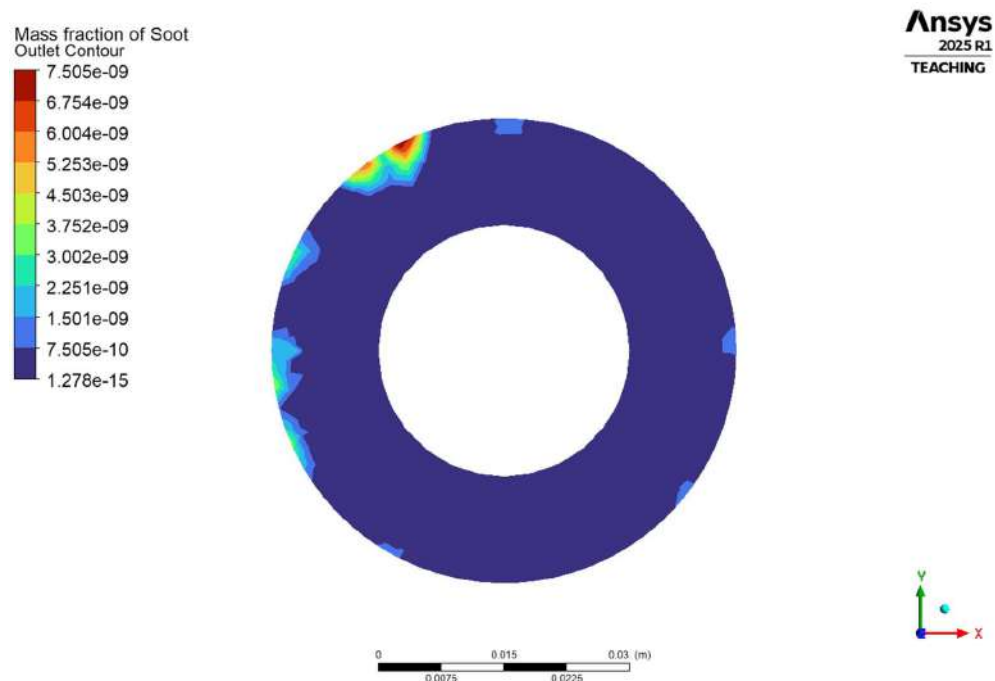
The combustion process becomes more intensified and soot forms in a concentrated area. These are elevated in regions with fuel-rich combustion where the reaction is not entirely complete.



4.1.6.1(c) Zone 2 Soot mass fraction

d. Outlet Soot mass fraction

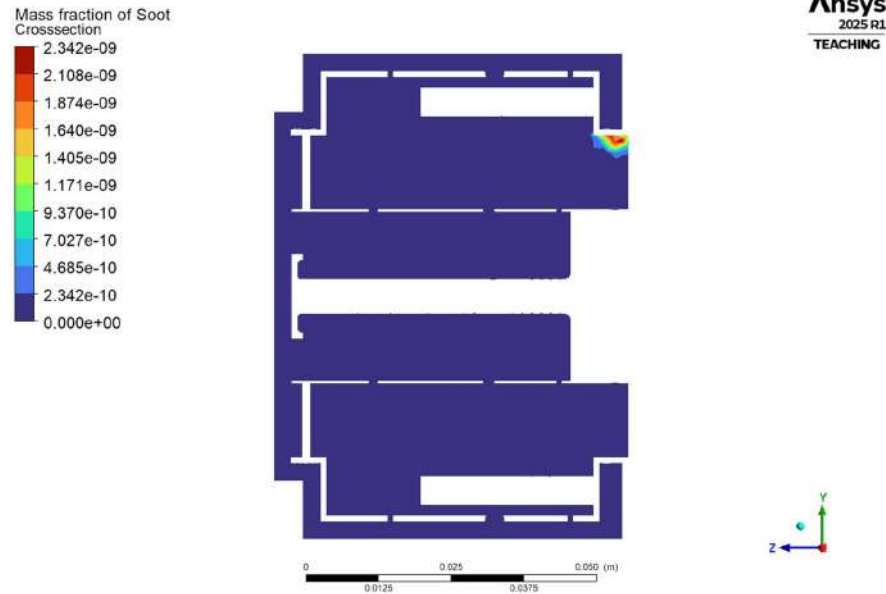
This demonstrates the particulate concentration at combustor exit. Overall, soot levels are very low in most of the chamber, indicating that a large percentage of soot particles accumulate in the table before it exits the exhaust.



4.1.6.1(d) Outlet Soot mass fraction

e. Cross-section Soot mass fraction

From the contour, it includes the dimensions of soot on combustor internal flow path. At these conditions, only small areas of the chamber have significant soot, while much of it remains low due to oxidation and mixing.

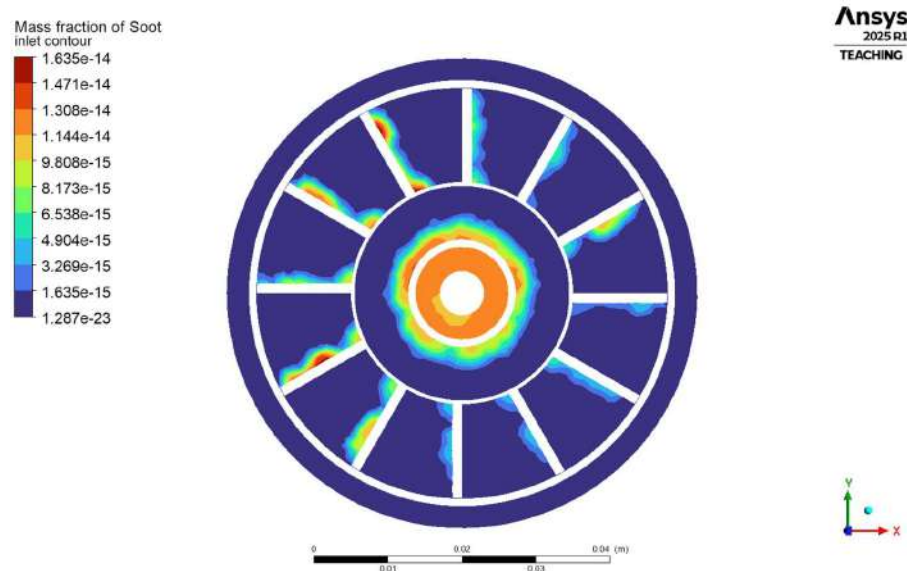


4.1.6.1(f) Cross-section Soot mass fraction

4.1.6.2 Soot contour 12 blade

a. Inlet Soot mass fraction

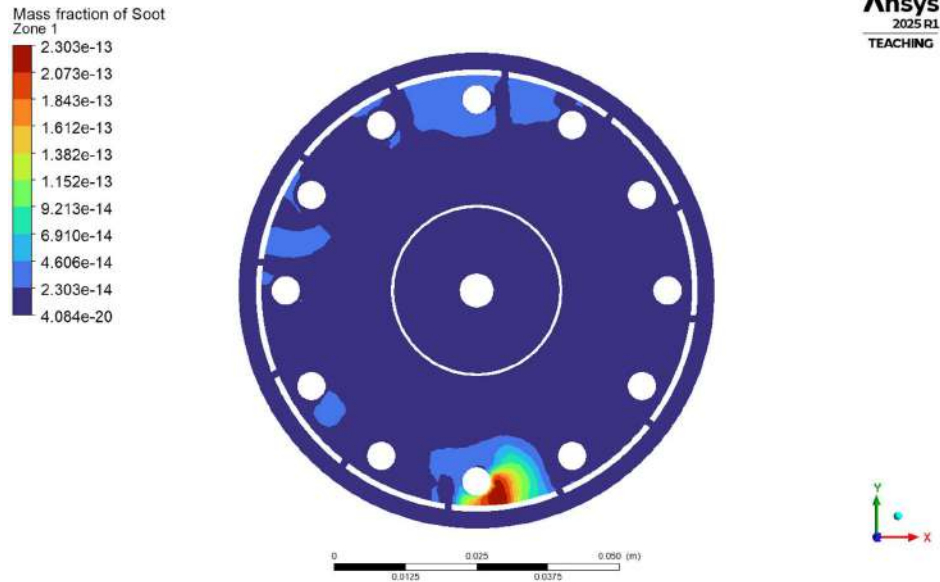
It Shows soot concentration along the combustor inlet region. The soot concentration is mainly concentrated in the central part, which gradually decreases to less than 0.1 g/m³ in the outer part of the domain showing very limited entry formation.



4.1.6.2(a) Inlet Soot mass fraction

b. Zone 1 Soot mass fraction

This contour is the first combustion zone where the fuel–air mixture starts to react and only small pockets of soot can be found in some areas, indicating that the process had just begun.



4.1.6.2(b) Zone 1 Soot mass fraction

c. Zone 2 Soot mass fraction

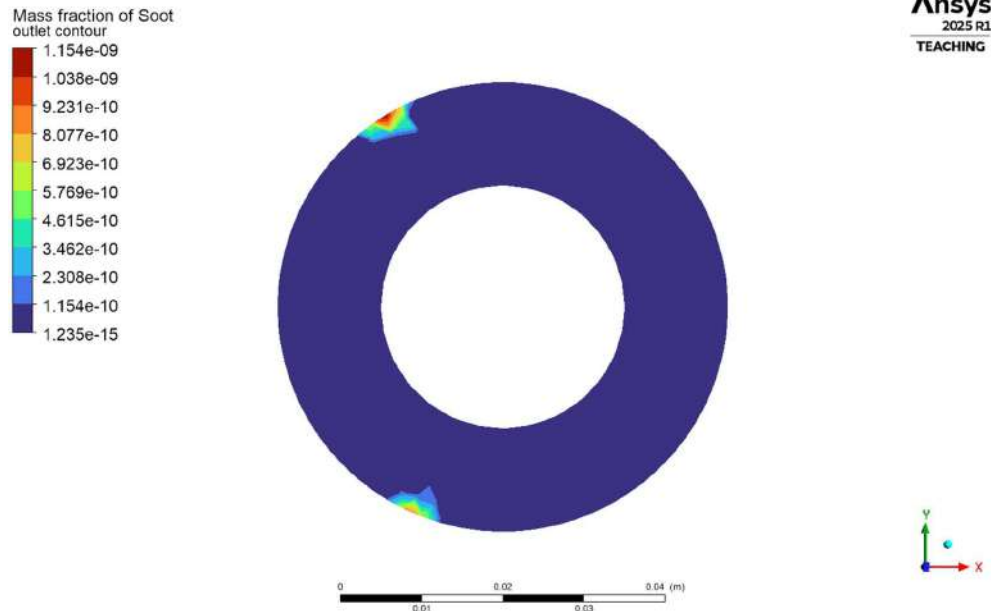
The combustion in the region continues and appears some soot particles for a few zones. The overall soot load is quite low due to better mixing and combustion.



4.1.6.2(c) Zone 2 Soot mass fraction

d. Outlet Soot mass fraction

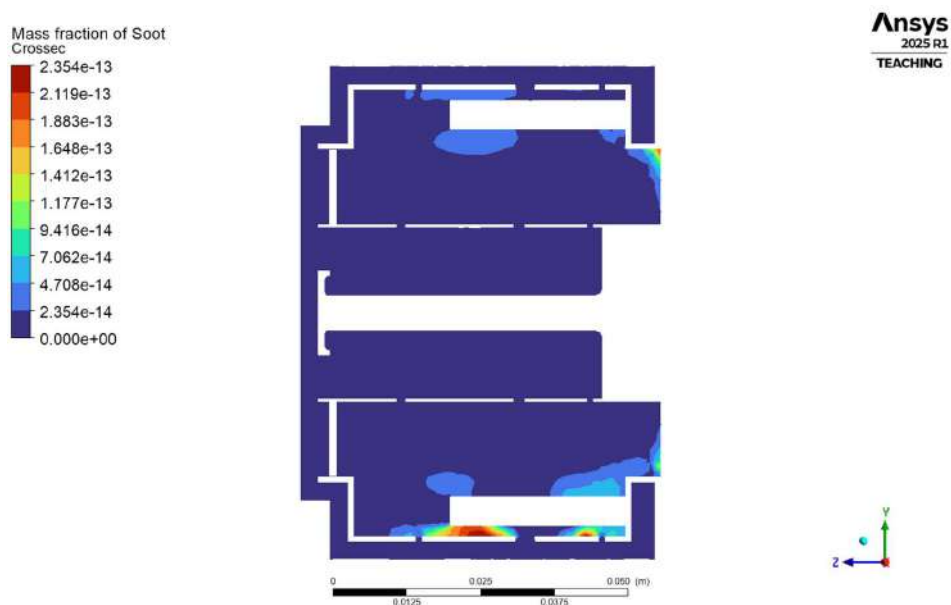
This is the soot concentration at the combustor outlet. The levels of soot are tiny throughout most of the area, which means that most soot particles get broken down before they escape through the exhaust gases.



4.1.6.2(d) Outlet Soot mass fraction

e. Cross-section Soot mass fraction

The Cross-section view showing soot along combustor internal path. Reaction zones contain small regions of soot, while the remaining regions are very low as a result of oxidation and mixing.

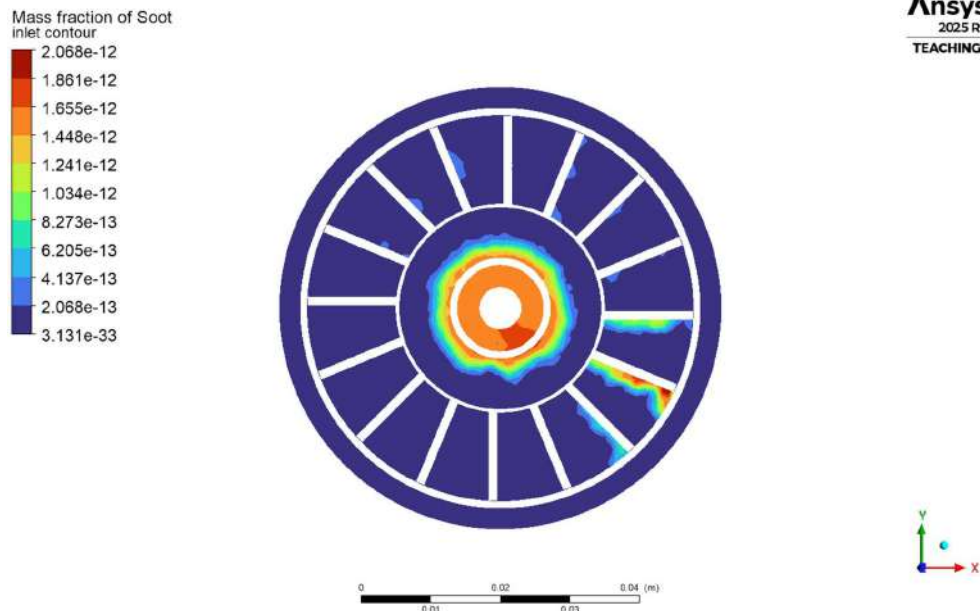


4.1.6.2(e) Cross-section Soot mass fraction

4.1.6.3 Soot contour 16 blade

a. Inlet Soot mass fraction

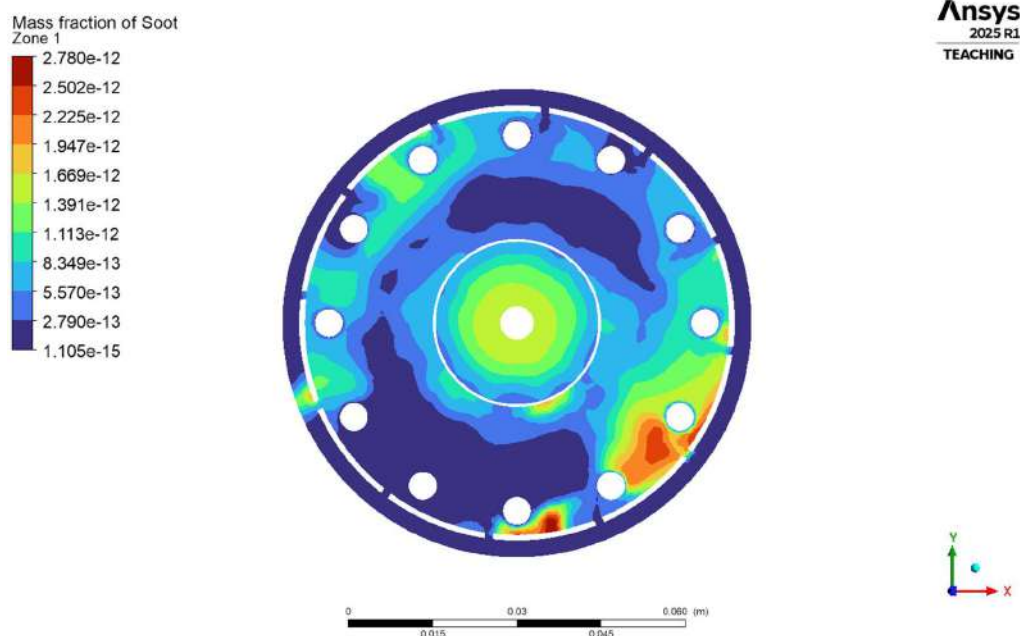
This plot represents the soot concentration at combustor inlet. The amount of soot formation lies very small and is concentrated near the center, revealing that there is little formation of soot in the entrance stage.



4.1.6.3(a) Inlet Soot mass fraction

b. Zone 1 Soot mass fraction

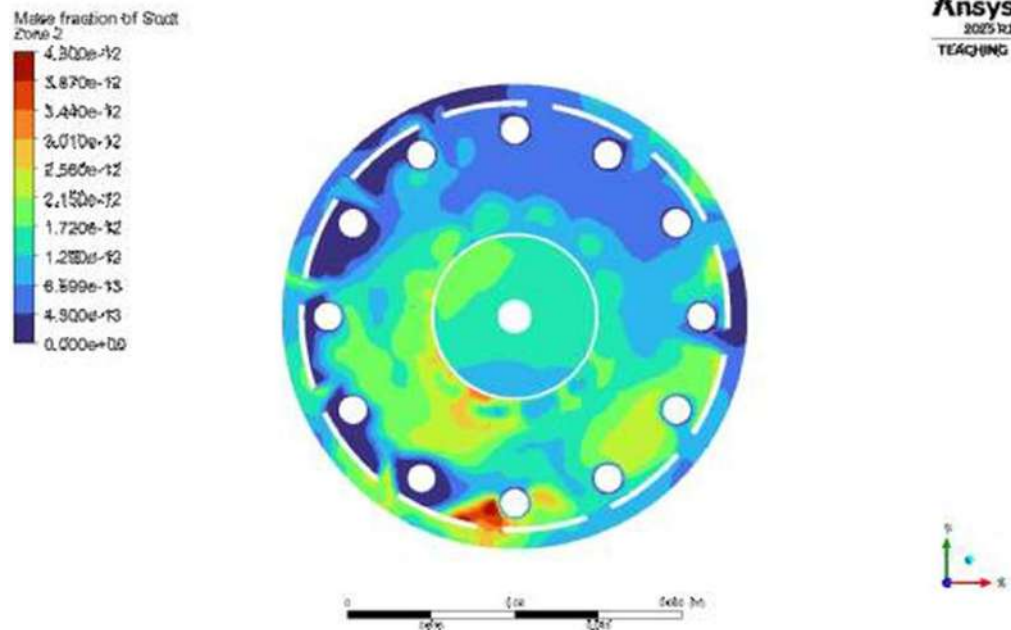
This contour is the first combustion zone, where fuel and air begin to mix and react. A few soot particles start to emerge in places where combustion is not quite complete.



4.1.6.3(b) Zone 1 Soot mass fraction

c. Zone 2 Soot mass fraction

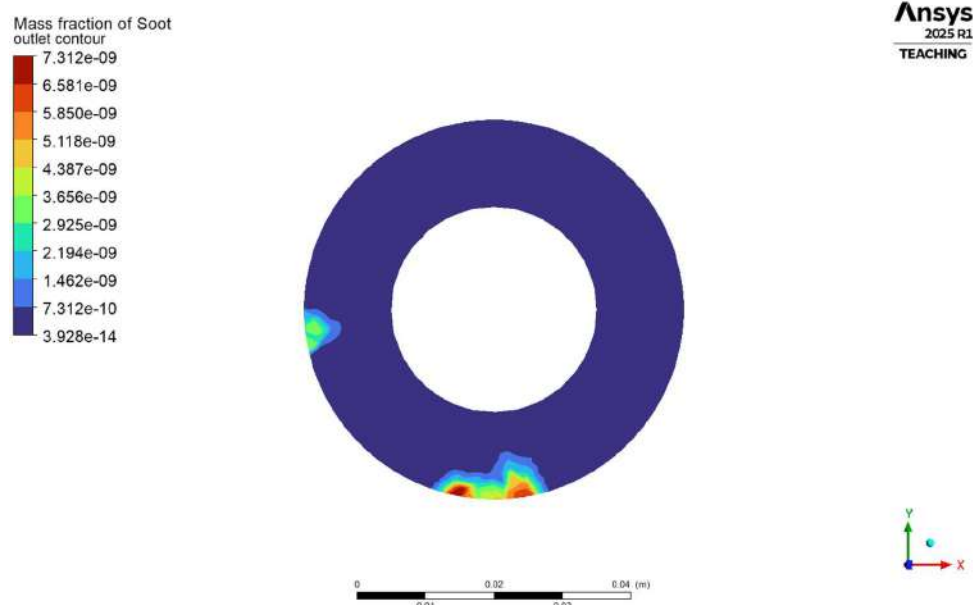
From the contour, within this zone the combustion process intensifies and the soot particles are dispersed over a larger extent. In some areas ones with high soot concentrations can be seen because of incomplete burning due to the fuel rich mixture.



4.1.6.3(c) Zone 2 Soot mass fraction

d. Outlet Soot mass fraction

The soot concentration at combustor outlet on the other hand, values across most areas are very low implying much soot is combusted and mixed in before effluent gasses escape.



4.1.6.3(d) Outlet Soot mass fraction

e. Cross-section Soot mass fraction

From this cross-section view showing where soot deposited along the internal path of the combustor. More specifically, the highest concentrations seem to be found around the combustion zones with the other regions kept low owing to enhanced mixing and oxidation.



4.1.6.3(f) Cross-section Soot mass fraction

4.2 Comparison of Flow and Combustion Characteristics for Different Swirler Vane Configurations

4.2.1 Number of Vanes vs Velocity (m/s)

No.of vanes	Inlet	Zone 1	Zone 2	Outlet
8	51.29	30.82	30.82	96.346
12	53.4113	29.3092	42.045	90.0766
16	64.9397	34.6582	33.5284	59.7101

Table.4.2.1 8,12, and 16 vanes Velocity values

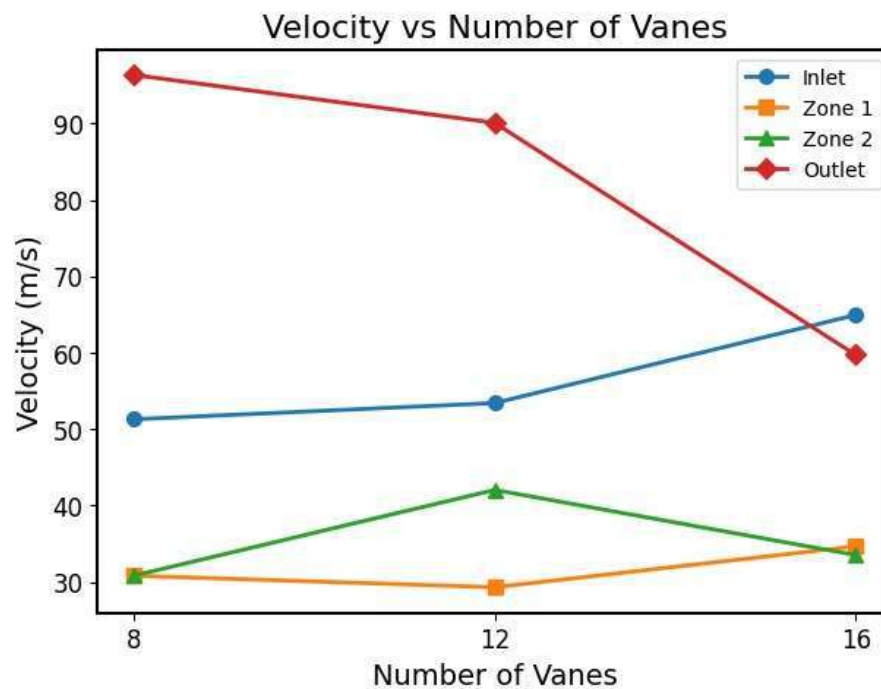


Fig.4.2.1 Number of Vanes vs Velocity (m/s)

Air Velocity Varying with No. of Vanes (8, 12 & 16) at Inlet, Zone 1, Zone 2 and Outlet With the increase in number of vanes, inlet velocity ranges from 51 m/s (8 vanes) to 65 m/s (16 vanes), signifies that, higher no of Van With respect to Velocity as they guide the incoming airflow more easily into system. For Zone 1, the speed is still relatively low and not very sensitive (29–35 m/s) thus confirming that flow in this area appears to be well-controlled with only a small sensitivity through variations in the vane number. This gradual increase in velocity indicates improved flow distribution and mixing in this zone, with the maximum velocity reaching about 42 m/s at 12 vanes and further decreasing slightly.

As it can be seen from the outlet, increasing the number of vanes reduces and

stabilizes exit velocity = 96 m/s (8 vanes) to ~60 m/s (16 vanes); a higher number of vanes results in more uniform airflow distribution inside the system.

In summary, the graphs reveal strong correlation between airflow distribution and even a slight change in the number of vanes; equilibrated flow in internal zones was obtained using 12 vanes, while the incorporation of 16 vanes produced lower outlet velocity and better control over the flow.

4.2.2 Number Of Vanes Vs Pressure (Pa)

No.of vanes	Inlet (Pa)	Zone 1 (Pa)	Zone 2 (Pa)	Outlet (Pa)
8	25198.5	18885	18590.9	311.944
12	21478.4	15468.8	15371	56.8676
16	13674.6	9063.71	11190.7	30.5436

Table.4.2.2 8,12, and 16 vanes Pressure values

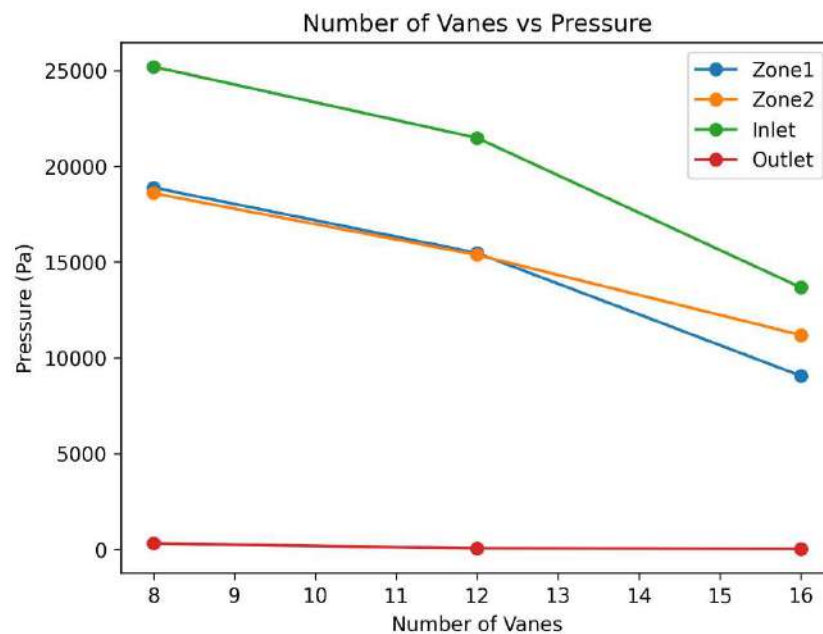


Fig.4.2.2 Number of Vanes vs Pressure (Pa)

The inlet drops from roughly 25,000 at 8 vanes to ~13500 at 16 vanes showing that with increasing number of vanes the intensity of incoming flow decreases as flow control/resistance increases. For this figure, the Zone 1 gradually reduced from nearly 18,800 to around at 9,000 indicates that due to the presence of extra vanes deviations are adjusted by better distribution across the pillar rabbits and energy or speed have decreased in this area. What is more, Zone 2 also has a decreasing trend from about around 18500 at 8 vanes to approximately 11200 at 16 vanes indicating that increasing the number of vanes help in stabilizing and controlling flow within the combustion or mixing region. So, at the outlet, that could potentially be a much smaller value than through all the other regions because

whatever is coming through the system has been dissipated, it has been controlled before it leaves.

4.2.3 Number Of Vanes Vs Temperature (K)

No.of vanes	Inlet (K)	Zone 1 (K)	Zone 2 (K)	Outlet (K)
8	1957.96	2189.82	1846.88	2039.47
12	1669.23	1584.07	1757.72	1792.26
16	1026.69	906.773	1031.51	759.231

Table.4.2.3 8,12, and 16 vanes Temperature values

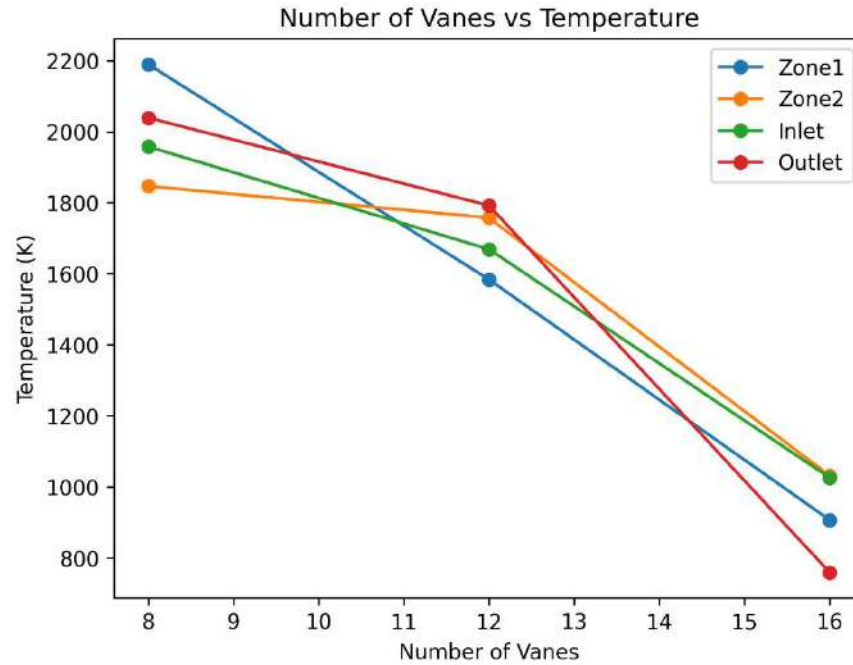


Fig.4.2.3 Number of Vanes vs Temperature (K)

The chart represents temperature fluctuation in each area (Inlet, Zone 1, Zone 2 and Outlet) between the increase of vanes from 8 to 16. At the inlet, temperature varies between 1950 K at eight vanes to approximately 1030 K for sixteen vanes, revealing that higher number of vanes helps slow down and therefore maintain control over the entering heat energy. The temperature at Zone 1 is highest for the 8 vanes similar to 2200 K and decreases dramatically as the number of vanes increases reaching 900 K at a configuration with 16 vanes, indicating that more vanes help improve distribution of thermal energy while reducing hot spot temperature. In Zone 2 is decreasing from around 1850 K to roughly 1030 K, indicating that with the increase in vane number, airflow tends to be more stable and mixing becomes significantly better. The outlet is a good reference point as it shows the relatively cool output, decreasing from approximately 2050 K (8 vanes) to almost 760 K (16 vanes), demonstrating higher vane quality plays an important role in cooling down the flow before its exit from the system.

In general, the graph indicates that increasing the vanes resulted in reduced temperature among all zones, which improved flow regulation and energy distribution as well as allowing for better thermal stabilization within the device.

4.2.4 Number Of Vanes Vs Turbulence Kinetic Energy (J/Kg)

No.of vanes	Inlet (J/Kg)	Zone 1 (J/Kg)	Zone 2 (J/Kg)	Outlet (J/Kg)
8	30.941	62.1698	79.9449	154.3
12	35.0833	69.9984	116.537	168.323
16	235.667	154.164	152.632	109.395

Table.4.2.4 8,12, and 16 vanes Turbulence Kinetic Energy values

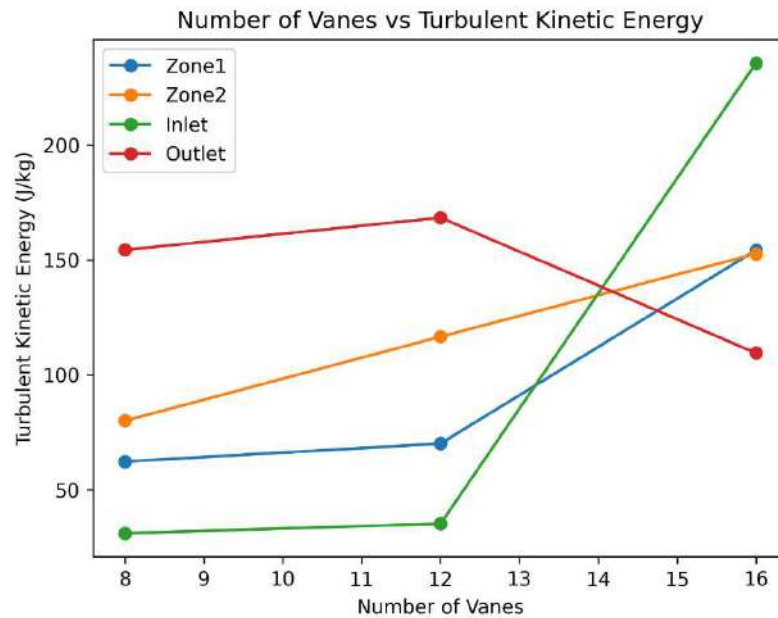


Fig.4.2.4 Number of Vanes vs Turbulence Kinetic Energy (J/Kg)

The extra turbulence induced in the incoming airflow is shown by the increase of turbulent kinetic energy at the inlet where it significantly raised from around 30 J/kg for 8 vanes up to close to 235 J/kg for each increased vanes. It was found in Zone 1 gradually increase from approximately 65 J/kg to the steadily turbulence kinetic energy up to about 153 J/kg this means that increasing vane number, improve the turbulent of this region and make large amount of distribution energy. In Zone 2, the turbulent kinetic energy increases from 80 J/kg at 8 vanes to roughly 152 J/kg at 16 vanes, indicating more efficient mixing and a better inner flow in the chamber. At the outlet, it can be seen from this figure that first turbine kinetic energy increases slightly (155 J/kg at 0 vanes -> approximately 168 J/kg 12 vanes) and then is decreased up to roughly 110/s in case of excess of 16 vanes, which suggests increased number if vanes contribute for reducing turbulence before leaving the system. So, the overall flow can be interpreted such that with increasing number of vanes leads to more turbulence and mixing in the system while less turbulence at outlet which in turn helps to maintain a controlled outflow from the exit.

4.2.5 Number Of Vanes Vs Mass Fraction of NO_x

No.of vanes	Inlet	Zone 1	Zone 2	Outlet
8	9.4734e-12	6.18525e-12	1.2983e-11	2.43391e-08
12	1.88178e-13	3.68883e-12	6.21259e-12	1.07774e-09
16	6.02408e-11	3.53241e-10	5.65307e-10	2.28061e-07

Table.4.2.5 8,12, and 16 vanes NO_x values

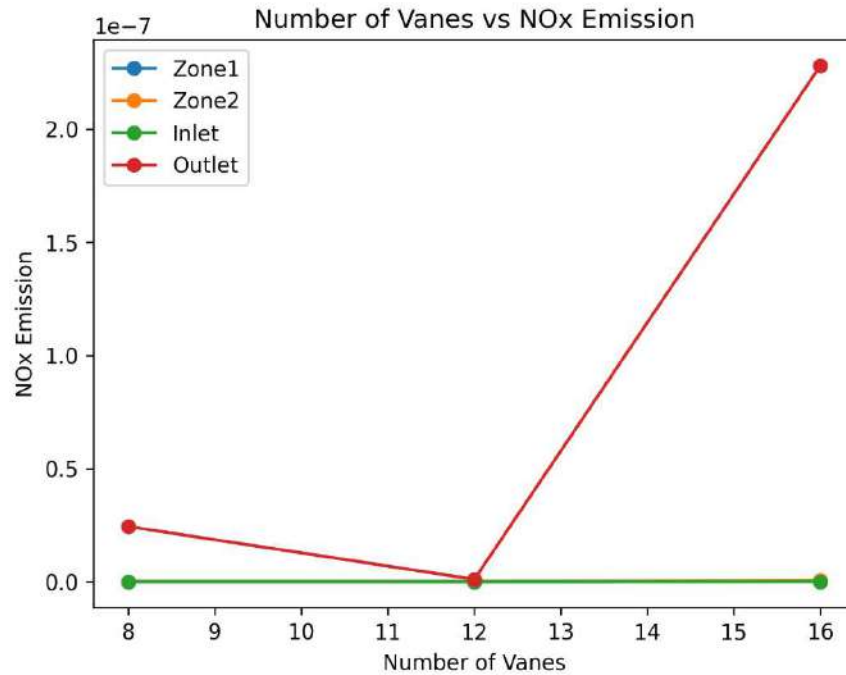


Fig.4.2.5 Number of Vanes vs NO_x

Variation of NO_x emission mass fraction throughout the mesh regions showing for 8, 12 and 16 vanes. It can be defined as that the mass fraction of NO_x for all vane numbers is nearly zero at the inlet, Zone 1 and Zone 2 as burnout occurs in latter stages of combustion rather than these initial areas. At 8 vanes the NO_x mass fraction at the outlet is approximately equal to 2.5×10^{-8} and drops almost to zero at 12 vanes, indicating that this vane configuration significantly reduces NO_x formation. For case with vane number of 16 however, NO_x mass fraction rises to nearly 2.3×10^{-7} due to enhanced turbulence levels and temperature gradients which leads to greater NO_x formation.

The graph shows that 12 vanes produce the minimum NO_x emission while 16 vanes cause increase in NO_x formation at outlet and confirms the effect of number of vanes on combustion characteristics and emission levels.

4.2.6 Number of Vanes Vs Mass Fraction of Soot

No.of vanes	Inlet	Zone 1	Zone 2	Outlet
8	2.26346e-14	2.06162e-14	4.7026e-14	3.2336e-10
12	1.89167e-15	1.28953e-14	1.10493e-14	1.9937e-11
16	2.04039e-13	5.94356e-13	1.24619e-12	1.87484e-10

Table.4.2.6 8,12, and 16 vanes Soot values

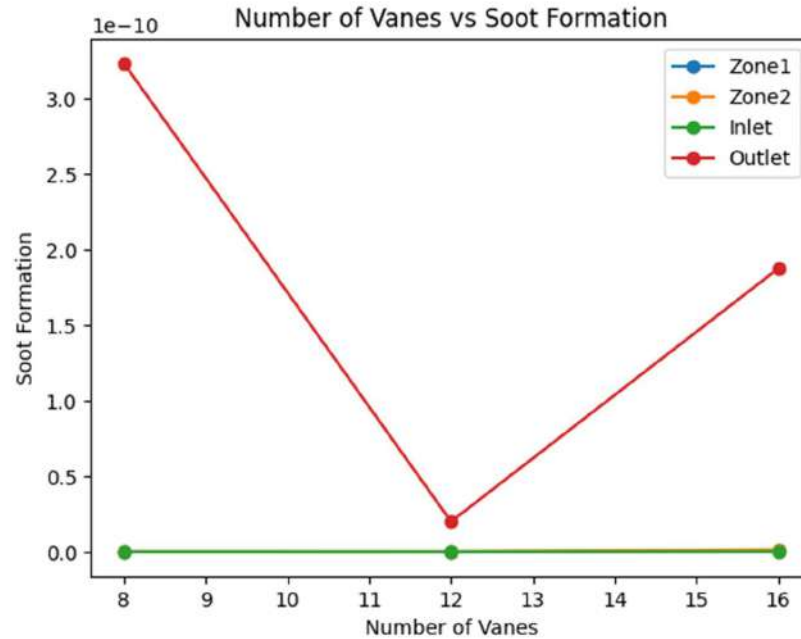


Fig.4.2.6 Number of Vanes vs Soot

The variation of mass fraction of soot with respect to number of vanes (8, 12, and 16) in different regions Inlet, Zone 1, Zone 2 and Outlet is shown as a graph in the Figure below. The mass fraction of soot is very close to ZERO at the inlet, Zone 1 and Zone 2 for all the numbers of vanes that soot formation only occurs after combustion with no presence in the inflow regions. At the outlet, there is more soot formed where volume fractions are highest at 8 vanes (around 3.2×10^{-10}), which indicates that combustion might not be complete or incomplete mixing of air and fuel.

As the number of vanes increases to 12, soot mass fraction reduces significantly to around 2.0×10^{-11} , which reveals the improved mixing and more complete combustion in the chamber. But for 16 vanes, the soot value increases back to about 1.9×10^{-10} , indicating that too many vanes may disrupt the optimal pattern of flow and slightly promote soot formation. The graph shows that 12 vanes are least soot generating which indicates better combustion efficiency and air – fuel mixing compared to both 8 and 16 vanes.

CHAPTER 5 Conclusion and Future Scope

5.1 CONCLUSION

The present works in this paper mainly focus on the procedure of designing axial swirler, and further investigation on flow characteristic of swirling flow generated by vane axial swirler by help of computational methods. Moreover, the effects of geometric parameters, namely vane angle and vane number and flow are also reported in the study.

Swirler design is carried out on the basis of semi-empirical equations. The design procedure using pressure drop method include three steps namely, mass flow calculation through air distribution, effective area calculation through pressure drops and sizing from the geometric relationship. The non-reacting swirling flow can be simulated very well through using $k-\epsilon$ realizable turbulence model provided in FLUENT that has been validated in this study.

Axial reverse flow velocity, turbulence along axial direction and pressure drop increases with the increase in angle of vanes. Namely the better fuel/air mixing and recirculation structure are obtained if the vane angle of swirler is larger but at the penalty of the pressure drop of combustors. Increasing the vane number increases the turbulence energy and decreases the axial flow velocity. For a given swirler, increasing mass flow at inlet decreases the pressure drop through the swirler and the pressure drop seems to keep constant if the mass flow is increased above a critical value. Thus, the mass flow through the swirler should be adjusted carefully when designing swirler. The results show that the axial velocity and radial velocity profiles have four different types of profiles.

5.2 Future Scope

This study primarily focused on analyzing axial swirler configurations with different blade counts using CFD simulations. While the results provide a clear understanding of how swirler geometry affects flow behavior and combustion performance, there are several areas that can be explored further to expand the findings of this work.

One possible direction for future work is the **experimental validation of the CFD results**. Performing laboratory experiments using a physical combustor model would help verify the numerical predictions and increase the reliability of the simulation outcomes.

Further studies could also investigate **alternative swirler designs**, such as curved vane swirlers, radial swirlers, or dual swirler configurations. These designs may generate different flow patterns and could potentially improve fuel–air mixing and flame stabilization inside the combustion chamber.

Another important aspect that can be explored is the **optimization of geometric parameters**, including blade angle, vane thickness, and hub diameter. Adjusting these parameters can significantly influence swirl intensity, turbulence levels, and pressure losses within the system. Optimizing these factors may lead to improved combustion efficiency while maintaining acceptable pressure drop.

In addition, future research may incorporate **more advanced combustion models and detailed chemical reaction mechanisms** to obtain more accurate predictions of pollutant formation, particularly nitrogen oxides (NO_x), carbon monoxide (CO), and unburned hydrocarbons.

Finally, the optimized swirler configurations developed through such studies could be applied to **practical combustion systems**, including gas turbine combustors, industrial burners, and propulsion systems. This would help improve system efficiency while minimizing harmful emissions.

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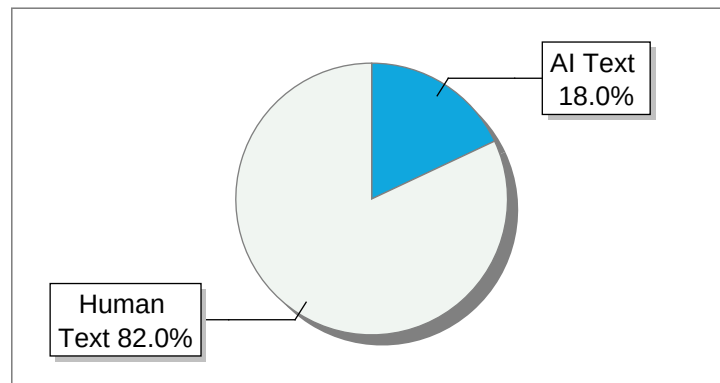
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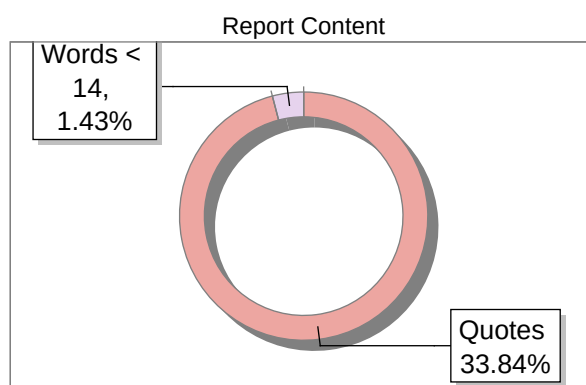
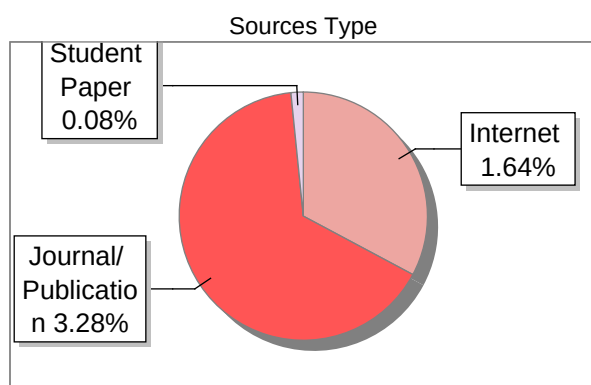
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